

From Rheology to Scaffold

*Engineering Principles and Operating Guide for Direct Ink Writing of CMC /
Sunflower-Oil Nanoemulsion Bioinks*

*Theory, dimensionless analysis, three worked-example inks, failure morphology, nanoemulsion
mechanics, biological considerations, lab protocols, and troubleshooting*

T. M. C. Rodrigues

R. Thiré

BioPoli – PEMM/COPPE/UFRJ

Version 3.0 June 8, 2026

Who this document is for and how to use it.

This is a self-contained engineering guide to printing CMC and CMC + sunflower-oil nanoemulsion bioinks on the piston-driven DIW gel printer. It is written for a new lab member or intern with an undergraduate engineering background (basic calculus, basic fluid mechanics) and no prior rheology experience. It is also a reference for experienced practitioners seeking a rigorous foundation for material selection, process parameter estimation, failure diagnosis, and regulatory justification.

The document is in four parts:

- **Part I – Theory** (Ch. 1–7): step-by-step engineering framework connecting rheological measurements to printing parameters. Includes failure morphology and dimensionless analysis.
- **Part II – Inks and Instruments** (Ch. 8–11): ink characterisation, nanoemulsion mechanics, rheological testing protocols, and extended material classes.
- **Part III – Protocols** (Ch. 12–17): MATLAB and Python workflows, calibration-cube sequence, and troubleshooting decision tree.
- **Part IV – Extended Topics** (Ch. 18–19): biological and regulatory considerations; statistical optimisation methods.

Navigation guide. If you only need to print and trust the upstream calibration, skip to Part III. If a print fails and you do not understand why, the answer is in Part I, Chapter 7. If you are preparing a manuscript or patent, Part IV is mandatory reading.

Contents

How to Read This Book — Three Paths	5
Conceptual Framework: DIW as a Coupled System	5
Part I – Theory	6
1 Why DIW is Not FFF	6
1.1 Three control modes, one printer family	7

2	Rheology in Depth	8
2.1	Stress, strain, and strain rate	8
2.2	Yield stress and the two-viscosity problem	8
2.3	Why DIW needs shear thinning	9
2.4	Two constitutive models you actually use	9
2.5	Viscoelasticity: G' and G''	10
2.6	Recovery: the time-domain check	12
2.7	Dimensionless rheological numbers	13
3	The Piston-Driven Hydraulic Loop	15
3.1	Mass balance: from V_p to flow	15
3.2	Velocity profile inside the needle	17
3.3	Mass balance: from V_p to print speed	17
4	From Flow to Pressure: Hagen–Poiseuille for Non-Newtonian Fluids	23
4.1	Newtonian baseline	24
4.2	Power-Law extension	24
4.3	Cross-model extension (Rabinowitsch–Weissenberg)	26
4.4	Which model to trust?	28
4.5	Wall shear rate and stress	28
5	Temperature and Process-Robustness	31
5.1	Temperature dependence of viscosity	32
5.2	Flow instabilities and the smooth-filament limit	33
5.3	Reproducibility checklist	33
6	Flow Factor, Die Swell, and the Slicer Translation	34
6.1	The crucial inversion: k_{flow} is not the number you type into the slicer	36
6.2	Die-swell proxy from Cross m	37
7	The Maximum Scaffold Height	40
7.1	Three heights, three scales	40
7.2	Deriving $h_{\text{scaffold}}^{\text{max}}$	40
7.3	What is σ_{max} ? Three criteria	41
7.4	Three numbers per ink	43
7.5	Lattices and unsupported spans – the Smay criterion	43
8	Failure Morphology: Physical Causes and Corrective Actions	48
8.1	Under-extrusion	48
8.2	Over-extrusion	48
8.3	Die swell excess	49
8.4	Scaffold collapse (base bulging)	49
8.5	Lattice strut buckling	49
8.6	Intermittent extrusion (motor skip)	49
8.7	Layer-by-layer fusion (loss of resolution)	50
8.8	Stringing during travel	50
8.9	Rough filament surface / extrudate distortion	50

Part II – Inks and Instruments

51

9 The Three Project Inks	51
9.1 C15 – CMC baseline	51
9.2 C15 Gira 5.5 – CMC + sunflower-oil nanoemulsion (NE)	51
9.3 Bozano – commercial hair gel	52
10 Nanoemulsion Mechanics	53
10.1 Nanoemulsion structure and droplet characteristics	53
10.2 How the NE stiffens the CMC matrix	53
10.3 Emulsion stability under shear	53
10.4 Structural recovery enhancement mechanism	54
10.5 Carrier model rationale	54
11 Rheological Tests: Instrument and Battery	54
11.1 Instrument	54
11.2 Standard test battery	55
11.3 Interpreting pre-shear effects	55
11.4 Interpreting the SAOS output: a field guide	55
12 Extended Material Classes	55
12.1 Cellulose-based inks	55
12.2 Alginate-based inks	56
12.3 Gelatin-based inks	56
12.4 Composite systems: natural + synthetic	56
Part III – Protocols	59
13 Workflow Overview	59
14 Step 2: Python Analysis	59
14.1 Pre-run checklist	59
14.2 Flow-curve fitting – <code>Fit_Muitos_Modelos_v4.py</code>	59
14.3 Recovery – <code>Recovery_v1.py</code>	60
14.4 Scaffold-height extraction – <code>extract_hmax_v2.py</code>	60
15 Step 3: MATLAB Simulation – <code>run_solver_v3.m</code>	60
16 Step 4: Calibration-Cube Protocol (C1–C5)	61
16.1 C1 – print-speed sweep	61
16.2 C2 – mass-per-line (k_{flow}) calibration	61
16.3 C3 – layer-height sweep	62
16.4 C4 – vertical-tower test	62
16.5 C5 (optional) – bridging test for lattice scaffolds	62
17 Step 5: Final Scaffold Print	62
18 Troubleshooting: Failure-Mode Decision Tree	63
19 The Friday Print – Closing the Loop	63
Part IV – Extended Topics	64

20 Biological and Regulatory Considerations	64
20.1 Cell viability and wall shear stress	64
20.2 Plug flow and cell protection	65
20.3 Long-duration print considerations	67
20.4 MAPA-oriented regulatory considerations	67
21 Statistical Optimisation: DOE and Response Surface Methodology	68
21.1 Overview	68
21.2 Continuous vs. categorical variables	68
21.3 Model selection recommendations	68
21.4 Responses and their transformations	69
A Glossary	69
B Notation and Unit Reference	71
C Worked End-to-End Example: C15 Gira 5.5 on 21G	71
D Engineering Design Rules – Consolidated Reference	72
E Single-Page Run Sheet	73
F Software Setup Guide	73
F.1 Python environment	73
F.2 MATLAB	73
F.3 LaTeX	73

How to Read This Book — Three Paths

This book teaches at three depths from one text. Read the *core spine* (plain text, worked examples, takeaways, design rules) on any path; then add the tiers your role needs.

Tier (badge)	What it gives you	Who should read it
Core spine	Intuition, the result, real numbers, takeaways	Everyone
IN PRACTICE (teal)	Bench-level “what you actually do”	Applied + everyone
DERIVATION (grey)	Full step-by-step proof of the result	Graduate-research ; skippable on first read
GOING DEEPER (in-indigo)	Advanced connections (Cox–Merz, N_1 /Tanner, structural kinetics)	Graduate-research / curious
Exercises ● ● ● ● ●	Self-check, by difficulty	By level; solutions in instructor edition

Profile	Suggested path
Applied / practitioner	Core spine + IN PRACTICE + ● exercises
Mixed (grad/adv. undergrad)	Core spine + IN PRACTICE + DERIVATION + ● ●
Graduate-research	All tiers + GOING DEEPER + ● ● ● ● ●

Instructors: compile the solutions edition with pdf_{latex} `"\def\SOLUTIONS{1}\input{Printing_Parameters`

Conceptual Framework: DIW as a Coupled System

Direct Ink Writing is not a single-physics problem. It is a **coupled system** in which formulation chemistry, network rheology, extrusion hydraulics, deposition geometry, and scaffold mechanics interact simultaneously. Every printing parameter is both a cause and a consequence in this web. The central thesis of this textbook is: *understand the coupling, and every slicer entry follows by algebra*.

Figure 1 shows the complete coupling map.

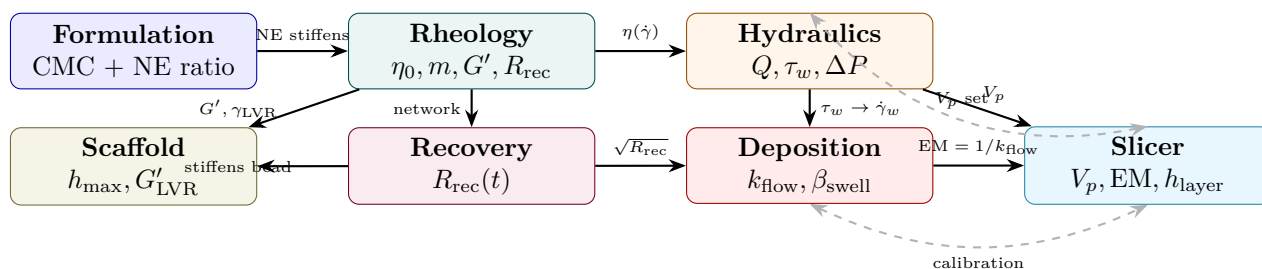


Figure 1: Systems-level coupling map for piston-driven DIW. Solid arrows: physical causation. Dashed: feedback via calibration or operator control. The operator directly commands only V_p (piston velocity); every other quantity is a consequence of the ink’s rheology and the system geometry.

PARAMETER INTERACTIONS

How parameters couple – the four most important interactions.

1. **Increasing V_p raises Q , which raises $\dot{\gamma}_w$, which lowers η (shear thinning), which raises τ_w sub-linearly.** Pressure does not scale linearly with V_p for shear-thinning inks: doubling V_p typically increases ΔP by a factor of 2^n ($n < 1$).

2. **Increasing V_p also raises v_{print} , which shortens dwell time per layer, which changes the SAOS criterion controlling $h_{\text{scaffold}}^{\text{max}}$.** Faster printing can mean the $G'(\omega = 1)$ criterion applies rather than the $G'(\omega = 0.01)$ one – apparently improving height capacity.
3. **Adding NE raises η_0 and G'_{LVR} but slightly lowers R_{rec} .** The first two effects improve scaffold height and resist gravity; the last increases the required Extrusion Multiplier. Both directions must be tracked simultaneously.
4. **Narrower needle reduces R_n , massively raising ΔP ($\propto R_n^{-4}$) and $\dot{\gamma}_w$ ($\propto R_n^{-3}$) at the same V_p .** Switching from 21G to 22G at fixed V_p raises wall shear stress and may breach cell-safety thresholds even though the ink parameters are unchanged.

RUNNING EXAMPLE – The Friday print

Monday, 9:00. You are Ana, a new intern at BioPoli. Your supervisor hands you a sealed jar of *C15 Gira 5.5* (CMC + sunflower-oil nanoemulsion ink, the **headline ink** of this project), a 21G blunt needle, and the following brief:

“By Friday at 16:00 I need a printed solid block, roughly $1 \times 1 \times 2 \text{ cm}^3$, of this ink. The shape doesn’t matter – any rectangular prism is fine. The Wednesday morning slot on the printer is yours. Don’t waste ink.”

Your job between now and Wednesday: extract, from the analysis scripts and this book, every number you must type into the slicer. Wednesday: calibrate at the bench. Thursday: print. Friday: document and deliver.

By the end of this book you will have answered, in order:

1. **Can this ink print at all?** (Ch. 2)
2. **What piston velocity V_p should I command?** (Ch. 3)
3. **Is the resulting extrusion pressure safe?** (Ch. 4)
4. **What flow factor k_{flow} (and slicer Extrusion Multiplier) do I enter?** (Ch. 6)
5. **Will a 20 mm-tall block survive its own weight?** (Ch. 7)
6. **What does my Wednesday calibration morning look like?** (Ch. 16)

After each chapter above, you will find a box titled “Running Example – Solution to QN”, solving one question with explicit numbers and marking what is still unknown.

Part I – Theory

1 Why DIW is Not FFF

Direct Ink Writing (DIW) extrudes a viscous fluid through a needle and relies on the fluid’s own rheology to retain its shape after deposition. Fused Filament Fabrication (FFF) melts a thermoplastic, extrudes it as a controlled-diameter molten strand, and solidifies it on contact with the cooler build plate. The two processes look similar (an extruder moves over a build plate), but the physics is fundamentally different, and the slicer firmware – written for FFF – does *not* translate cleanly.

PHYSICAL INTUITION

In FFF, the filament has a fixed diameter (1.75 or 2.85 mm). The extruder pushes a known length of filament, giving a known volume of melt. “How much material was deposited?” is answered by E-axis geometry alone.

In DIW, the ink is a free fluid in a syringe. “How much material was deposited?” depends on: (i) how fast the piston moves; (ii) how the ink responds to shear in the needle; (iii) how the deposited filament swells or relaxes after leaving the needle; (iv) how the network recovers in the time between deposition and the next layer.

The slicer knows none of that. It still thinks it is FFF. Our job is to compute the right numbers *outside* the slicer and feed them in as if they were FFF parameters.

1.1 Three control modes, one printer family

DIW printers differ in what the operator commands:

1. **Pneumatic DIW.** Operator commands ΔP (the gas pressure driving the syringe). Flow rate Q is the consequence. Dominant in academic literature; simpler hardware and lower cell stress at constant pressure.
2. **Screw / progressive-cavity DIW.** Operator commands auger rotation, which translates to a quasi-constant Q . Robust against air bubbles; more expensive; common in commercial bioprinting.
3. **Piston / volumetric DIW.** Operator commands piston linear velocity V_p (via a stepper-motor lead screw). Volumetric flow Q is fixed by syringe geometry. ΔP is whatever the rheology demands. **This is the lab’s printer.**

CONTROL PARADIGM

The lab DIW printer is **piston-driven / volumetric**. The single flow input is the piston velocity V_p (mm/s), entered into the slicer firmware as the E-axis feedrate. Everything downstream – Q , $\bar{v}_n$, τ_w , ΔP , v_{print} – follows from V_p and the ink rheology. They are **outputs and diagnostics**, not knobs.

Implication for literature reading. A paper that says “we extruded at 100 kPa” must be translated to a V_p on our printer before the number is useful. The translation goes through the rheological model (Ch. 4).

KEY TAKEAWAYS

- DIW is a rheology-dependent process; FFF slicers must be “tricked” with pre-computed parameters.
- On the lab printer, V_p is the only flow control input. ΔP is a diagnostic, never a knob.
- Comparing across literature requires knowing the control mode (pneumatic, screw, or piston).

ENGINEERING DESIGN RULES

DR 1.1. Before reading any DIW paper, identify the control mode. Pneumatic data give pressure–flowrate relationships; piston data give velocity–flowrate relationships. They are not interchangeable without going through the constitutive model.

DR 1.2. Never enter ΔP directly into the slicer – it does not exist as a slicer input on this printer. If a calibration strategy asks for a pressure setting, it was written for pneumatic DIW.

2 Rheology in Depth

Rheology is the science of how matter deforms and flows. For DIW we need seven concepts: stress, strain rate, viscosity, shear thinning, yield stress, viscoelasticity, and structural recovery.

2.1 Stress, strain, and strain rate

Motivation. Every pressure, velocity, and shape prediction in this book is built on three quantities; confuse stress with strain rate, or report a viscosity without the shear rate it was measured at, and every downstream number is meaningless.

Place a fluid between two parallel plates separated by a gap h . Hold the bottom plate fixed; slide the top plate at velocity V under a tangential force F over area A . The fluid is subjected to:

- a **shear stress** $\tau = F/A$, in pascals (Pa),
- a **shear strain rate** $\dot{\gamma} = V/h$, in inverse seconds (s^{-1}).

For a Newtonian fluid the two are linearly related:

$$\tau = \eta \dot{\gamma} \quad (1)$$

where η (Pa s) is the dynamic viscosity. Water has $\eta \approx 10^{-3}$ Pa s; honey, $\eta \approx 10$ Pa s; our CMC inks span $\eta \sim 100$ Pa s at high shear to $\eta \sim 10^3$ Pa s at rest – four orders of magnitude.

2.2 Yield stress and the two-viscosity problem

Motivation. Get this wrong and the ink either won't hold its shape after deposition (no yield stress, so it slumps) or won't start flowing at a safe pressure (too high a yield stress for the syringe). The yield stress is the single number that decides which of those two failures you get.

Many structured fluids behave like solids at low stress and like viscous fluids above a critical threshold called the **yield stress** τ_y . The simplest model is the Bingham fluid:

$$\tau = \tau_y + \eta_p \dot{\gamma} \quad (\tau > \tau_y); \quad \dot{\gamma} = 0 \quad (\tau \leq \tau_y). \quad (2)$$

For DIW, a non-zero τ_y is advantageous: once extruded, the material stops flowing at rest and holds its shape. The Herschel-Bulkley model generalises this to power-law viscosity above yield:

$$\tau = \tau_y + K \dot{\gamma}^n \quad (\tau > \tau_y). \quad (3)$$

In practice, the inks in this project behave as *apparent* yield stress fluids: the zero-shear plateau of the Cross model (η_0) is so high that the material appears solid on lab timescales. We do not formally fit τ_y but we extract an equivalent from SAOS data (Ch. 7).

DERIVATION

Why $\tau_y \approx G'_{\text{LVR}} \gamma_{\text{LVR}}$. In the linear viscoelastic regime the elastic stress stored at strain γ is $\tau_{\text{el}} = G' \gamma$. The network yields where the microstructure can no longer store additional elastic energy — i.e. at the end of the LVR, $\gamma = \gamma_{\text{LVR}}$, where G' first departs from its plateau. Evaluating the stored elastic stress there,

$$\tau_y \approx G'_{\text{LVR}} \gamma_{\text{LVR}}. \quad (4)$$

This is a *lower-bound, oscillatory* estimate; flow-onset (dynamic) yield stresses from a stress ramp are typically larger.

Three ways to measure τ_y (and why they disagree).

1. **Steady stress ramp.** Ramp τ up; τ_y is where the apparent viscosity peaks then collapses (the “viscosity-divergence” yield point). Gives the *dynamic* yield stress.
2. **Flow-curve model extrapolation.** Fit Herschel–Bulkley or Casson and read the intercept τ_y at $\dot{\gamma} \rightarrow 0$ (Eq. 3). Model-dependent; HB and Casson can differ by tens of percent.
3. **Oscillatory amplitude sweep.** Use Eq. 4, or take the $G'-G''$ crossover stress. Gives the *static/elastic* yield stress and is what this project uses (no formal τ_y fit).

A factor-of-two spread between methods is normal — always state which method produced a quoted τ_y .

WORKED EXAMPLE – the three inks

The oscillatory yield-stress proxy for the three inks. Applying Eq. 4 to the SAOS table below (with γ_{LVR} in absolute strain):

- C15 Gira 5.5: $916 \times 0.609 = 558 \text{ Pa}$,
- C15: $270 \times 0.64 = 173 \text{ Pa}$,
- Bozano: $348 \times 0.137 = 48 \text{ Pa}$.

The NE composite has the largest proxy yield stress (it both stiffens the network and keeps a wide LVR); Bozano, despite its high rest viscosity, has the smallest because its LVR is narrow. These are *lower-bound, oscillatory* values — a stress-ramp τ_y would read higher.

IN PRACTICE

On the MCR 501, run the *amplitude sweep first*: read G'_{LVR} off the low-strain plateau and take γ_{LVR} where G' has dropped 10% from it; their product (Eq. 4) is your working yield stress. If you only have a steady flow curve, fit Herschel–Bulkley and use its intercept instead — and expect a larger number than the oscillatory proxy. Always note which method produced a quoted τ_y .

2.3 Why DIW needs shear thinning

If our inks were Newtonian, we would face a brutal compromise: a viscosity high enough to keep the printed filament from slumping (say, 1000 Pa·s) would require extrusion pressures orders of magnitude above any safe syringe rating. The only way out is to design an ink whose viscosity drops under shear and recovers at rest:

$$\eta(\dot{\gamma}) = \text{decreasing function of } \dot{\gamma}. \quad (5)$$

PHYSICAL INTUITION

A good DIW ink has two effective viscosities:

- a *low* in-needle viscosity, where $\dot{\gamma} \sim 10\text{--}10^3 \text{ s}^{-1}$, so ΔP stays manageable;
- a *high* post-deposition viscosity, where $\dot{\gamma} \rightarrow 0$, so the filament resists gravity.

The printability problem reduces to: *does the ink switch fast enough between these two regimes, and does it recover fast enough to hold successive layers?*

2.4 Two constitutive models you actually use

Motivation. A pressure-drop calculation cannot be fed a table of raw data points; it needs a closed-form $\eta(\dot{\gamma})$. Pick the wrong model – or use it outside the shear-rate range it was fitted on – and the predicted extrusion pressure can be off by orders of magnitude.

The viscosity-shear-rate curve $\eta(\dot{\gamma})$ is fitted to a mathematical form so it can be plugged into pressure-drop calculations. The two models used in this project are the Power Law and the Cross model.

Power Law.

$$\eta(\dot{\gamma}) = K \dot{\gamma}^{n-1}, \quad \tau = K \dot{\gamma}^n. \quad (6)$$

Two parameters: K (Pas ^{n}) sets the viscosity scale; n ($0 < n < 1$ for shear thinning) controls how rapidly the viscosity drops. Newtonian is recovered at $n = 1$.

Virtues: analytical extrusion solutions (Eq. 30), two-parameter simplicity, easy to invert.

Limitations: predicts $\eta \rightarrow \infty$ at $\dot{\gamma} \rightarrow 0$ and $\eta \rightarrow 0$ at $\dot{\gamma} \rightarrow \infty$ – both unphysical. Valid only within a single decade of the fitted $\dot{\gamma}$ range.

Cross model.

$$\eta(\dot{\gamma}) = \eta_\infty + \frac{\eta_0 - \eta_\infty}{1 + (\lambda \dot{\gamma})^m}. \quad (7)$$

Four parameters: η_0 (zero-shear rest viscosity), η_∞ (infinite-shear viscosity), λ (transition time constant – $1/\lambda$ is the centre-of-transition shear rate), and m (the sharpness exponent, $0 < m \leq 1$).

Virtues: finite rest and high-shear viscosities, smooth transition, physically motivated, extends naturally across many decades of $\dot{\gamma}$.

Limitations: four parameters increase fitting sensitivity; η_∞ is often poorly constrained at high $\dot{\gamma}$.

Interpreting m physically. $m \rightarrow 1$ means the ink transitions sharply from η_0 to η_∞ – almost a binary switch. $m \approx 0.5$ means a gradual decline. A high m implies the ink stores very little elastic energy during shear-thinning (less die swell on exit). A low m implies broader energy storage and larger die swell.

WORKED EXAMPLE – the three inks

Cross parameters for the three inks (Ramp 1, no pre-shear, Fit_Muitos_Modelos_v4):

Ink	η_0 [Pa s]	η_∞ [Pa s]	λ [s]	m [-]	R^2
C15	578.3	0.0163	3.71	0.811	0.9956
C15 Gira 5.5	2439.5	0.001	4.10	0.988	0.9882
Bozano	6354.6	0.932	73.05	0.866	0.9937

Reading: η_0 ranks Bozano $\gg$ NE $\gg$ C15 at rest. The NE adds roughly a factor of four to the base CMC. Bozano is in a different class. λ tells you where the transition is centred: $1/\lambda \approx 0.25 \text{ s}^{-1}$ for C15 and NE vs. $\approx 0.014 \text{ s}^{-1}$ for Bozano. $m = 0.988$ for the NE means an almost-binary switch – minimal elastic energy storage – hence minimal die swell.

2.5 Viscoelasticity: G' and G''

The flow curve tells us how the ink behaves under *continuous* shear. But during printing, deposition is followed by rest, then by the next layer; the ink must also respond well under *small, oscillating* strains. That is the domain of small-amplitude oscillatory shear (SAOS).

Apply a sinusoidal strain $\gamma(t) = \gamma_0 \sin(\omega t)$ at fixed amplitude γ_0 and angular frequency ω . The resulting stress is also sinusoidal but phase-shifted:

$$\tau(t) = \gamma_0 [G'(\omega) \sin \omega t + G''(\omega) \cos \omega t]. \quad (8)$$

We extract two moduli:

- $G'(\omega)$ – the **storage modulus**: in-phase with strain, captures the elastic / solid-like part.

- $G''(\omega)$ – the **loss modulus**: $\pi/2$ out of phase, captures the viscous / liquid-like part.

The ratio $\tan \delta = G''/G'$ (loss tangent) summarises the character: $\tan \delta < 1$ means gel-like; $\tan \delta > 1$ means liquid-like.

DERIVATION

The single-mode Maxwell model and the relaxation time. A spring (modulus G) in series with a dashpot (viscosity η) gives relaxation time $\lambda_M = \eta/G$. Under oscillatory strain $\gamma = \gamma_0 e^{i\omega t}$ the Maxwell constitutive equation $\tau + \lambda_M \dot{\tau} = \eta \dot{\gamma}$ yields the complex modulus $G^* = G \frac{i\omega\lambda_M}{1 + i\omega\lambda_M}$; separating real and imaginary parts gives

$$G'(\omega) = G \frac{(\omega\lambda_M)^2}{1 + (\omega\lambda_M)^2}, \quad G''(\omega) = G \frac{\omega\lambda_M}{1 + (\omega\lambda_M)^2}. \quad (9)$$

G' and G'' cross ($G' = G''$, i.e. $\tan \delta = 1$) exactly at $\omega_c \lambda_M = 1$. Hence the material relaxation time is read directly off the crossover:

$$\lambda_{\text{relax}} = 1/\omega_c. \quad (10)$$

The Kelvin–Voigt arrangement (spring *parallel* to dashpot) instead describes a solid that creeps to a finite strain — the right picture for a *deposited, resting* bead, whereas Maxwell suits the *flowing* ink.

GOING DEEPER

You now have two “ λ ”s — they are not the same number. The Cross fit gives a λ whose inverse $1/\lambda$ is the shear rate at which the *steady* flow curve transitions from η_0 to η_∞ ; it is a property of the nonlinear, large-deformation flow. The SAOS crossover gives $\lambda_{\text{relax}} = 1/\omega_c$ (Eq. 10), the *linear*-viscoelastic relaxation time measured at vanishingly small strain. They probe different physics — one the breakdown of structure under flow, the other the slow rearrangement of an undisturbed network — so there is no reason they must coincide, and for structured inks they typically differ by an order of magnitude. When the Deborah number is evaluated for post-deposition stability ($\text{De} = \lambda_{\text{relax}} \omega$, Sec. 2.7), it is the SAOS λ_{relax} — not the Cross λ — that belongs in the formula, because shape retention is a small-strain, linear-regime question.

Connecting G'/G'' to print outcomes.

Rheological feature	What it predicts	Failure if violated
High G'_{LVR}	Filament stiff after deposition	Layer collapse; gravity-driven sag
Wide γ_{LVR}	Tolerant to deformation	None – wider is better
Flat $G'(\omega)$ (true gel)	Stable across all print timescales	Quasi-static collapse on slow prints
Sloped $G'(\omega)$ (critical gel)	Modulus drops at long timescale	Collapse during slow or paused prints
High G'' / low $\tan \delta$	Energy dissipation after deposition	Elastic rebound, bouncing filament

Two SAOS tests, two pieces of information.

1. **Amplitude sweep.** Fix ω , sweep γ_0 small to large. At small strain, G' is constant – the *linear viscoelastic regime* (LVR). At the critical strain γ_{LVR} (defined where G' has dropped 10% from its plateau), the network is permanently damaged. Beyond γ_{LVR} the response is nonlinear.

2. **Frequency sweep.** Fix γ_0 inside the LVR, sweep ω . $G'(\omega)$ encodes the network's timescale response: high ω probes fast (in-needle) dynamics; low ω probes slow (post-deposition, layer-stacking) dynamics.

WORKED EXAMPLE – the three inks

SAOS-derived numbers for the three inks:

Ink	G'_{LVR} [Pa]	γ_{LVR} [%]	$G'(\omega = 1 \text{ rad s}^{-1})$ [Pa]
C15	270	64.0	114
C15 Gira 5.5	916	60.9	228
Bozano	348	13.7	346

Reading: Adding NE to C15 raises G'_{LVR} from 270 to 916 Pa ($3.4\times$ stiffening) without narrowing the LVR (61% vs. 64%). The NE acts as a *reinforcing filler*, not a crosslinker. A crosslinker would narrow the LVR. Bozano has a comparable absolute modulus at $\omega = 1$ (346 Pa) but a far narrower LVR (13.7%): it is stiff but brittle to shear deformation.

GOING DEEPER

Cox–Merz: one curve from two experiments. For many polymer solutions the steady-shear viscosity and the magnitude of the complex viscosity coincide when shear rate equals angular frequency:

$$\eta(\dot{\gamma})|_{\dot{\gamma}=\omega} = |\eta^*(\omega)| = \frac{\sqrt{G'^2 + G''^2}}{\omega}. \quad (11)$$

This is why the steady flow curve (stress domain) and the SAOS sweep (oscillatory domain) are not independent stories — they should overlay. *Filled and strongly structured systems routinely violate Cox–Merz:* a measurable gap between $\eta(\dot{\gamma})$ and $|\eta^*(\omega)|$ is itself evidence of a yield stress or particle network — expected for the CMC+NE composite. Checking the overlay is a free consistency test on your two fits (Cox and Merz, 1958).

2.6 Recovery: the time-domain check

The amplitude sweep tells us *when* the network breaks; the recovery test tells us *how fast it heals*. Protocol (three-interval thixotropy test, `Recovery_v1.py`):

1. **Interval 1 (rest reference):** low shear $\dot{\gamma}_1 = 0.1 \text{ s}^{-1}$; record η_1 .
2. **Interval 2 (disruption):** high shear $\dot{\gamma}_2 \in \{50, 100, 150\} \text{ s}^{-1}$ (matched to the needle wall shear rate of the intended print), 60 s. Network is broken.
3. **Interval 3 (recovery):** return to $\dot{\gamma}_1$. Watch $\eta(t)$ rebuild. Define $R_{\text{rec}}(\%) = \eta_3(t = 30 \text{ s})/\eta_1 \times 100$.

Higher R_{rec} means the network heals faster, reducing sag of the deposited filament before the next layer arrives.

Thixotropy vs. rheopexy. Compare the fitted consistency index K from the Flow (no pre-shear) and Flow Sheared curves:

- $K_{\text{sheared}} < K$: structure broken by pre-shear, recovers incompletely $\Rightarrow$ **thixotropic**.
- $K_{\text{sheared}} > K$: pre-shear has organised the network $\Rightarrow$ **rheopexic** (shear-induced structuring).

This distinction is mechanistically significant and must be discussed explicitly in manuscripts.

WORKED EXAMPLE – the three inks

Ink	R_{rec} at 50 s^{-1}	100 s^{-1}	150 s^{-1}
C15	75%	60%	55%
C15 Gira 5.5	60%	45%	40%
Bozano	99%	98%	96%

Bozano recovers almost completely at every shear rate – consistent with its true-gel character. The NE recovers *less* than pure C15: the NE stiffens the rest network (higher G') but the network bonds are slightly more brittle to high-shear disruption. This trade-off must be compensated at the slicer with a higher Extrusion Multiplier.

IN PRACTICE

Match interval 2 of the recovery test to the *wall shear rate of the needle you will actually print with* ($\dot{\gamma}_w \approx 50\text{--}150 \text{ s}^{-1}$ for 21–22G), not an arbitrary high shear. Read R_{rec} at $t = 30 \text{ s}$ — the layer-dwell time of a typical print — so the number you report is the recovery the next layer actually sees.

2.7 Dimensionless rheological numbers

Three dimensionless groups govern whether a DIW ink will print, store shape, and survive long prints.

DIMENSIONLESS ANALYSIS

Deborah number $De = t_{\text{relax}}/t_{\text{process}}$: ratio of the material’s relaxation time to the characteristic process time. For SAOS: $De = \lambda_{\text{relax}} \omega$ (the linear relaxation time $\lambda_{\text{relax}} = 1/\omega_c$ from the crossover, Eq. 10 — *not* the Cross λ). If $De \gg 1$, the material is elastic at that timescale; if $De \ll 1$, it is viscous.

- In the needle, $t_{\text{process}} \sim R_n/\bar{v}_n \sim \text{ms}$: $De \gg 1$ for all three inks. Elastic behaviour dominates $\Rightarrow$ die swell is expected.
- After deposition, $t_{\text{process}} \sim \text{layer-dwell time}$ (1–100 s): De decreases. If $De < 1$ at this timescale, the material has relaxed and lost memory – it is susceptible to sag.

Deborah number, worked. In the needle $t_{\text{process}} = R_n/\bar{v}_n \sim \text{ms}$, so even a sub-second relaxation time gives $De \gg 1$: the ink is elastic there and die swell is expected for all three inks. After deposition t_{process} is the layer-dwell time ($\sim 1\text{--}100 \text{ s}$), and the relevant timescale is the *linear* relaxation time $\lambda_{\text{relax}} = 1/\omega_c$ from the SAOS crossover (Eq. 10), *not* the Cross λ . For a relaxation time of order seconds, De falls from $\sim O(1)$ at a 1 s dwell to $\ll 1$ at a 100 s dwell — slow or paused prints give the network time to relax and sag. Measure ω_c on the frequency sweep to pin this down for a given ink.

Bingham number $Bi = \tau_y/(\eta_p \dot{\gamma}_w)$ (for yield-stress fluids, or analogously $\eta_0/(\eta_\infty \cdot (\lambda \dot{\gamma}_w)^m)$ for Cross): measures whether viscosity or yield stress dominates in the needle.

- $Bi \gg 1$: plug flow in the centre with thin shear layers at the wall (preferred for cell viability – most cells are in the low-shear core).
- $Bi \ll 1$: Newtonian-like parabolic profile.

Numeric line. For the NE, take the oscillatory yield proxy $\tau_y \approx 558 \text{ Pa}$ (computed above), the high-shear viscosity $\eta_p \approx \eta_\infty$ from the Cross table, and a representative wall shear rate $\dot{\gamma}_w = 100 \text{ s}^{-1}$ from the recovery battery: $Bi = \tau_y/(\eta_p \dot{\gamma}_w)$. Even with a generous $\eta_p \sim 1 \text{ Pa s}$

this gives $Bi \sim 558/(1 \times 100) \approx 5.6 > 1$ — yield stress dominates and a central plug is expected; using the fitted $\eta_\infty = 10^{-3}$ Pa s makes Bi larger still.

Capillary number $Ca = \eta \dot{\gamma}/(2\sigma/R)$: ratio of viscous to surface tension forces at the needle exit. If $Ca \gg 1$, the filament is controlled by viscosity and recovers slowly; die swell is purely elastic in origin. If $Ca \sim 1$, surface tension is non-negligible and affects bead shape on the substrate. For hydrogels ($\sigma \sim 0.07 \text{ N m}^{-1}$) and $R \sim 0.26 \text{ mm}$, $Ca \gg 1$ in the printable $\dot{\gamma}$ range — surface tension is secondary. *Numeric line.* With a representative in-needle viscosity $\eta \sim 100 \text{ Pa s}$ (the high-shear value quoted in Sec. 2) at $\dot{\gamma} = 100 \text{ s}^{-1}$, the surface-tension stress scale is $2\sigma/R = 2(0.07)/(2.6 \times 10^{-4}) \approx 540 \text{ Pa}$, so $Ca = \eta \dot{\gamma}/(2\sigma/R) = 10^4/540 \approx 19 \gg 1$ — viscous forces dominate and bead spreading is set by rheology, not surface tension.

Forward reference. A fourth group, the generalized (Metzner–Reed) Reynolds number Re_{gen} , decides whether the in-needle flow is laminar (the assumption behind every pressure formula here); it is derived for power-law and Cross fluids in the Hydraulic Loop chapter (Sec. 3).

RUNNING EXAMPLE – Solution to Q1

Question. Can the C15 Gira 5.5 ink print at all?

Answer. Ana reads the parameter tables above and notes three signals:

1. Cross $m = 0.988$. Nearly binary shear thinning: $\eta_0 \approx 2440 \text{ Pa s}$ at rest; collapses sharply above $\dot{\gamma} \approx 0.25 \text{ s}^{-1}$. *Excellent printability signature.*
2. $\gamma_{\text{LVR}} = 60.9\%$. Wide linear regime: the deposited filament tolerates up to 60% strain before yielding. *Tolerant to small perturbations during deposition.*
3. $R_{\text{rec}} \approx 45\%$ at 100 s^{-1} . Each layer regains roughly half its rest modulus within 30 s. *Manageable – compensated by the Extrusion Multiplier.*

Verdict. Rheologically well-suited to DIW. Proceed.

Still unknown: what V_p to enter into the slicer. Q2.

Check Yourself

Exercise 2.1 ◦ Water has $\eta = 10^{-3} \text{ Pa s}$; the NE ink reaches $\eta_0 = 2440 \text{ Pa s}$. Express the rest-state contrast as orders of magnitude.

SOLUTION

$\log_{10}(2440/10^{-3}) \approx 6.4$ — about six and a half decades.

Exercise 2.2 ◦ An ink has $G' = 400 \text{ Pa}$ and $G'' = 520 \text{ Pa}$ at $\omega = 1 \text{ rad s}^{-1}$. Is it gel-like or liquid-like there? Compute $\tan \delta$.

SOLUTION

$\tan \delta = 520/400 = 1.3 > 1$ — liquid-like at this timescale.

Exercise 2.3 ◦◦ Using Eq. 4 and the SAOS table, rank the three inks by oscillatory yield-stress proxy.

SOLUTION

NE $916 \times 0.609 = 558 \text{ Pa} >$ C15 $270 \times 0.64 = 173 \text{ Pa} >$ Bozano $348 \times 0.137 = 48 \text{ Pa}$.

Exercise 2.4 ○○ A frequency sweep crosses ($G' = G''$) at $\omega_c = 0.2$ rad/s. What is the relaxation time, and is the material elastic or viscous at the 1 s layer-dwell timescale?

SOLUTION

$\lambda_{\text{relax}} = 1/0.2 = 5$ s. At dwell $\omega \sim 1$ rad/s, $De = 5 > 1$ — elastic; it holds shape.

Exercise 2.5 ○○○ The CMC+NE composite shows $|\eta^*(\omega)| > \eta(\dot{\gamma})$ at matched $\dot{\gamma} = \omega$. What does the Cox–Merz violation tell you physically, and which design rule does it support?

SOLUTION

A particle network / apparent yield stress: steady shear breaks structure that the small-amplitude oscillation leaves intact, so the steady-shear viscosity falls below the complex viscosity (a positive Cox–Merz deviation). Supports treating the NE as a reinforcing filler with a yield-stress proxy (filler vs. crosslinker design rule).

KEY TAKEAWAYS

- Shear thinning is *necessary* for DIW: low in-needle viscosity, high post-deposition viscosity.
- The Cross model ($\eta_0, \eta_\infty, \lambda, m$) is preferred over the Power Law for the full $\dot{\gamma}$ range.
- G'_{LVR} and γ_{LVR} from SAOS predict scaffold height and tolerance to deformation.
- Recovery R_{rec} directly feeds the deposition efficiency k_{flow} and the required Extrusion Multiplier.
- Dimensionless numbers (De, Bi) explain velocity profiles inside the needle and post-deposition stability.

ENGINEERING DESIGN RULES

DR 2.1. Always report both G'_{LVR} and γ_{LVR} — neither alone is sufficient. $G'_{\text{LVR}} \cdot \gamma_{\text{LVR}}$ is the yield-stress proxy that controls scaffold height.

DR 2.2. Prefer the Cross model for printability prediction. Only use Power Law for single-decade fitting when analytical expressions are needed and the Cross fit is poorly constrained.

DR 2.3. Structural recovery is more important than static viscosity for scaffold quality. An ink that recovers slowly will produce under-built beads even if η_0 is high.

DR 2.4. When $m \approx 1$ (NE), expect minimal die swell and minimal β_{swell} . When $m < 0.7$, measure die swell directly in calibration test C1.

3 The Piston-Driven Hydraulic Loop

Why this chapter matters. A DIW printer is **piston-controlled**: the only setpoint the operator commands is the piston velocity V_p . Pressure, flow rate, mean needle velocity, wall shear stress, and print speed are all *consequences* of V_p and the ink's rheology — they are diagnostics, not inputs. The classic mistake is to treat ΔP as a control variable (as one might in a pressure-driven syringe pump) and then wonder why the deposited line width is unpredictable. Get the mass-balance chain $V_p \rightarrow Q \rightarrow \bar{v}_n \rightarrow v_{\text{print}}$ wrong, and every slicer entry is wrong before the first layer is deposited.

We now connect the ink rheology to the printer hardware. The system is a simple hydraulic circuit: piston $\rightarrow$ syringe barrel $\rightarrow$ conical transition $\rightarrow$ needle.

3.1 Mass balance: from V_p to flow

Motivation. If you skip this step and guess Q from the slicer's extrusion rate setting (which presumes FFF-style filament diameter, not a piston), you will be off by a factor of $\sim 10^3$. The calculation is two lines of algebra — there is no excuse for guessing.

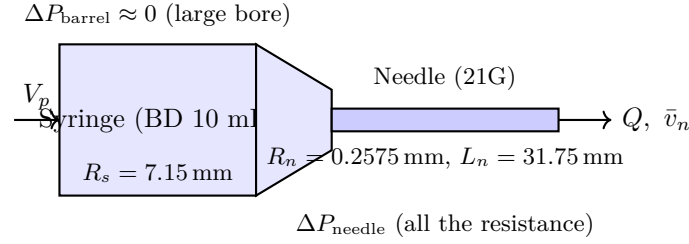


Figure 2: The DIW hydraulic loop. The operator commands V_p ; syringe geometry fixes Q ; the ink rheology and needle geometry fix τ_w and ΔP . The syringe barrel contributes negligible pressure drop because $(R_s/R_n)^4 \approx 5 \times 10^5$ – almost all resistance is in the needle.

The piston advances at V_p , displacing fluid at volumetric rate

$$Q = \pi R_s^2 V_p. \quad (12)$$

The same Q passes through the needle (continuity):

$$\bar{v}_n = \frac{Q}{\pi R_n^2} = \left(\frac{R_s}{R_n} \right)^2 V_p. \quad (13)$$

For our hardware ($R_s = 7.15 \text{ mm}$, 21G $R_n = 0.2575 \text{ mm}$), the area ratio $(R_s/R_n)^2 \approx 771$. **A 1 mm/s piston velocity becomes a 771 mm/s mean needle velocity.**

DERIVATION — Mass balance: $V_p \rightarrow Q \rightarrow \bar{v}_n$

Assumptions. (i) The ink is incompressible ($\rho = \text{const}$). (ii) Piston sealing is perfect ($f_{\text{slip}} = 1$; relaxed in Ch. 6). (iii) The barrel diameter is uniform over the piston stroke ($A_{\text{piston}} = \text{const}$).

Step 1 – Volumetric flow from the piston.

In time dt the piston face advances by $V_p dt$, sweeping a volume $d\mathcal{V} = A_{\text{piston}} V_p dt$. Because the fluid is incompressible and the barrel walls do not move, the same volume must leave through the needle exit per unit time:

$$Q = \frac{d\mathcal{V}}{dt} = A_{\text{piston}} V_p = \frac{\pi D_{\text{syringe}}^2}{4} V_p = \pi R_s^2 V_p. \quad (12 \text{ restated})$$

Step 2 – Mean needle velocity by continuity.

Continuity for steady, incompressible, one-dimensional flow between any two cross-sections i and j requires $A_i \bar{v}_i = A_j \bar{v}_j = Q$. Applying this between the piston face (cross-section s , area $A_s = \pi R_s^2$, velocity V_p) and the needle interior (cross-section n , area $A_n = \pi R_n^2$, mean velocity $\bar{v}_n$):

$$A_s V_p = A_n \bar{v}_n \implies \bar{v}_n = \frac{A_s}{A_n} V_p = \frac{\pi R_s^2}{\pi R_n^2} V_p = \left(\frac{R_s}{R_n} \right)^2 V_p. \quad (13 \text{ restated})$$

Step 3 – Hardware numbers (BD 10 mL + 21G).

- Syringe: $D_{\text{syringe}} = 14.3 \text{ mm}$, so $R_s = 7.15 \text{ mm}$, $A_s = \pi(7.15)^2 \approx 160.7 \text{ mm}^2$.
- 21G needle: $D_{\text{needle}} = 0.515 \text{ mm}$, so $R_n = 0.2575 \text{ mm}$, $A_n = \pi(0.2575)^2 \approx 0.2083 \text{ mm}^2$.
- Area ratio: $A_s/A_n = 160.7/0.2083 \approx 771$.

Therefore:

$$\bar{v}_n = 771 V_p. \quad (14)$$

At $V_p = 0.01$ mm/s: $\bar{v}_n = 771 \times 0.01 = 7.71$ mm/s; $Q = 160.7 \times 0.01 = 1.607$ mm³/s = 96.4 μ L/min.

For 22G ($D_{\text{needle}} = 0.413$ mm, $R_n = 0.2065$ mm, $A_n = \pi(0.2065)^2 \approx 0.1340$ mm²): $A_s/A_n \approx 1199$, so $\bar{v}_n \approx 1199 V_p$.

Key insight. V_p is the operator’s lever; $\bar{v}_n$ and Q follow deterministically from geometry alone – no rheological input needed at this stage.

3.2 Velocity profile inside the needle

Motivation. Knowing the *mean* needle velocity $\bar{v}_n$ is not enough for cell-safety evaluation: a cell suspended near the needle wall experiences a local shear rate that may be ten to a hundred times the tube average. The velocity profile shape – controlled by the Bingham number – tells you whether your cells spend the transit in a low-shear core (plug flow, high Bi) or are exposed to full parabolic shear (Newtonian limit, low Bi). Getting this wrong means under- or over-estimating cell damage.

The shape of the velocity profile inside the needle depends on the Bingham number. Figure 3 shows the two limits.

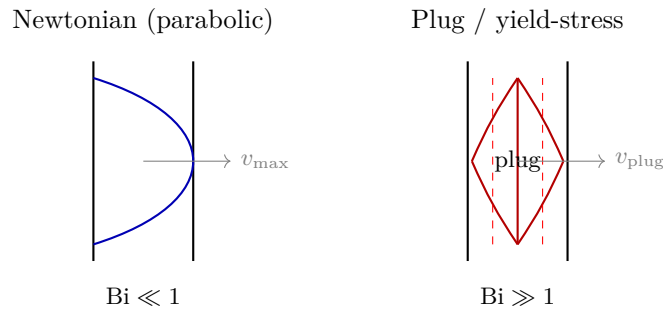


Figure 3: Velocity profiles in the needle cross-section. Left: Newtonian / power-law fluid – maximum shear at the wall, zero at the centreline. Right: plug-flow limit for a high-yield-stress fluid – most of the cross-section moves as a solid plug; shear is concentrated in thin wall layers. Our CMC inks lie between these limits; high-modulus inks (Bozano) approach the plug-flow regime.

Biological significance. Cells suspended in the ink experience the *local* shear rate, not the wall average. In plug flow, cells in the core see very low shear stress. For bioprinting with live cells, inks with partial plug-flow behaviour (high Bi) are preferable because the cell-damaging high-shear zone is confined to a thin wall layer.

3.3 Mass balance: from V_p to print speed

Motivation. If you set V_p and v_{print} independently – the classic error of separating “extrusion speed” from “travel speed” – you will either over-extrude (swell, clogged layers) or under-extrude (broken filaments). The relationship is not a tuning knob; it is a consequence of volume conservation, and it fixes the ratio v_{print}/V_p from geometry alone.

The print head deposits a filament of cross-section $w h_{\text{layer}}$ as it moves over the build plate at speed v_{print} :

$$Q = v_{\text{print}} w h_{\text{layer}}. \quad (15)$$

With $w \approx 2R_n(1 + \beta_{\text{swell}})$ and $h_{\text{layer}} = h_{\text{factor}} \cdot 2R_n$ this becomes

$$v_{\text{print}} \approx \frac{\pi(R_s/R_n)^2}{4 h_{\text{factor}}(1 + \beta_{\text{swell}})} V_p \approx \frac{700-865}{(1 + \beta_{\text{swell}})} V_p \quad (21\text{G}). \quad (16)$$

DERIVATION — Volume conservation: $V_p \rightarrow v_{\text{print}}$

Assumptions. (i) Incompressible ink ($f_{\text{slip}} = 1$, $k_{\text{flow}} = 1$; corrections deferred to Ch. 6). (ii) The deposited filament cross-section is a rectangle of width w and height h_{layer} (rounded cross-sections are accommodated by adjusting h_{factor}). (iii) No voids, no over-compression: volume extruded equals volume deposited.

Step 1 – Equate extruded and deposited volume rates.

The extruded volumetric rate is $Q = \pi R_s^2 V_p$ (from Eq. 12). The deposited volumetric rate equals the deposited cross-sectional area swept by the print head:

$$Q = v_{\text{print}} A_{\text{bead}}, \quad A_{\text{bead}} = w h_{\text{layer}}. \quad (17)$$

Step 2 – Parameterise the bead geometry.

The filament width is set by the needle exit diameter, amplified by die swell:

$$w = 2R_n(1 + \beta_{\text{swell}}) = D_{\text{needle}}(1 + \beta_{\text{swell}}). \quad (18)$$

The layer height is a controlled fraction of the needle diameter (typical $h_{\text{factor}} \approx 0.7$ –1.0 in DIW):

$$h_{\text{layer}} = h_{\text{factor}} \cdot 2R_n. \quad (19)$$

Step 3 – Solve for v_{print} .

Substituting Eqs. 18–19 into Eq. 17 and using $Q = \pi R_s^2 V_p$:

$$\begin{aligned} v_{\text{print}} &= \frac{Q}{w h_{\text{layer}}} = \frac{\pi R_s^2 V_p}{2R_n(1 + \beta_{\text{swell}}) \cdot h_{\text{factor}} \cdot 2R_n} \\ &= \frac{\pi R_s^2}{4 R_n^2 h_{\text{factor}}(1 + \beta_{\text{swell}})} V_p = \frac{\pi(R_s/R_n)^2}{4 h_{\text{factor}}(1 + \beta_{\text{swell}})} V_p. \end{aligned} \quad (20)$$

Step 4 – Hardware numbers (21G, $\beta_{\text{swell}} = 0$).

$(R_s/R_n)^2 \approx 771$; $h_{\text{factor}} \in [0.7, 1.0]$:

$$v_{\text{print}} = \frac{\pi \cdot 771}{4 h_{\text{factor}}} V_p \approx \frac{606}{h_{\text{factor}}} V_p \in [606, 865] V_p,$$

recovering Eq. 16. The range 700–865 in the main text corresponds to $h_{\text{factor}} \approx 0.87$ –0.70 (a larger layer height gives a lower print speed for the same flow).

Physical check. Dimensional consistency: $[v_{\text{print}}] = [\pi(R_s/R_n)^2 V_p / (4 h_{\text{factor}})] = (\text{dimensionless})(\text{mm/s}) = \text{mm/s}$. ✓

PHYSICAL INTUITION

The piston is the slowest thing. On 21G, V_p in mm/s maps to v_{print} in mm/s with a multiplier of ≈ 700 –865:

- $V_p = 0.005 \Rightarrow v_{\text{print}} \approx 4 \text{ mm/s}$
- $V_p = 0.010 \Rightarrow v_{\text{print}} \approx 8 \text{ mm/s}$
- $V_p = 0.030 \Rightarrow v_{\text{print}} \approx 25 \text{ mm/s}$
- $V_p = 0.050 \Rightarrow v_{\text{print}} \approx 40 \text{ mm/s}$

The mechanical envelope of the lab gantry is roughly $v_{\text{print}} \in [3, 30] \text{ mm/s}$, so $V_p \in [\sim 0.004, \sim 0.04] \text{ mm/s}$.

DIMENSIONLESS ANALYSIS

Generalised Reynolds number for power-law flow in a tube:

$$\text{Re}_{\text{gen}} = \frac{\rho \bar{v}_n^{2-n} (2R_n)^n}{K 8^{n-1}} \left(\frac{4n}{3n+1} \right)^n. \quad (21)$$

For laminar flow (the fundamental assumption of all our ΔP equations), $\text{Re}_{\text{gen}} \ll 2100$. For our operating conditions:

Ink (PL fit)	K [Pas ^{n}]	Re_{gen} at $V_p = 0.01$ mm/s
C15	78	$\ll 1$ (deeply laminar)
C15 Gira 5.5	267	$\ll 1$
Bozano	1086	$\ll 1$

All three inks are deeply laminar across the entire printable V_p range. The fully-developed laminar flow assumption is valid. No turbulence correction is needed.

DERIVATION — Generalised (Metzner–Reed) Reynolds number

Context. The classical Hagen–Poiseuille and Stokes formulae were derived for Newtonian fluids; the Reynolds number $\text{Re} = \rho \bar{v} D / \eta$ uses a single, constant viscosity. For a power-law fluid the apparent viscosity depends on shear rate, so “which η to use?” is ill-defined. Metzner and Reed (1955) resolved this by constructing a Reynolds number whose laminar-to-turbulent threshold remains at $\text{Re}_{\text{gen}} \approx 2100$ regardless of the flow index n , provided the correct generalised denominator is used.

Step 1 – Laminar wall shear rate for a power-law fluid.

For steady, fully developed, laminar flow of a power-law fluid ($\tau = K \dot{\gamma}^n$) in a tube of radius R , integration of the velocity profile gives the Newtonian wall shear rate corrected by the Rabinowitsch–Weissenberg factor:

$$\dot{\gamma}_w = \frac{4\bar{v}}{R_n} \cdot \frac{3n+1}{4n} = \frac{8\bar{v}}{D} \cdot \frac{3n+1}{4n}, \quad (22)$$

where $D = 2R_n$ is the tube diameter and $\bar{v} \equiv \bar{v}_n$ is the cross-sectional mean velocity. The factor $(3n+1)/4n$ corrects for the non-parabolic profile: for $n = 1$ (Newtonian) it equals 1, recovering the classical Newtonian result $\dot{\gamma}_w = 8\bar{v}/D$; for $n < 1$ (shear thinning) it is greater than 1, reflecting the steeper gradient near the wall.

Step 2 – Wall shear stress in terms of $\bar{v}$.

Substituting Eq. 22 into the power-law constitutive equation $\tau_w = K \dot{\gamma}_w^n$:

$$\begin{aligned} \tau_w &= K \left[\frac{8\bar{v}}{D} \cdot \frac{3n+1}{4n} \right]^n \\ &= K \left(\frac{8\bar{v}}{D} \right)^n \left(\frac{3n+1}{4n} \right)^n. \end{aligned} \quad (23)$$

This may be rewritten in the Metzner–Reed form by introducing the *generalised viscosity* μ' :

$$\tau_w = \mu' \left(\frac{8\bar{v}}{D} \right), \quad \mu' = K \left(\frac{8\bar{v}}{D} \right)^{n-1} \left(\frac{3n+1}{4n} \right)^n. \quad (24)$$

Step 3 – Construct Re_{gen} by analogy with the Newtonian case.

For a Newtonian fluid, $\tau_w = \eta \cdot (8\bar{v}/D)$, and the Reynolds number is $\text{Re} = \rho\bar{v}D/\eta$. By dimensional analogy, replace η with μ' :

$$\text{Re}_{\text{gen}} = \frac{\rho \bar{v}_n D}{\mu'} = \frac{\rho \bar{v}_n D}{K \left(\frac{8\bar{v}_n}{D} \right)^{n-1} \left(\frac{3n+1}{4n} \right)^n}. \quad (25)$$

Step 4 – Simplify to the standard form.

Collect powers of $\bar{v}_n$ and D in the denominator: $(8\bar{v}_n/D)^{n-1} = 8^{n-1}\bar{v}_n^{n-1}/D^{n-1}$. Then:

$$\begin{aligned} \text{Re}_{\text{gen}} &= \frac{\rho \bar{v}_n D}{K \cdot 8^{n-1} \bar{v}_n^{n-1} D^{-(n-1)} \cdot \left(\frac{3n+1}{4n} \right)^n} \\ &= \frac{\rho \bar{v}_n^{1-(n-1)} D^{1+(n-1)}}{K \cdot 8^{n-1} \left(\frac{3n+1}{4n} \right)^n} \\ &= \frac{\rho \bar{v}_n^{2-n} D^n}{K 8^{n-1} \left(\frac{3n+1}{4n} \right)^n}. \end{aligned} \quad (26)$$

Using $D = 2R_n$ and noting $\left(\frac{4n}{3n+1} \right)^n = 1 / \left(\frac{3n+1}{4n} \right)^n$:

$$\boxed{\text{Re}_{\text{gen}} = \frac{\rho \bar{v}_n^{2-n} (2R_n)^n}{K 8^{n-1} \left(\frac{3n+1}{4n} \right)^n} = \frac{\rho \bar{v}_n^{2-n} D^n}{K 8^{n-1} \left(\frac{3n+1}{4n} \right)^n}.} \quad (27)$$

This is the Metzner–Reed generalised Reynolds number (Metzner and Reed, 1955). For Newtonian flow ($n = 1$, $K = \eta$) it reduces exactly to $\text{Re} = \rho\bar{v}D/\eta$. The laminar criterion is $\text{Re}_{\text{gen}} \lesssim 2100$; turbulent transition occurs above $\text{Re}_{\text{gen}} \approx 3000$ –4000 for power-law fluids.

Step 5 – Qualitative evaluation for our inks.

For the three inks in this project, the key parameters are $n < 0.5$ (strong shear thinning) and $K \gg 1 \text{ Pas}^n$ (high consistency). The operating needle diameter is $D \sim 0.5 \text{ mm}$ and the mean needle velocity (from Eq. 14) at $V_p = 0.01 \text{ mm/s}$ is $\bar{v}_n \approx 7.7 \text{ mm/s} = 7.7 \times 10^{-3} \text{ m/s}$. Taking $\rho \approx 1000 \text{ kg/m}^3$ (water-based ink), $n = 0.3$, $K = 100 \text{ Pas}^n$:

$$\begin{aligned} 8^{n-1} &= 8^{-0.7} \approx 0.233, \\ \left(\frac{3(0.3) + 1}{4(0.3)} \right)^{0.3} &= \left(\frac{1.9}{1.2} \right)^{0.3} = (1.583)^{0.3} \approx 1.148, \\ \bar{v}_n^{2-n} &= (7.71 \times 10^{-3})^{1.7} \approx 2.56 \times 10^{-4} \text{ m}^{1.7}/\text{s}^{1.7}, \\ D^n &= (5.15 \times 10^{-4})^{0.3} \approx 0.103 \text{ m}^{0.3}, \\ \text{Re}_{\text{gen}} &= \frac{1000 \cdot 2.56 \times 10^{-4} \cdot 0.103}{100 \cdot 0.233 \cdot 1.148} \approx \frac{2.64 \times 10^{-2}}{26.7} \approx 9.8 \times 10^{-4}. \end{aligned}$$

$\text{Re}_{\text{gen}} \approx 9.8 \times 10^{-4} \approx 10^{-3} \ll 1 \ll 2100$: the flow is **deeply laminar** by more than three orders of magnitude. This confirms that every analytical velocity-profile result in this book – the Hagen–Poiseuille pressure formulae, the Rabinowitsch integral, the wall-shear-stress expressions – rests on solid ground for all three inks across the entire printable V_p range.

IN PRACTICE

Pre-print flow-regime check (takes <5 minutes). Before loading any new ink or switching needle gauge, compute two dimensionless numbers:

1. Re_{gen} (Eq. 27): use the Power-Law K and n from `Fit_Muitos_Modelos_v4.py`, $\bar{v}_n$ from Eq. 14 at your target V_p , and $\rho \approx 1000 \text{ kg/m}^3$ for water-based inks. If $\text{Re}_{\text{gen}} < 1$, all pressure formulae are valid. If it approaches 100, add a note in your lab record.
2. Bi (Bingham number): use $\tau_y \approx G'_{\text{LVR}} \gamma_{\text{LVR}}$ (Eq. 4) and estimate $\eta_p \approx \eta_\infty$ from the Cross fit. If $\text{Bi} > 1$, expect a partial plug at the centre – plan for the cell-viability benefit and check that the plug radius does not cause clogging at the needle entrance.

Rule: If $\text{Re}_{\text{gen}} < 1$ and $\text{Bi} > 1$, the ink is in the preferred bioprinting regime: analytically tractable laminar flow with a low-shear cell-protective core. Record both numbers in the print log alongside V_p and needle gauge.

GOING DEEPER

Why the 8^{n-1} and $((3n+1)/4n)^n$ factors? Both arise from the Rabinowitsch correction to the Newtonian wall-shear-rate formula $\dot{\gamma}_w^N = 8\bar{v}/D$. For a general generalised Newtonian fluid the true wall shear rate is $\dot{\gamma}_w = \dot{\gamma}_w^N \cdot \frac{3n'+1}{4n'}$, where $n' = d \log \tau_w / d \log (8\bar{v}/D)$ is the local log-log slope of the flow curve. For a power-law fluid n' is constant, giving the exact result used in Step 1. For a Cross fluid n' varies with $\bar{v}$, and the Rabinowitsch correction must be applied numerically via the integral in Eq. 37. The Metzner–Reed group then becomes an *effective* Reynolds number; its laminar threshold still lies near 2100 for most generalised Newtonian models, making it the universal laminar-flow diagnostic in non-Newtonian pipe flow (Metzner and Reed, 1955).

Check Yourself

Exercise 3.1 ◦ At $V_p = 0.010 \text{ mm/s}$ on a 21G needle (inner radius $R_n = 0.2575 \text{ mm}$, syringe inner radius $R_s = 7.15 \text{ mm}$):

- (a) Compute the volumetric flow Q in mm^3/s and $\mu\text{L}/\text{min}$.
- (b) Compute the mean needle velocity $\bar{v}_n$ in mm/s .
- (c) Repeat (b) for a 22G needle ($R_n = 0.2065 \text{ mm}$).

SOLUTION

- (a) $Q = \pi R_s^2 V_p = \pi (7.15)^2 (0.010) = \pi (51.12) (0.010) \approx 1.607 \text{ mm}^3/\text{s} = 1.607 \times 60 \approx 96.4 \mu\text{L}/\text{min}$.
- (b) $\bar{v}_n = (R_s/R_n)^2 V_p = (7.15/0.2575)^2 \times 0.010 = (27.77)^2 \times 0.010 = 771.2 \times 0.010 \approx 7.71 \text{ mm/s}$.
- (c) $(R_s/R_n)^2 = (7.15/0.2065)^2 = (34.63)^2 = 1199.2$; $\bar{v}_n = 1199.2 \times 0.010 \approx 12.0 \text{ mm/s}$.
Note: switching from 21G to 22G at the same V_p raises $\bar{v}_n$ by a factor of $1199/771 \approx 1.56$ and raises ΔP by roughly $(R_{21\text{G}}/R_{22\text{G}})^4 = (0.2575/0.2065)^4 \approx 2.4$.

Exercise 3.2 ◦◦ For an ink with Power-Law parameters $K = 100 \text{ Pas}^n$, $n = 0.30$, density $\rho = 1000 \text{ kg/m}^3$, printed through a 21G needle ($D = 0.515 \text{ mm} = 5.15 \times 10^{-4} \text{ m}$) at $V_p = 0.01 \text{ mm/s}$:

- (a) Compute $\bar{v}_n$ in m/s .
- (b) Evaluate each factor in Eq. 27 explicitly.
- (c) State whether fully-developed laminar flow is a valid assumption.

(Use the assumed round values given in the stem; do not substitute fitted parameters from the appendices.)

SOLUTION

(a) $\bar{v}_n = 771 \times 0.010 = 7.71 \text{ mm/s} = 7.71 \times 10^{-3} \text{ m/s}$.

(b) Factor-by-factor:

$$\rho \bar{v}_n^{2-n} = 1000 \times (7.71 \times 10^{-3})^{1.70} = 1000 \times 2.56 \times 10^{-4} = 0.256 \text{ kgm}^{n-3}\text{s}^{n-2},$$

$$D^n = (5.15 \times 10^{-4})^{0.30} = e^{0.30 \ln(5.15 \times 10^{-4})} = e^{0.30 \times (-7.571)} = e^{-2.271} \approx 0.103 \text{ m}^n,$$

$$\text{numerator} = 0.256 \times 0.103 \approx 2.64 \times 10^{-2},$$

$$8^{n-1} = 8^{-0.70} = e^{-0.70 \ln 8} = e^{-1.456} \approx 0.233,$$

$$\left(\frac{3n+1}{4n}\right)^n = \left(\frac{1.90}{1.20}\right)^{0.30} = (1.583)^{0.30} = e^{0.30 \times 0.460} = e^{0.138} \approx 1.148,$$

$$\text{denominator} = 100 \times 0.233 \times 1.148 \approx 26.7,$$

$$\text{Re}_{\text{gen}} = 2.64 \times 10^{-2} / 26.7 \approx 9.8 \times 10^{-4}.$$

(c) $\text{Re}_{\text{gen}} \approx 9.8 \times 10^{-4} \ll 1 \ll 2100$: the flow is deeply laminar. Fully-developed laminar flow is an excellent assumption; no turbulence correction is needed and the analytical ΔP formulae are valid.

Exercise 3.3 ○ ○ ○ A cell-loaded ink has a Bingham number $\text{Bi} = 12$ at the wall shear rate imposed by a 21G needle at the nominal print speed.

- Sketch the expected velocity profile qualitatively. What fraction of the cross-sectional area is occupied by the plug (unyielded core)? The plug radius for a Bingham fluid is $r_{\text{plug}} = R_n \cdot \tau_y / \tau_w$; express this as a fraction of R_n in terms of Bi for $\text{Bi} \gg 1$.
- What does a high Bi imply about the shear rate that cells in the core of the needle experience? Relate this to cell viability literature ($\tau_w < 5 \text{ kPa}$ for human cells).
- If you switch to a 22G needle at the same V_p , qualitatively how does Bi change, and is the change beneficial or detrimental for cell viability? (Hint: τ_w scales as $R_n^{-(3+n)}$ for a power-law fluid at fixed V_p ; $\dot{\gamma}_w$ scales as R_n^{-3} .)

SOLUTION

(a) The plug radius is $r_{\text{plug}}/R_n = \tau_y/\tau_w$. By definition $\text{Bi} = \tau_y/(\eta_p \dot{\gamma}_w)$ and $\tau_w = \tau_y + \eta_p \dot{\gamma}_w$, so for the Bingham model $\tau_w/\tau_y = 1 + 1/\text{Bi}$. Therefore $r_{\text{plug}}/R_n = \tau_y/\tau_w = 1/(1 + 1/\text{Bi}) = \text{Bi}/(\text{Bi} + 1)$. At $\text{Bi} = 12$: $r_{\text{plug}}/R_n = 12/13 \approx 0.923$. The plug occupies $(r_{\text{plug}}/R_n)^2 \approx 0.85$ – about 85% of the cross-sectional area. The velocity profile shows a broad flat plug with thin shear layers of thickness $\sim 0.077 R_n$ at the wall.

(b) In the plug, $\dot{\gamma} = 0$: cells in the core experience zero shear rate and zero shear stress during transit. Only cells within the thin wall layer are exposed to high shear. For $\text{Bi} = 12$, 85% of the cells are in zero-shear conditions; only the outer 15% of the *area* (but a thin annulus) experiences wall-level shear. The cell-safety criterion $\tau_w < 5 \text{ kPa}$ still applies to the wall annulus, but the *majority* of cells are well-protected. This is why high-Bi inks are preferred for cell-loaded bioprinting.

(c) Switching 21G $\rightarrow$ 22G at fixed V_p : R_n decreases ($0.2575 \rightarrow 0.2065 \text{ mm}$), so $\dot{\gamma}_w \propto R_n^{-3}$ increases, and $\tau_w \propto R_n^{-(3+n)}$ increases even faster (for $n = 0.3$: factor $(0.2575/0.2065)^{3.3} \approx 2.0$ increase in τ_w , but τ_y is unchanged). Therefore $\text{Bi} = \tau_y/(\eta_p \dot{\gamma}_w)$ decreases: the ink is less plug-like in 22G than in 21G at the same V_p . The change is **detrimental** for cell viability: the protective plug shrinks (lower Bi) *and* the absolute wall stress increases. For cell-loaded prints, the 21G is the preferred gauge unless resolution requirements demand 22G.

RUNNING EXAMPLE – Solution to Q2

Question. What piston velocity V_p should Ana command?

Reasoning. The mechanical envelope is $v_{\text{print}} \in [3, 30]$ mm/s. Ana picks a target of $v_{\text{print}} \approx 8$ mm/s – slow enough to forgive minor mis-calibration, fast enough to finish the production print in under an hour.

For 21G with $\beta_{\text{swell}} \approx 0$ (justified for the NE, see Q4), Eq. 16 gives amplification ≈ 865 :

$$V_p = \frac{8}{865} \approx 0.0093 \text{ mm/s} \rightarrow \text{rounded to } \mathbf{0.01} \text{ mm/s.}$$

Sanity check. $Q = \pi(7.15)^2 \cdot 0.01 = 1.6 \text{ mm}^3/\text{s} \approx 100 \mu\text{L}/\text{min}$ – typical for syringe bioprinting.

Still unknown: is the resulting ΔP within hardware limits? Q3.

KEY TAKEAWAYS

- V_p is the only setpoint. ΔP , Q , $\bar{v}_n$, τ_w , and v_{print} are all consequences of V_p plus geometry. Treating ΔP as a control input is the classic DIW error.
- $V_p \rightarrow Q \rightarrow \bar{v}_n$: a factor of ~ 771 velocity amplification on 21G. The piston is slow; the needle is fast.
- v_{print} is set by V_p and geometry alone – it is not an independent knob unless the slicer X/Y feedrate is decoupled from E.
- The Metzner–Reed Re_{gen} (Eq. 27) generalises the Newtonian Reynolds number for power-law fluids; its laminar threshold is still ≈ 2100 . $\text{Re}_{\text{gen}} \ll 2100$ for all printable conditions: laminar flow is always valid, and every analytical ΔP formula in this book is sound.
- Plug flow in high-Bi inks protects cells in the core from damaging shear: $\sim 85\%$ of the cross-sectional area is zero-shear at $\text{Bi} = 12$.

ENGINEERING DESIGN RULES

DR 3.1. Always verify $\text{Re}_{\text{gen}} \ll 2100$ when changing needle gauge or ink. Turbulent transition would invalidate all ΔP and τ_w calculations. Use Eq. 27 with the Power-Law K and n from `Fit_Muitos_Modelos_v4.py`.

DR 3.2. Do not change needle gauge (R_n) without recomputing all downstream quantities – the R_n^{-4} scaling of ΔP makes gauge changes the largest single-factor lever in the whole workflow. Switching 21G $\rightarrow$ 22G at fixed V_p also raises $\dot{\gamma}_w \propto R_n^{-3}$, potentially breaching the $\tau_w < 5 \text{ kPa}$ cell-safety threshold.

DR 3.3. For cell-loaded inks, prefer inks with high Bi (approaching plug flow) to confine high shear to thin wall layers. Compute Bi from the oscillatory yield-stress proxy and the Cross η_∞ before committing to a needle gauge.

DR 3.4. Always compute the $V_p \rightarrow v_{\text{print}}$ ratio (Eq. 16) analytically before entering any print speed into the slicer. If the slicer decouples travel speed from extrusion, the extrusion multiplier must absorb the difference.

4 From Flow to Pressure: Hagen–Poiseuille for Non-Newtonian Fluids

Given Q from the mass-balance of the previous chapter, we need ΔP across the needle on two grounds: as a mechanical safety check (does the required pressure stay inside the syringe rating?) and as the upstream quantity from which wall shear stress and wall shear rate are derived. Get ΔP wrong and you either exceed the syringe burst rating or, worse, exceed the cell-safe wall-shear

ceiling without realising it – two independent failure modes from a single mis-predicted number.

4.1 Newtonian baseline

Motivation. The Newtonian formula sets the physical intuition and the scaling laws before we extend to non-Newtonian fluids. Every exponent and denominator in the generalised formulae traces back to this starting point; understanding the baseline makes the extensions transparent rather than magic.

For a Newtonian fluid in a straight cylinder of radius R and length L , Hagen–Poiseuille gives

$$\Delta P = \frac{8 \eta L Q}{\pi R^4}. \quad (28)$$

The R^{-4} dependence is severe: halving the needle radius multiplies the pressure 16-fold. This is the dominant reason needle gauge matters so much.

PHYSICAL INTUITION

Think of the R^{-4} scaling in two steps. Halving R doubles the wall shear rate (the fluid must deform twice as fast in the same time), which doubles τ_w ; and it simultaneously halves the moment arm R over which that stress acts, so $\Delta P = 2L\tau_w/R$ doubles again. Two doublings give a factor of four – and that is just the shear-rate/moment-arm argument. The fourth power includes the $Q \propto R^2 \bar{v}_n$ reduction in cross-section as well.

4.2 Power-Law extension

Motivation. Our CMC inks are strongly shear-thinning; treating them as Newtonian at a single representative viscosity can produce ΔP errors of an order of magnitude. The Power-Law model provides an exact closed-form correction for any shear-thinning index n – but only if the derivation is done rigorously, which is what this subsection supplies.

For a Power-Law fluid, the analytical solution for steady laminar fully- developed flow in a cylinder gives wall shear stress:

$$\tau_w = K \left(\frac{(3n+1) Q}{\pi R^3 n} \right)^n, \quad (29)$$

and pressure drop from a force balance:

$$\Delta P = \frac{2L\tau_w}{R}. \quad (30)$$

DERIVATION — Power-Law pipe flow: velocity profile, $Q(\tau_w)$, and inversion to $\tau_w(Q)$

Assumptions. (i) Steady, fully-developed, laminar flow in a straight circular cylinder of radius R and length L . (ii) The fluid obeys the Power-Law constitutive model $\tau = K\dot{\gamma}^n$ ($n < 1$ for shear thinning), with the convention that $\dot{\gamma} = -dv/dr > 0$ for the decreasing velocity profile (v maximal at the centreline, zero at the wall). (iii) No wall slip; no-slip boundary condition $v(R) = 0$. (iv) Gravity and inertia are negligible (verified by $\text{Re}_{\text{gen}} \ll 1$; see Eq. 27).

Step 1 – Universal linear stress profile.

Consider a fluid element in the shape of a coaxial cylinder of radius r ($0 \leq r \leq R$) and length L . In steady flow, Newton’s second law applied to this element along the axial direction balances the net pressure force on the two end faces against the integrated viscous shear on the curved surface:

$$\underbrace{\pi r^2 \Delta P}_{\text{pressure}} = \underbrace{2\pi r L \tau(r)}_{\text{shear on curved surface}}.$$

Solving for $\tau(r)$:

$$\tau(r) = \frac{\Delta P}{2L} r. \quad (31)$$

This is a central result: *the shear stress profile is linear in r , regardless of the constitutive model*. It is maximum at the wall ($r = R$):

$$\tau_w \equiv \tau(R) = \frac{\Delta P R}{2L}, \quad \Longleftrightarrow \quad \Delta P = \frac{2L \tau_w}{R}. \quad (30 \text{ derived})$$

Only the velocity profile $v(r)$ – derived in the next step – depends on the constitutive model.

Step 2 – Velocity profile for a Power-Law fluid.

Substitute the Power-Law relation into Eq. (31):

$$K \left(-\frac{dv}{dr} \right)^n = \frac{\Delta P}{2L} r.$$

(The sign convention $-dv/dr > 0$ makes $\dot{\gamma} > 0$ since velocity decreases toward the wall.) Solving for $-dv/dr$:

$$-\frac{dv}{dr} = \left(\frac{\Delta P}{2LK} \right)^{1/n} r^{1/n}.$$

Integrate from r outward to the wall at R , applying no-slip $v(R) = 0$:

$$\int_r^R \left(-\frac{dv}{dr'} \right) dr' = \left(\frac{\Delta P}{2LK} \right)^{1/n} \int_r^R r'^{1/n} dr'.$$

The left side evaluates to $v(R) - v(r) = -v(r)$ (since $v(R) = 0$). The right side integrates as a power of r' :

$$\int_r^R r'^{1/n} dr' = \left[\frac{r'^{1/n+1}}{1/n+1} \right]_r^R = \frac{n}{n+1} \left(R^{(n+1)/n} - r^{(n+1)/n} \right).$$

Substituting back (and reversing the sign on the left):

$$v(r) = \left(\frac{\Delta P}{2LK} \right)^{1/n} \frac{n}{n+1} \left(R^{(n+1)/n} - r^{(n+1)/n} \right). \quad (32)$$

Sanity checks. At $r = 0$: $v(0) = (\Delta P/2LK)^{1/n} \frac{n}{n+1} R^{(n+1)/n}$ – the centreline maximum. At $r = R$: $v(R) = 0$ – no-slip satisfied. As $n \rightarrow 1$ (Newtonian): the profile recovers the standard parabola $v(r) = (\Delta P/4\eta L)(R^2 - r^2)$. As $n \rightarrow 0$: the exponent $(n+1)/n \rightarrow \infty$ and the profile becomes flat – a “plug-like” distribution with all shear concentrated near the wall.

Step 3 – Volumetric flow rate Q .

Integrate $v(r)$ over the circular cross-section:

$$Q = \int_0^R v(r) 2\pi r dr = 2\pi \left(\frac{\Delta P}{2LK} \right)^{1/n} \frac{n}{n+1} \int_0^R \left(R^{(n+1)/n} - r^{(n+1)/n} \right) r dr.$$

Split the integral into two terms:

$$I_1 = \int_0^R R^{(n+1)/n} r dr = R^{(n+1)/n} \cdot \frac{r^2}{2} \Big|_0^R = \frac{1}{2} R^{(n+1)/n+2} = \frac{1}{2} R^{(3n+1)/n}, \quad (33)$$

$$\begin{aligned} I_2 &= \int_0^R r^{(n+1)/n} r dr = \int_0^R r^{(n+1)/n+1} dr = \left[\frac{r^{(n+1)/n+2}}{(n+1)/n+2} \right]_0^R \\ &= \frac{R^{(3n+1)/n}}{(n+1)/n+2}. \end{aligned} \quad (34)$$

Simplify the denominator in Eq. (34):

$$\frac{n+1}{n} + 2 = \frac{n+1+2n}{n} = \frac{3n+1}{n},$$

so

$$I_2 = \frac{n}{3n+1} R^{(3n+1)/n}. \quad (35)$$

The bracket $I_1 - I_2$ is then:

$$I_1 - I_2 = R^{(3n+1)/n} \left(\frac{1}{2} - \frac{n}{3n+1} \right) = R^{(3n+1)/n} \frac{3n+1-2n}{2(3n+1)} = R^{(3n+1)/n} \frac{n+1}{2(3n+1)}.$$

Substituting back:

$$Q = 2\pi \left(\frac{\Delta P}{2LK} \right)^{1/n} \frac{n}{n+1} \cdot R^{(3n+1)/n} \frac{n+1}{2(3n+1)},$$

the $(n+1)$ cancels:

$$Q = \frac{\pi n}{3n+1} \left(\frac{\Delta P}{2LK} \right)^{1/n} R^{(3n+1)/n} = \frac{\pi n R^3}{3n+1} \left(\frac{\tau_w}{K} \right)^{1/n}, \quad (36)$$

where the second form uses $\tau_w = \Delta P R / (2L)$ to replace $(\Delta P / 2LK)^{1/n} R^{(3n+1)/n} = (\tau_w / K)^{1/n} R^3 / R^0$.

Step 4 – Inversion: τ_w as a function of Q .

Raise both sides of Eq. (36) to the power n and solve for τ_w :

$$Q^n = \left(\frac{\pi n R^3}{3n+1} \right)^n \frac{\tau_w^n}{K}, \quad \implies \quad \tau_w = K \left(\frac{(3n+1)Q}{\pi n R^3} \right)^n. \quad (29 \text{ derived})$$

This is the closed-form result for wall shear stress. Combined with $\Delta P = 2L\tau_w/R$, it completes the derivation.

Summary and physical interpretation.

- The universal linear stress profile $\tau(r) = (\Delta P / 2L)r$ (Step 1) is *model-independent*; it is an exact consequence of Newton's second law alone.
- Only the velocity profile $v(r)$ (Step 2) and hence Q (Step 3) depend on the constitutive model. For Newtonian fluids $n = 1$, $K = \eta$, and Eq. (36) reduces exactly to the classical Hagen–Poiseuille result.
- The factor $(3n+1)/(4n)$ that appears in the Rabinowitsch wall-shear-rate correction (see §4.3) is a direct consequence of this derivation: it converts the apparent Newtonian wall shear rate $4Q/\pi R^3$ to the true value for a power-law fluid.

4.3 Cross-model extension (Rabinowitsch–Weissenberg)

Motivation. The Cross model gives a physically correct rest viscosity η_0 that the Power Law cannot represent. At very slow piston velocities – the safe operating end of the window for cell-loaded inks – the Cross model can predict pressures and wall shear rates that differ by an order of magnitude from the Power-Law estimate. Accepting the extra numerical effort of the Rabinowitsch–Weissenberg integral is therefore not optional for production-quality predictions.

The Cross model has no closed-form $Q(\tau_w)$ expression. We use the Rabinowitsch–Weissenberg

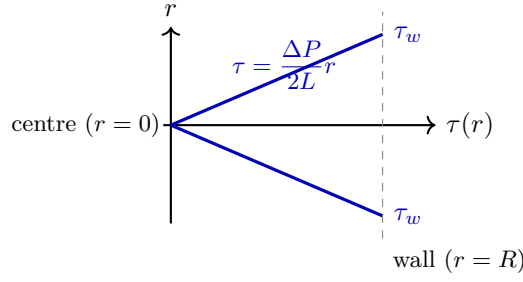


Figure 4: Shear stress profile inside the needle. The stress is always linear in r , peaking at the wall with value $\tau_w = \Delta P R/2L$ and zero at the centreline. This universal result holds for any constitutive model.

integral:

$$Q(\tau_w) = \frac{\pi R^3}{\tau_w^3} \int_0^{\tau_w} \tau^2 \dot{\gamma}(\tau) d\tau, \quad (37)$$

where $\dot{\gamma}(\tau)$ is obtained by inverting Eq. 7 (solve $\eta(\dot{\gamma})\dot{\gamma} = \tau$ for $\dot{\gamma}$ numerically – a single `fzero` call per point). The MATLAB code at `bioprinting_algorithm_v4.m` runs this on a 400-point grid for each V_p .

Why this integral is exact (for any constitutive model). The key insight is that the universal linear stress profile $\tau(r) = (\Delta P/2L)r$ allows a change of integration variable from r to τ . Writing $r = 2L\tau/\Delta P$ and $dr = 2L/\Delta P d\tau$, and noting that $Q = 2\pi \int_0^R v(r)r dr$ can be integrated by parts (with $v(R) = 0$):

$$Q = \pi \int_0^R r^2 \dot{\gamma}(r) dr = \frac{\pi R^3}{\tau_w^3} \int_0^{\tau_w} \tau^2 \dot{\gamma}(\tau) d\tau.$$

No assumption about the form of $\dot{\gamma}(\tau)$ has been made; the identity holds for *any* generalised Newtonian fluid.

The Weissenberg–Rabinowitsch wall-shear-rate correction. Differentiating Eq. (37) with respect to τ_w yields the true wall shear rate from the apparent Newtonian value $\dot{\gamma}_a \equiv 4Q/\pi R^3$:

$$\dot{\gamma}_w = \frac{1}{\pi R^3} \left(3Q + \tau_w \frac{dQ}{d\tau_w} \right) = \frac{3n' + 1}{4n'} \cdot \frac{4Q}{\pi R^3}, \quad (38)$$

where

$$n' \equiv \frac{d \ln \tau_w}{d \ln \dot{\gamma}_w} \quad (39)$$

is the **local log-log slope** of the flow curve τ_w vs $\dot{\gamma}_w$ evaluated at the operating point. Three observations follow immediately:

1. **For the Power-Law model**, $n' = n$ everywhere (constant slope in log-log), so Eq. (38) reduces to the exact result from the derivation above: $\dot{\gamma}_w = \frac{3n+1}{4n} \cdot \frac{4Q}{\pi R^3}$.
2. **For the Cross model**, n' varies with $\dot{\gamma}_w$ (from $n' \rightarrow 1$ at very low $\dot{\gamma}_w$ in the Newtonian plateau to $n' \rightarrow 1 - m$ in the power-law region, where $\eta \sim \dot{\gamma}^{-m}$ so $\tau \sim \dot{\gamma}^{1-m}$; for the NE, $m = 0.988$ gives $n' \approx 0.01$). The correction factor $(3n' + 1)/(4n')$ must therefore be re-evaluated at each operating point; this is equivalent to evaluating the integral in Eq. (37) numerically.
3. **Connection to the generalised Reynolds number.** The same local slope n' enters the Metzner–Reed Re_{gen} (Eq. 27): both the laminar-flow validation and the wall-shear-rate correction share the same local-slope characterisation of the flow curve. Note that this n' (slope of τ_w vs. the true wall shear rate $\dot{\gamma}_w$) and the Metzner–Reed n' used in Re_{gen} (slope vs. the nominal rate $8\bar{v}/D$, Hydraulic-Loop chapter) are the same local log-log slope of the flow curve; for a power-law fluid they coincide exactly and the two uses are interchangeable.

4.4 Which model to trust?

Motivation. Using the Power-Law model outside its fitted shear-rate range is a leading source of over-predicted pressures and spurious safety failures. Knowing *when* the Cross model differs materially from the Power Law – and why – prevents the common mistake of dismissing an ink as “unprintable” because the Power-Law ΔP estimate exceeds the syringe ceiling at a V_p where the true Cross-model value is perfectly safe.

Both models are available in the MATLAB output, and they are compared side-by-side.

- **In the printable range** ($\dot{\gamma}_w \sim 50\text{--}1000\text{ s}^{-1}$ for 21G), both agree within 20% for our inks.
- **At low $\dot{\gamma}_w$** (very slow prints or 22G at low V_p): the Power Law over-estimates ΔP because it has no η_0 plateau. The Cross model is correct.
- **Preferred:** Cross for all production runs. Power Law reserved for quick back-of-envelope estimates.

4.5 Wall shear rate and stress

Motivation. ΔP and τ_w are both required, and they are not interchangeable: ΔP is the mechanical load on the hardware; τ_w is the biological stress on the cells. A print can be mechanically safe ($\Delta P \ll$ syringe ceiling) yet biologically damaging ($\tau_w > 5\text{ kPa}$) – or vice versa. Reporting either without the other gives an incomplete safety picture.

Once τ_w is known, wall shear rate follows by inverting the constitutive model:

$$\dot{\gamma}_w = (\tau_w/K)^{1/n} \text{ (PL); } \quad \dot{\gamma}_w \text{ by } \mathbf{fzero} \text{ on } \eta(\dot{\gamma})\dot{\gamma} = \tau_w \text{ (Cross).} \quad (40)$$

$\dot{\gamma}_w$ is the shear rate cells see at the needle wall, and is the disruption shear rate to use when evaluating the recovery test result.

Cell-safety threshold. Literature consensus places the upper bound for human cells at $\tau_w \sim 5\text{ kPa}$ (Blaeser et al., 2016). Above this, cell viability drops sharply. Both τ_w and cumulative shear exposure (Ch. 20) must be checked for cell-loaded prints.

WORKED EXAMPLE – the three inks

Hagen–Poiseuille on 21G at $V_p = 0.01\text{ mm/s}$ (Cross model):

Ink	Q [mm ³ /s]	τ_w [Pa]	$\dot{\gamma}_w$ [s ⁻¹]	ΔP [kPa]
C15	1.61	564	220	139
C15 Gira 5.5	1.61	663	60	164
Bozano	1.61	469	8	116

All three inks deliver the same Q (geometry-fixed), but at very different τ_w and $\dot{\gamma}_w$ – the rheology decides. Pressures sit at 100–170 kPa, well below the 1500 kPa syringe ceiling and the 5 kPa cell-safe wall-shear ceiling.

RUNNING EXAMPLE – Solution to Q3

Question. At $V_p = 0.01\text{ mm/s}$ on 21G, is the extrusion pressure safe for C15 Gira 5.5?

From the worked example above:

$$\Delta P_{\text{Cross}} \approx 164\text{ kPa}, \quad \tau_w \approx 663\text{ Pa}, \quad \dot{\gamma}_w \approx 60\text{ s}^{-1}.$$

Safety check:

- Syringe ceiling 1500 kPa: $164/1500 = 11\%$. Comfortable.

- Cell-safety $\tau_w < 5$ kPa: 663 Pa is far below.
- Motor stall (estimated ~ 500 – 800 kPa): comfortable.

Verdict. Safe to print. At $V_p = 0.03$ mm/s, ΔP rises to ~ 400 kPa (scaling as $V_p^{0.24}$ for $n = 0.24$) – still safe but with smaller margin.

Still unknown: the slicer Extrusion Multiplier. Q4.

IN PRACTICE

Reading the MATLAB output: always check the trio ΔP , τ_w , $\dot{\gamma}_w$.

After running `run_solver_v4.m`, the output file and summary plot report all three quantities for every (V_p , needle) combination. A safe operating point requires passing *two independent* checks:

1. **Mechanical:** $\Delta P < 1500$ kPa (BD 10 mL syringe rating). Most DIW inks clear this easily at bench-realistic V_p .
2. **Biological:** $\tau_w < 5$ kPa (cell-safety ceiling, Eq. 4.5). This is the binding constraint for cell-loaded inks; note that $\tau_w = \Delta P R / 2L$ grows with decreasing R , so a narrow needle can be mechanically fine yet biologically dangerous.

Do not stop at ΔP . A syringe pressure of 300 kPa is perfectly safe for hardware, but for a 22G needle at that pressure $\tau_w = \Delta P R / (2L) = 300 \times 10^3 \times 0.2065 \times 10^{-3} / (2 \times 25.4 \times 10^{-3}) = 1.22$ kPa – safely below 5 kPa in this case, but always verify explicitly; do not rely on intuition. If τ_w exceeds the ceiling, the corrective action is to increase R (use a wider needle) or decrease V_p , not merely to check the syringe rating.

GOING DEEPER

Detecting wall slip (Mooney analysis). Real DIW inks can slip at the needle wall, inflating apparent flow. Mooney’s test: measure the apparent wall shear rate $\dot{\gamma}_a = 4Q/\pi R^3$ at fixed wall stress τ_w for needles of *different* radii R and plot $\dot{\gamma}_a$ vs. $1/R$. A non-zero slope $= 4v_s(\tau_w)$ reveals a slip velocity v_s ; zero slope means no slip. Slip masquerades as shear thinning, so unflagged slip corrupts K, n . This is the physical basis of the f_{slip} factor in k_{flow} (Eq. 4.5) (Mooney, 1931).

Check Yourself

Exercise 4.1 ◦ A 21G needle has inner radius $R = 0.2575$ mm and length $L = 31.75$ mm. The Cross-model MATLAB solver predicts a wall shear stress of $\tau_w = 500$ Pa at a given V_p .

- Use Eq. 30 to compute the pressure drop ΔP in kPa.
- Is this ΔP within the syringe ceiling of 1500 kPa?
- Is τ_w within the cell-safety ceiling of 5 kPa?

SOLUTION

(a) $\Delta P = 2L\tau_w/R = 2 \times 31.75 \text{ mm} \times 500 \text{ Pa} / 0.2575 \text{ mm}$. Convert all lengths to metres: $L = 0.03175 \text{ m}$, $R = 2.575 \times 10^{-4} \text{ m}$.

$$\Delta P = \frac{2 \times 0.03175 \times 500}{2.575 \times 10^{-4}} = \frac{31.75}{2.575 \times 10^{-4}} = 1.233 \times 10^5 \text{ Pa} \approx 123.3 \text{ kPa}.$$

- 123.3 kPa < 1500 kPa: within the syringe ceiling (8.2% of rating). Comfortable.
- 500 Pa = 0.5 kPa < 5 kPa: within the cell-safety ceiling by a factor of 10.

Exercise 4.2 ○○ A bioink follows the Power-Law model with $K = 120 \text{ Pas}^n$ and $n = 0.25$. It is printed through a 21G needle ($R = 0.2575 \text{ mm}$, $L = 31.75 \text{ mm}$) at a piston velocity $V_p = 0.01 \text{ mm/s}$ ($Q \approx 1.61 \text{ mm}^3/\text{s}$; use this value directly).

- Apply Eq. 29 to compute τ_w in Pa.
- Compute ΔP in kPa via Eq. 30.
- Compute the true wall shear rate $\dot{\gamma}_w$ in s^{-1} by inverting the Power-Law (Eq. 40).
(Use the stated round values; do not substitute the project-ink parameters from the appendices.)

SOLUTION

Step (a) – Wall shear stress.

First evaluate the argument of the Power-Law:

$$\frac{(3n+1)Q}{\pi n R^3} = \frac{(3 \times 0.25 + 1) \times 1.61}{\pi \times 0.25 \times (0.2575)^3}.$$

Numerator: $(0.75 + 1) \times 1.61 = 1.75 \times 1.61 = 2.8175 \text{ mm}^3/\text{s}$.

Denominator: $\pi \times 0.25 \times (0.2575)^3$. $(0.2575)^3 = 0.2575 \times 0.2575 \times 0.2575$. $0.2575^2 = 0.066306$; $0.066306 \times 0.2575 = 0.017074 \text{ mm}^3$. $\pi \times 0.25 \times 0.017074 = 0.78540 \times 0.017074 = 0.013413 \text{ mm}^3/\text{s}$ (the Q units cancel, leaving s^{-1} after division).

$$\frac{(3n+1)Q}{\pi n R^3} = \frac{2.8175}{0.013413} = 210.1 \text{ s}^{-1}.$$

Now raise to the power $n = 0.25$: $(210.1)^{0.25} = \sqrt{\sqrt{210.1}}$. $\sqrt{210.1} \approx 14.50$; $\sqrt{14.50} \approx 3.808$.

$$\tau_w = K \times (210.1)^{0.25} = 120 \times 3.808 = 456.9 \text{ Pa} \approx 457 \text{ Pa}.$$

Step (b) – Pressure drop.

$$\Delta P = \frac{2L\tau_w}{R} = \frac{2 \times 31.75 \text{ mm} \times 457 \text{ Pa}}{0.2575 \text{ mm}} = \frac{29,017}{0.2575} \text{ Pa} = 112.689 \text{ Pa} \approx 112.7 \text{ kPa}.$$

Step (c) – Wall shear rate (PL inversion).

$$\dot{\gamma}_w = \left(\frac{\tau_w}{K} \right)^{1/n} = \left(\frac{457}{120} \right)^{1/0.25} = (3.808)^4.$$

$(3.808)^2 = 14.50$; $(3.808)^4 = (14.50)^2 = 210.3 \text{ s}^{-1}$.

As expected, $\dot{\gamma}_w \approx 210 \text{ s}^{-1}$ – this is consistent with the argument evaluated in (a), confirming the algebra is self-consistent.

Exercise 4.3 ○○○ A researcher suspects wall slip in a CMC hydrogel and performs a Mooney test at fixed wall stress $\tau_w = 400 \text{ Pa}$. Measuring the apparent wall shear rate $\dot{\gamma}_a = 4Q/\pi R^3$ at two needle radii gives:

Needle	R [mm]	$\dot{\gamma}_a$ [s^{-1}]
21G	0.2575	180
22G	0.2065	260

- Plot $\dot{\gamma}_a$ vs. $1/R$ (identify the two data points and their coordinates) and compute the slope.
- Extract the slip velocity v_s from the slope and comment on its magnitude relative to typical mean needle velocities.
- If wall slip is confirmed, does the fitted Power-Law index n overestimate or underestimate shear thinning? Explain.

SOLUTION**Step (a) – Mooney plot coordinates.**

$$1/R_1 = 1/0.2575 = 3.883 \text{ mm}^{-1}, \dot{\gamma}_{a1} = 180 \text{ s}^{-1}.$$

$$1/R_2 = 1/0.2065 = 4.842 \text{ mm}^{-1}, \dot{\gamma}_{a2} = 260 \text{ s}^{-1}.$$

Slope:

$$m_{\text{Mooney}} = \frac{\dot{\gamma}_{a2} - \dot{\gamma}_{a1}}{1/R_2 - 1/R_1} = \frac{260 - 180}{4.842 - 3.883} = \frac{80}{0.959} = 83.4 \text{ s} \cdot \text{mm} = 83.4 \text{ mm/s} \cdot \text{mm}^{-1} \cdot \text{mm} = 83.4 \text{ mm/s}.$$

(Units: $[\dot{\gamma}_a] = \text{s}^{-1}$, $[1/R] = \text{mm}^{-1}$, so slope has dimensions of mm/s.)

Step (b) – Slip velocity.

The Mooney relation is $\dot{\gamma}_a(\tau_w) = \dot{\gamma}_{\text{true}}(\tau_w) + 4v_s(\tau_w)/R$, so a plot of $\dot{\gamma}_a$ vs. $1/R$ at fixed τ_w has slope $= 4v_s$:

$$v_s = \frac{m_{\text{Mooney}}}{4} = \frac{83.4}{4} = 20.8 \text{ mm/s}.$$

At $V_p = 0.01 \text{ mm/s}$ the mean needle velocity for 21G is $\bar{v}_n = 771 \times 0.01 = 7.71 \text{ mm/s}$. A slip velocity of 20.8 mm/s is nearly three times this mean velocity – a very large fraction of the measured flow rate is wall slip, not true bulk shear. The apparent flow curve is severely contaminated.

Step (c) – Effect on fitted n .

Wall slip adds an R -dependent excess to the apparent $\dot{\gamma}_a$ that scales as $\sim 1/R$ rather than the $\sim 1/R^3$ dependence of the true wall shear rate. At small R (narrow needles, high $1/R$), the slip contribution proportionally inflates $\dot{\gamma}_a$ more, making $\tau_w(\dot{\gamma}_a)$ appear to flatten more quickly with $\dot{\gamma}_a$ than it would in a true bulk measurement. Fitting $\tau = K\dot{\gamma}^n$ to the slip-contaminated data therefore yields a *lower* apparent n (more shear-thinning) than the true bulk behaviour – i.e., n is *underestimated*. Equivalently, K is overestimated. Both errors inflate the predicted ΔP at high $\dot{\gamma}_w$.

KEY TAKEAWAYS

- The shear stress profile $\tau(r) = (\Delta P/2L)r$ is universal and linear – wall stress is always the maximum.
- $\Delta P \propto R^{-4}$: needle gauge is the dominant lever. Cross-model prediction is preferred over Power Law for all multi-decade ranges.
- $\tau_w < 5 \text{ kPa}$ is the cell-safety constraint. All three inks satisfy it comfortably at the nominal V_p .

ENGINEERING DESIGN RULES

DR 4.1. Always report ΔP , τ_w , and $\dot{\gamma}_w$ as a trio when characterising a print condition. ΔP alone is insufficient – τ_w determines cell survival.

DR 4.2. Use the Cross model for ΔP prediction. Report both Cross and Power Law outputs so inconsistencies can be flagged.

DR 4.3. For any new ink or needle combination, always run Re_{gen} and Bi estimates *before* the first print to confirm laminar flow and to anticipate the velocity profile type.

5 Temperature and Process-Robustness

Motivation. Everything so far assumed a fixed temperature and an ideal wall. Two real-world levers move the rheology under your feet: *temperature* and *flow instabilities*. This short chapter gives the theory you need to keep a print reproducible across a warm afternoon and a cold lab.

5.1 Temperature dependence of viscosity

PHYSICAL INTUITION

Viscosity falls as temperature rises: molecules jump their neighbours' cages more easily. A few degrees of drift can change ΔP and bead width enough to fail a print — so either control T or measure its slope.

For most aqueous gels well above any glass transition, the **Arrhenius** form holds:

$$\eta(T) = \eta_{\infty} \exp\left(\frac{E_a}{RT}\right), \quad (41)$$

where E_a (J/mol) is the flow activation energy and $R = 8.314$ J/mol/K. A plot of $\ln \eta$ vs. $1/T$ is linear with slope E_a/R .

DERIVATION — Arrhenius: two-point method and small- ΔT sensitivity

Two-point method. Write Eq. 41 at two temperatures T_1 and T_2 :

$$\ln \eta_1 = \ln \eta_{\infty} + \frac{E_a}{RT_1}, \quad \ln \eta_2 = \ln \eta_{\infty} + \frac{E_a}{RT_2}.$$

Subtract to eliminate η_{∞} :

$$\ln \frac{\eta_1}{\eta_2} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right).$$

Solving for E_a :

$$E_a = \frac{R \ln(\eta_1/\eta_2)}{1/T_1 - 1/T_2}. \quad (42)$$

This requires only two viscosity measurements at two temperatures; no pre-knowledge of η_{∞} is needed.

Small-drift sensitivity. Differentiate Eq. 41 with respect to T :

$$\frac{d\eta}{dT} = -\frac{E_a}{RT^2} \eta(T).$$

Divide both sides by $\eta(T)$:

$$\frac{d \ln \eta}{dT} = -\frac{E_a}{RT^2}.$$

For a small temperature excursion ΔT (a few degrees), the fractional viscosity change is therefore:

$$\frac{\Delta \eta}{\eta} \approx e^{-(E_a/RT^2) \Delta T} - 1 \approx -\frac{E_a}{RT^2} \Delta T \quad (\text{first-order}). \quad (43)$$

The same proportional change in η propagates into ΔP via Hagen–Poiseuille, so a ± 1 °C drift at $T = 293$ K causes a fractional ΔP shift of $\approx \pm(E_a/RT^2)$, independent of needle gauge or V_p .

GOING DEEPER

When Arrhenius fails: WLF. Near a glass/gel transition ($T < T_g + 100$ K) the temperature dependence is super-Arrhenius and the Williams–Landel–Ferry equation applies to the shift factor a_T :

$$\log_{10} a_T = \frac{-C_1 (T - T_{\text{ref}})}{C_2 + (T - T_{\text{ref}})}, \quad (44)$$

with “universal” $C_1 \approx 17.4$, $C_2 \approx 51.6$ K when $T_{\text{ref}} = T_g$. For the dilute aqueous CMC/NE inks here, Arrhenius (Eq. 41) is the practical choice; WLF matters for concentrated or near-

gelation formulations (Williams et al., 1955).

IN PRACTICE

Measure η at $T_{\text{lab}} \pm 5^\circ\text{C}$, fit E_a via Eq. 42, and report the %-change in ΔP per $^\circ\text{C}$ using Eq. 43. If it exceeds your pressure margin, thermostat the syringe.

5.2 Flow instabilities and the smooth-filament limit

Even laminar, isothermal extrusion has an upper usable rate. Above a critical wall stress the extrudate stops being smooth: *sharkskin* (surface mattness), then *gross melt fracture* (helical/chaotic distortion). The onset is an *elastic* instability, governed by the Weissenberg number $Wi = N_1/(2\tau_w) = S_R$ (Eq. 55) and the Deborah number rather than by Reynolds number.

IN PRACTICE

The “rough filament / extrudate distortion” failure mode (Ch. 8) is this instability. Corrective action: lower V_p (reduce τ_w and S_R), widen the needle, or raise T to drop viscosity — all reduce the stored elastic strain at exit.

5.3 Reproducibility checklist

1. Record lab T at print time; keep the syringe within $\pm 2^\circ\text{C}$.
2. Re-fit K, n if T drifts more than your measured E_a tolerance allows (Eq. 43).
3. Stay below the onset of surface distortion (keep S_R modest; see Eq. 56).
4. Re-run recovery if the batch or storage time changed (thixotropy ages: Barnes, 1997; Mewis and Wagner, 2009).

Check Yourself

Exercise 5.1 ◦ An ink has $E_a = 30\text{ kJ/mol}$ at a lab temperature of $T = 293\text{ K}$. Estimate the fractional viscosity change for a $+5^\circ\text{C}$ rise using the small-drift approximation (Eq. 43).

SOLUTION

$$E_a/R = 30000/8.314 = 3608\text{ K}.$$

$$\Delta \ln \eta \approx -(E_a/RT^2) \Delta T = -3608 \times 5/293^2 = -18040/85849 = -0.210.$$

Fractional change $= e^{-0.210} - 1 = 0.810 - 1 = -0.190$, i.e., viscosity drops $\approx 19\%$ per 5°C rise.

Note: the first-order approximation $\approx -E_a\Delta T/(RT^2) = -0.210$ agrees well because $|\Delta \ln \eta| \ll 1$.

Exercise 5.2 ◦◦ Two-point Arrhenius. An ink has $\eta = 100\text{ Pa s}$ at $T_1 = 293\text{ K}$ and $\eta = 70\text{ Pa s}$ at $T_2 = 303\text{ K}$. Use Eq. 42 to find E_a .

SOLUTION

$$\ln(100/70) = \ln(10/7) = 0.3567.$$

$$1/T_1 - 1/T_2 = 1/293 - 1/303 = (303 - 293)/(293 \times 303) = 10/88779 = 1.126 \times 10^{-4}\text{ K}^{-1}.$$

$$E_a = R \ln(\eta_1/\eta_2) / (1/T_1 - 1/T_2) = 8.314 \times 0.3567 / 1.126 \times 10^{-4} = 2.965 / 1.126 \times 10^{-4} = 2.63 \times 10^4\text{ J/mol} \approx 26.3\text{ kJ/mol}.$$

(Some rounding conventions give 26.4 kJ/mol ; the spread reflects significant-figure choices in $\ln(10/7)$ and the denominator.)

Exercise 5.3 ◦◦◦ Explain why extrudate distortion (melt fracture) at the needle exit scales with the Weissenberg number rather than the Reynolds number for CMC/NE inks.

SOLUTION

The generalised Reynolds number $\text{Re}_{\text{gen}} \ll 1$ for these high-viscosity inks (typically $\mathcal{O}(10^{-3})$ – $\mathcal{O}(10^{-1})$), so inertial forces are negligible and the flow is deeply laminar. Extrudate distortion is therefore not an inertial (Kelvin–Helmholtz-type) instability but an *elastic* one: at high V_p the polymer chains are highly stretched inside the needle, storing elastic energy characterised by the first normal stress difference N_1 . At the exit the confinement is removed and the stored elastic strain $S_R = N_1/(2\tau_w)$ (Eq. 55) is released non-uniformly across the cross-section, destabilising the free surface. The Weissenberg number $\text{Wi} = S_R$ is the dimensionless ratio of elastic to viscous stresses; once S_R exceeds an $\mathcal{O}(1)$ threshold the surface buckles into sharkskin or chaotic fracture. Corrective actions that lower S_R — reducing V_p , switching to a wider needle, or raising T to decrease viscosity and relax the chains faster — all delay or prevent the onset.

KEY TAKEAWAYS

- Viscosity is exponentially sensitive to temperature (Arrhenius); even $\pm 2^\circ\text{C}$ drift can shift ΔP and bead width measurably if E_a is large.
- The two-point method (Eq. 42) gives E_a from just two flow-curve runs at different temperatures; report it for every ink used outside $20 \pm 2^\circ\text{C}$.
- Extrudate distortion is an elastic instability (Wi , not Re): lower V_p or raise T to reduce the recoverable strain S_R at the exit.
- Thixotropy ages with batch and storage: re-run recovery if the ink is more than a day old.

ENGINEERING DESIGN RULES

DR T.1. For any ink used outside the characterisation temperature $T_0 \pm 2^\circ\text{C}$, measure η at two temperatures, compute E_a (Eq. 42), and report the fractional ΔP sensitivity per $^\circ\text{C}$ to the print log.

DR T.2. Keep the recoverable strain S_R (Eq. 55) below the surface-distortion onset. If rough filaments appear, reduce V_p by 20% increments or switch to the next wider needle before adjusting any slicer compensation.

6 Flow Factor, Die Swell, and the Slicer Translation

Why this chapter matters. A conventional FFF slicer assumes that the Extrusion Multiplier is close to 1.0 and that deposited filament width equals the nozzle diameter. Neither assumption holds for DIW: three independent physical mechanisms shift the deposited volume away from the commanded volume, and ignoring them produces systematically wrong line widths and layer heights that accumulate across every layer of a multi-layer scaffold. This chapter derives each mechanism, assembles them into k_{flow} (Eq. 45), and shows how to translate that diagnostic into the single number the slicer needs.

The slicer treats the printer as FFF. It multiplies the commanded volumetric flow by an Extrusion Multiplier $\text{EM}_{\text{slicer}}$, which in FFF normally sits near 1.0. In DIW, three independent physical deviations change the effective deposited volume:

1. **Die swell** (β_{swell}): after leaving the needle, viscoelastic ink expands radially as stored elastic stresses relax. The deposited filament is wider than the needle ID. $w_{\text{line}}/\text{ID} = 1 + \beta_{\text{swell}}$.
2. **Syringe-wall slip** ($f_{\text{slip}} \leq 1$): imperfect piston sealing or ink-wall slip reduces the actual volume leaving the needle. Default 1.0 until C2 calibration.
3. **Viscoelastic recovery penalty** ($\sqrt{R_{\text{rec}}}$): if the network has only partially recovered when

the next layer arrives, the deposited bead has not stiffened back up; material spreads and effective cross-sectional volume is lost.

$$k_{\text{flow}} = (1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}. \quad (45)$$

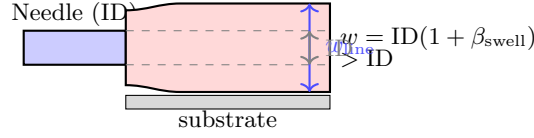


Figure 5: Die swell: the deposited filament is wider than the needle ID because elastic stresses stored during confinement in the needle relax on exit. Inks with $m \rightarrow 1$ (narrow transition) store little elastic energy and swell minimally. Inks with $m < 0.7$ may swell by $\beta > 0.15$ and require direct measurement in calibration test C1.

DERIVATION — Each factor in $k_{\text{flow}} = (1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}$

We derive each of the three dimensionless factors in Eq. 45 from first principles and then assemble the full expression.

Factor 1 — Die-swell area term $(1 + \beta_{\text{swell}})^2$.

Setup. Define the swell ratio so that the deposited bead has width

$$w = \text{ID} (1 + \beta_{\text{swell}}), \quad \beta_{\text{swell}} \geq 0. \quad (46)$$

The slicer models a cylindrical filament of cross-sectional area $A_{\text{ref}} = \pi(\text{ID}/2)^2$ per unit length of travel.

Actual cross-section. For a bead whose cross-section scales with the square of its lateral width (the bead spreads uniformly in both transverse directions), the deposited cross-sectional area is

$$A_{\text{dep}} = A_{\text{ref}} \left(\frac{w}{\text{ID}} \right)^2 = A_{\text{ref}} (1 + \beta_{\text{swell}})^2. \quad (47)$$

Volume per unit length. Volume deposited per unit nozzle travel $= A_{\text{dep}} \cdot \ell = A_{\text{ref}} (1 + \beta_{\text{swell}})^2 \cdot \ell$, while the slicer assumes $A_{\text{ref}} \cdot \ell$. The ratio of deposited to commanded volume per unit travel is therefore:

$$\left. \frac{V_{\text{dep}}}{V_{\text{cmd}}} \right|_{\text{swell}} = (1 + \beta_{\text{swell}})^2. \quad (48)$$

This factor exceeds 1 whenever $\beta_{\text{swell}} > 0$: swell deposits *more* material per unit length than commanded, so k_{flow} is driven above the slip and recovery baseline.

Factor 2 — Slip factor $f_{\text{slip}} \leq 1$.

Definition. In an ideal piston-driven system (no seal leak, no ink-barrel slip), a piston displacement δx sweeps a volume $A_{\text{piston}} \delta x$ that exits entirely through the needle. In practice, imperfect sealing or slip at the ink-barrel interface means only a fraction of that swept volume actually exits:

$$V_{\text{exit}} = f_{\text{slip}} A_{\text{piston}} \delta x, \quad 0 < f_{\text{slip}} \leq 1. \quad (49)$$

Equivalently,

$$f_{\text{slip}} = \frac{V_{\text{exit}}}{V_{\text{swept}}}. \quad (50)$$

A value $f_{\text{slip}} < 1$ reduces the actual flow rate below the slicer's assumed value, causing under-extrusion. Experimentally, f_{slip} is determined from the C2 calibration (mass-per-line test).

For well-sealed syringes with low-viscosity inks, $f_{\text{slip}} \approx 1.0$; for high-viscosity bioinks with marginal sealing, values as low as 0.8 have been reported.

Factor 3 — Viscoelastic recovery penalty $\sqrt{R_{\text{rec}}}$.

Physical argument (semi-empirical). When a hydrogel bead is deposited, its microstructure has been disrupted by the high shear inside the needle. If the inter-layer dwell time is shorter than the network’s relaxation time, the network is only partially recovered (recovery fraction $R_{\text{rec}} < 1$) when the next layer arrives.

A bead with $R_{\text{rec}} < 1$ has a lower effective elastic modulus $G'_{\text{eff}} \approx R_{\text{rec}} G'_0$. The restorative elastic stress that opposes lateral spreading of the bead scales with G'_{eff} . Under a gravitational or contact load, the equilibrium lateral deformation (and hence the bead width w) scales as $(G'_{\text{eff}})^{-1/2}$ — a softer bead deforms more in proportion to the square root of its compliance. Because bead width $\propto (R_{\text{rec}})^{-1/2}$ and the locked-in area scales with w^2 , the ratio of *effectively retained* bead volume to a fully recovered ($R_{\text{rec}} = 1$) bead scales as

$$\left. \frac{V_{\text{retained}}}{V_{\text{ref}}} \right|_{\text{rec}} \propto \sqrt{R_{\text{rec}}} . \quad (51)$$

Note: this is a physically motivated scaling argument, not an exact theorem. The exponent $\frac{1}{2}$ captures the dominant dependence observed across hydrogel systems; materials with strongly non-affine network response may deviate. Direct C2 calibration is preferred over purely analytic estimates when $R_{\text{rec}} < 50\%$.

Assembly. Multiplying the three independent contributions, the overall deposition efficiency is

$$k_{\text{flow}} = \underbrace{(1 + \beta_{\text{swell}})^2}_{\text{swell area}} \cdot \underbrace{f_{\text{slip}}}_{\text{seal/slip}} \cdot \underbrace{\sqrt{R_{\text{rec}}}}_{\text{recovery}} . \quad (45 \text{ restated})$$

The slicer Extrusion Multiplier is the inverse of this efficiency:

$$\text{EM}_{\text{slicer}} = \frac{1}{k_{\text{flow}}} = \frac{1}{(1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}} . \quad (53 \text{ restated})$$

This inversion is the step most often missed in practice: k_{flow} is a diagnostic; $\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$ is the control input.

6.1 The crucial inversion: k_{flow} is not the number you type into the slicer

Motivation. Every DIW slicer that has an “Extrusion Multiplier” field is asking for the factor by which to *scale up* the commanded volumetric flow to achieve the target deposited volume. Because k_{flow} is the *efficiency* (deposited/commanded), the multiplier must be its reciprocal. Entering k_{flow} directly is the most common single-number error in DIW printing, and it compounds: if $k_{\text{flow}} = 0.67$ and the operator types 0.67, the under-build worsens by a further factor of 0.67 on top of the original deficit.

k_{flow} (Eq. 45) is the **deposition efficiency**: the fraction of the slicer-commanded volume that ends up locked in the bead:

$$k_{\text{flow}} = \frac{\text{volume locked in bead}}{\text{volume slicer was told to deliver}} . \quad (52)$$

The slicer’s Extrusion Multiplier is the *control knob* the operator uses to compensate for this inefficiency:

$$\text{EM}_{\text{slicer}} = \frac{1}{k_{\text{flow}}} = \frac{1}{(1 + \beta_{\text{swell}})^2 f_{\text{slip}} \sqrt{R_{\text{rec}}}} . \quad (53)$$

SAFETY / DATA INTEGRITY

The single most common operational error in DIW. Reading $k_{\text{flow}} = 0.67$ from the CSV and typing 0.67 into the slicer *worsens* under-build by another factor of $1/0.67 \approx 1.5$. The slicer multiplier is **always the inverse**: $\text{EM} = 1/k_{\text{flow}}$. When in doubt, verify from first principles: if the bead is too thin (under-extrusion), EM must go *up*.

Physical interpretation:

- $k_{\text{flow}} < 1$: bead loses material to swell-then-spread or recovery-limited slump. Slicer under-builds at nominal volume; raise EM above 1.
- $k_{\text{flow}} > 1$: bead locks in more volume than nominal (high recovery + large swell combine). Slicer over-builds; lower EM below 1.
- $k_{\text{flow}} = 1$: perfect FFF analogy; $\text{EM} = 1$.

6.2 Die-swell proxy from Cross m

Motivation. Die swell requires knowledge of β_{swell} , which is strictly a post-extrusion measurement. When no direct measurement is available (e.g. during ink design or simulation before printing), the Cross sharpness exponent m provides a physically grounded first estimate: a high m (sharp, cooperative transition) corresponds to low elastic energy storage during flow, and hence minimal swell.

Die swell is driven by stored elastic energy. An ink with $m \rightarrow 1$ (sharp transition) stores very little elastic energy during shear-thinning – hence minimal swell. Empirical proxy:

$$\beta_{\text{swell}} \approx 0.30 (1 - m). \quad (54)$$

This spans $\beta = 0$ ($m = 1$, no swell) to $\beta = 0.30$ ($m = 0$, 30% swell). Direct measurement in C1 is preferred when $m < 0.7$.

GOING DEEPER

Where die swell comes from: the first normal-stress difference. In shear, a viscoelastic fluid develops a normal-stress difference $N_1 = \tau_{xx} - \tau_{yy} > 0$ that stores elastic energy. The recoverable shear strain is

$$S_R = \frac{N_1}{2\tau_w}. \quad (55)$$

On exit, that stored strain relaxes and the extrudate swells. Tanner’s relation links the swell ratio $B = D_{\text{extrudate}}/D_{\text{die}}$ to S_R :

$$B = 0.13 + \left[1 + \frac{1}{2} S_R^2 \right]^{1/6}, \quad (56)$$

where the 0.13 is the inertia-less Newtonian baseline swell. So $\beta_{\text{swell}} = B - 1$ is *elastic* in origin. A high Cross m (sharp, cooperative thinning — NE $m = 0.988$) stores little elastic strain $\Rightarrow$ small $S_R \Rightarrow$ minimal swell, which is why the empirical Cross- m proxy used above works (Tanner, 1970).

WORKED EXAMPLE – the three inks

Predicted deposition efficiency and slicer Extrusion Multiplier ($f_{\text{slip}} = 1$, R_{rec} at $\dot{\gamma}_w \sim 100 \text{ s}^{-1}$):

Ink	m	β_{swell}	R_{rec} [%]	k_{flow}	$\text{EM}_{\text{ slicer}} = 1/k_{\text{flow}}$
C15	0.811	0.057	60	0.86	1.17
C15 Gira 5.5	0.988	0.004	45	0.67	1.49
Bozano	0.866	0.040	98	1.07	0.93

The NE is the most demanding ink because of its low recovery: it requires $\sim 49\%$ more material at the slicer than the nominal FFF command. Bozano, with near-full recovery, actually asks for a $\sim 7\%$ reduction.

RUNNING EXAMPLE – Solution to Q4

Question. What Extrusion Multiplier should Ana enter for C15 Gira 5.5?

Step 1 – compute k_{flow} . $\beta_{\text{swell}} = 0.30(1 - 0.988) = 0.004$. $f_{\text{slip}} = 1.0$. $R_{\text{rec}} \approx 45\%$ at $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$:

$$k_{\text{flow}} = (1.004)^2 \cdot 1.0 \cdot \sqrt{0.45} \approx 0.67.$$

Step 2 – invert. $\text{EM}_{\text{ slicer}} = 1/0.67 \approx 1.49$.

Why $\text{EM} > 1$? The NE has the lowest recovery (45%) of the three inks. Material leaves the needle but does not stiffen quickly enough before the next layer arrives; it spreads and loses effective cross-sectional volume. Pushing 49% more compensates.

Still unknown: will a 20-mm block survive its own weight? Q5.

IN PRACTICE

Calibrating k_{flow} at the bench (C2 protocol).

- Measure β_{swell} directly (C1).** Extrude a single line at your target V_p onto a hydrophobic surface. After $\sim 30 \text{ s}$, image the free-standing bead from above and measure its width w with a calibrated micrograph. Then $\beta_{\text{swell}} = (w/\text{ID}) - 1$. Repeat 3 times and average.
- Measure R_{rec} .** Read the structural recovery percentage from the 3-interval thixotropy test (Rheocompass output at the shear rate nearest your computed $\dot{\gamma}_w$).
- Compute k_{flow} .** Use Eq. 45 with $f_{\text{slip}} = 1.0$ as first estimate.
- Set $\text{EM}_{\text{ slicer}} = 1/k_{\text{flow}}$.** Enter this value in the slicer’s “Extrusion Multiplier” field. *Never type k_{flow} itself.*
- Print a C2 calibration line.** Weigh the deposited line on a microbalance and compute $k_{\text{flow}}^{\text{exp}}$ directly from the deposited mass vs. expected mass (see Eq. 52). If $k_{\text{flow}}^{\text{exp}}$ deviates from the analytic prediction by more than 5%, update f_{slip} to absorb the discrepancy.
- Diagnostic guide.**
 - Lines too wide after setting EM: k_{flow} was under-estimated; measure β_{swell} at a finer resolution or check for partial seal failure ($f_{\text{slip}} < 1$).
 - Lines too narrow: k_{flow} was over-estimated; verify that R_{rec} was read at the correct shear rate.

Expected range: $k_{\text{flow}}^{\text{exp}} \in [0.60, 0.85]$ for typical CMC/NE hydrogels; $\text{EM}_{\text{ slicer}} \in [1.17, 1.67]$.

Exercise 6.1 ◦ A 21G needle has an inner diameter of 0.515 mm. Measured die swell gives $\beta_{\text{swell}} = 0.15$. (a) Compute the deposited bead width w . (b) What is the die-swell area term $(1 + \beta_{\text{swell}})^2$?

SOLUTION

(a) $w = \text{ID}(1 + \beta_{\text{swell}}) = 0.515 \text{ mm} \times 1.15 = 0.592 \text{ mm}$.

(b) $(1.15)^2 = 1.3225 \approx \mathbf{1.32}$.

The bead is 15% wider in each transverse direction than the needle ID, which means 32% more cross-sectional area per unit length: a slicer that ignores this will under-extrude by 24% ($\text{EM} = 1/1.32 \approx 0.76$ from swell alone, before recovery and slip are included).

Exercise 6.2 ○○ The first normal-stress difference of an ink at the needle wall is $N_1 = 1000 \text{ Pa}$ and the wall shear stress is $\tau_w = 500 \text{ Pa}$. (a) Compute the recoverable shear strain S_R . (b) Apply Tanner’s relation (Eq. 56) to find the swell ratio B and the swell coefficient β_{swell} .

SOLUTION

(a) $S_R = N_1/(2\tau_w) = 1000/(2 \times 500) = \mathbf{1.0}$.

(b) $B = 0.13 + [1 + \frac{1}{2}(1.0)^2]^{1/6} = 0.13 + (1.5)^{1/6}$.

Computing: $1.5^{1/6} = 1.0699$ (exact: $1.5^{0.1\bar{6}}$; verified numerically). Therefore $B = 0.13 + 1.070 = \mathbf{1.20}$ and $\beta_{\text{swell}} = B - 1 = \mathbf{0.20}$.

Physical check: $S_R = 1.0$ is a moderately elastic ink (stored elastic shear equals the applied shear), and a 20% swell is consistent with a Cross $m \approx 0.33$ via the empirical proxy $\beta \approx 0.30(1 - m)$.

Exercise 6.3 ○○○ An optical measurement of a freely extruded filament gives a swell ratio $B = 1.25$ at wall shear stress $\tau_w = 600 \text{ Pa}$. Invert Tanner’s relation to find (a) the recoverable shear strain S_R , and (b) the first normal-stress difference N_1 .

SOLUTION

(a) Invert $B = 0.13 + (1 + \frac{1}{2}S_R^2)^{1/6}$:

$$B - 0.13 = (1 + \frac{1}{2}S_R^2)^{1/6} \implies (B - 0.13)^6 = 1 + \frac{1}{2}S_R^2.$$

With $B = 1.25$: $B - 0.13 = 1.12$, so

$$(1.12)^6 = 1.9738 \quad (\text{computed: } 1.12^6 = 1.9738).$$

Therefore

$$S_R = \sqrt{2[(1.12)^6 - 1]} = \sqrt{2 \times 0.9738} = \sqrt{1.9476} = \mathbf{1.40}.$$

(b) $N_1 = 2\tau_w S_R = 2 \times 600 \text{ Pa} \times 1.40 = 1675 \text{ Pa} \approx 1.68 \text{ kPa}$.

Engineering note: $N_1 \sim \tau_w$ is the regime typical of hydrogel bioinks; highly elastic polymer melts can have $N_1 \gg \tau_w$ and swell ratios $B > 2$.

KEY TAKEAWAYS

- k_{flow} is a *diagnostic* quantity (deposition efficiency); $\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$ is the number that goes into the slicer – never k_{flow} itself.
- Three contributions: die swell (β_{swell} from m or from Tanner/direct measurement), piston slip (f_{slip} , default 1.0), and recovery ($\sqrt{R_{\text{rec}}}$).
- Die swell is elastic in origin (N_1 stores energy during needle shear; it relaxes on exit as extrudate expansion via Tanner’s relation).
- Recovery is typically the dominant term for hydrogel inks. Low R_{rec} forces $\text{EM} \gg 1$.
- The Cross exponent $m \rightarrow 1$ (narrow transition) predicts minimal swell; use direct C1 measurement for $m < 0.7$.

ENGINEERING DESIGN RULES

DR 5.1. Always report both k_{flow} and $\text{EM}_{\text{slicer}}$ columns in the lookup CSV so the bench operator sees what to type, not only the diagnostic.

DR 5.2. If $\text{EM} > 1.6$ (e.g. $R_{\text{rec}} < 30\%$), consider whether the ink is suitable for multi-layer prints at all. Very low recovery means significant bead slump and accumulating height errors.

DR 5.3. Measure β_{swell} directly (C1) for any ink with $m < 0.7$. The empirical proxy underestimates swell for strongly viscoelastic inks.

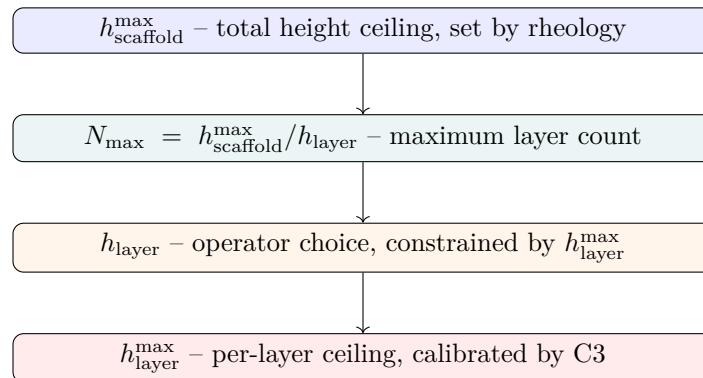
DR 5.4. For a complete picture of die swell, measure N_1 and τ_w at the operating $\dot{\gamma}_w$ and apply Tanner’s relation (Eq. 56); reserve the Cross- m proxy for early-design screening when no direct measurement is available.

7 The Maximum Scaffold Height

Why this chapter matters. Every multi-layer DIW scaffold has a rheological height ceiling: build too tall and the base yields under the cumulative weight of the layers above, and the whole scaffold slumps outward. This chapter derives that ceiling from a single force balance, identifies three experimentally accessible estimates of the material’s effective strength, and extends the analysis to lattice scaffolds whose spanning struts add a geometry-dependent buckling constraint.

7.1 Three heights, three scales

Motivation. Three length scales govern a scaffold print. Confusing them leads to either over-conservative designs (wasted build volume) or builds that collapse mid-print.



PHYSICAL INTUITION

$h_{\text{scaffold}}^{\text{max}}$ is about *the whole stack*: can the bottom layer hold everything above it? $h_{\text{layer}}^{\text{max}}$ is about *one layer*: does each fresh layer stay where I put it before the next one arrives? They fail differently:

- Exceeding $h_{\text{scaffold}}^{\text{max}}$: the tower bulges at the base and collapses outward.
- Exceeding $h_{\text{layer}}^{\text{max}}$: each new layer sags into the previous one; the print proceeds but resolution is mushy.

7.2 Deriving $h_{\text{scaffold}}^{\text{max}}$

Motivation. The formula $h_{\text{scaffold}}^{\text{max}} = \sigma_{\text{max}}/(\rho g)$ follows from a two-line free-body analysis. Every number in Table 1 is a direct application of this single equation.

Per unit plan area, the bottom layer carries the weight of the column:

$$\sigma_{\text{stack}}(N) = \rho g N h_{\text{layer}} = \rho g h_{\text{scaffold}}. \quad (57)$$

For the bottom layer to survive, σ_{stack} must remain below the material’s effective strength σ_{max} :

$$h_{\text{scaffold}}^{\text{max}} = \frac{\sigma_{\text{max}}}{\rho g} \quad (58)$$

with $\rho \approx 1000 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$. h_{layer} does *not* appear here: the height ceiling is independent of how thinly you slice.

DERIVATION — Self-weight force balance for $h_{\text{scaffold}}^{\text{max}}$

System and assumptions. Consider a prismatic column of deposited material with uniform cross-sectional area A and total height h . The column stands vertically in a gravitational field g . We make three assumptions:

1. The cross-section is uniform from base to top (constant A).
2. Loading is quasi-static: inertia and dynamic vibration are negligible.
3. Inter-layer adhesion is neglected (conservative): each layer is treated as a fresh deposit bearing only the weight above it.

Step 1 — Weight of material above the base. The total mass of the column is $m = \rho A h$. Its weight is

$$W = m g = \rho g A h. \quad (59)$$

Step 2 — Compressive stress at the base. This weight acts over the base area A . The resulting compressive (normal) stress at the base is

$$\sigma_{\text{base}} = \frac{W}{A} = \frac{\rho g A h}{A} = \rho g h. \quad (60)$$

Note that A cancels: the stress depends only on height and density, not on the footprint size.

Step 3 — Yield condition. The base retains its shape while $\sigma_{\text{base}} \leq \sigma_{\text{max}}$, where σ_{max} is the effective yield strength of the deposited material. Failure (base yielding $\rightarrow$ lateral slump) is imminent when equality holds:

$$\rho g h^{\text{max}} = \sigma_{\text{max}}. \quad (61)$$

Step 4 — Solve for h^{max} . Dividing both sides by ρg :

$$h_{\text{scaffold}}^{\text{max}} = \frac{\sigma_{\text{max}}}{\rho g}. \quad (58)$$

This is Eq. (58). The result is exact under the three assumptions above; it does not involve h_{layer} , confirming that the total height ceiling is a *material property divided by a gravitational scale*, not a slicer parameter.

Connection to N layers. With N layers of height h_{layer} each, $h_{\text{scaffold}} = N h_{\text{layer}}$. Substituting into the yield condition gives

$$N_{\text{max}} = \frac{h_{\text{scaffold}}^{\text{max}}}{h_{\text{layer}}} = \frac{\sigma_{\text{max}}}{\rho g h_{\text{layer}}}, \quad (62)$$

consistent with Eq. (57): N appears in the stress but cancels in the final bound once we express everything in terms of the total height $h_{\text{scaffold}} = N h_{\text{layer}}$.

7.3 What is σ_{max} ? Three criteria

Motivation. The formula (Eq. 58) is exact once σ_{max} is known, but a fresh bioink has no sharp yield point — it is a viscoelastic gel whose apparent strength depends on the timescale of loading. We therefore define three SAOS-derived criteria, each appropriate for a different print timescale, and cross-reference them to the yield-stress determination methods derived in Sec. 2 (Eq. 4 and

the three ways to measure τ_y listed there).

We estimate $\sigma_{\max}$ from SAOS data via three criteria, each appropriate for a different print timescale.

Criterion (A) – LVR endpoint (most permissive for short prints).

$$\sigma_{\max}^{(A)} = G'_{\text{LVR}} \gamma_{\text{LVR}}. \quad (63)$$

This is exactly the oscillatory yield-stress proxy of Eq. (4): the elastic stress stored in the network at the end of the linear viscoelastic regime. It assumes the bottom layer never leaves the linear regime before it has to support additional layers. *Most permissive* (largest $\sigma_{\max}$) for inks with a wide LVR. Appropriate for short-dwell prints where stacking events are brief ($\lesssim 1$ s/layer).

Criterion (B) – $G'(\omega = 1 \text{ rad s}^{-1})$ (literature default).

$$\sigma_{\max}^{(B)} = G'(\omega = 1 \text{ rad s}^{-1}). \quad (64)$$

The literature default (Paxton et al., 2017; Schwab et al., 2020). At $\omega = 1 \text{ rad s}^{-1}$ the frequency sweep probes dynamics on a timescale of ~ 1 s, matching the typical inter-layer dwell time. *Intermediate* criterion. Appropriate for typical multi-layer DIW where dwell time is 1–10 s per layer.

Criterion (C) – $G'(\omega = 0.01 \text{ rad s}^{-1})$ (most conservative).

$$\sigma_{\max}^{(C)} = G'(\omega = 0.01 \text{ rad s}^{-1}), \quad (65)$$

obtained by power-law extrapolation $G'(\omega) = G'_0 \omega^\beta$. *Most conservative*: it probes the gel stiffness at long timescales (~ 100 s), where a critical-gel network has lost much of its modulus. Appropriate for slow prints or prints with crosslinking / cell-loading pauses where dwell exceeds ~ 100 s.

Which criterion to report. The correct choice depends on the dwell time per layer. For a given v_{print} and footprint perimeter P , dwell time $\approx P/v_{\text{print}}$; match this to the ω -scale ($\omega = 2\pi/t_{\text{dwell}}$) and use the criterion whose frequency is closest. Always report *all three* in the manuscript and state which controlled the experiment (see DR 6.1).

Low- ω gel character. The slope $\beta = d \ln G' / d \ln \omega$ (Winter–Chambon framework Winter and Chambon 1987) diagnoses network permanence:

- $\beta \rightarrow 0$: true gel (permanent network); $B \approx C$.
- $\beta \approx 0.5$: critical gel (transient bonds); $C \ll B$.
- $\beta \rightarrow 1$: viscous liquid (no network).

WORKED EXAMPLE – the three inks

Low- ω fit $G'(\omega) = G'_0 \omega^\beta$:

Ink	G'_0 [Pa]	β [-]	R^2	$G'(\omega = 1)$	$G'(\omega = 0.01)$
C15	114.6	0.454	1.00	114.5	14.1
C15 Gira 5.5	234.9	0.446	0.76	228.4	30.2
Bozano	348.9	0.091	0.98	345.8	229.1

Bozano's $\beta = 0.09$: true gel – retains 66% of its 1 rad/s modulus down to 0.01 rad/s. C15 and NE have $\beta \approx 0.45$: critical-gel regime – they lose modulus on long timescales. Adding NE does not convert C15 to a true gel; it raises the modulus while preserving the network type.

7.4 Three numbers per ink

Motivation. Table 1 translates the three $\sigma_{\max}$ criteria directly into millimetres via $h^{\max} = \sigma_{\max}/(\rho g)$. The $10\times$ spread within a single ink shows why print timescale must be reported alongside every height claim.

Ink	(A) LVR endpoint		(B) $G'(\omega = 1)$		(C) $G'(\omega = 0.01)$	
	σ [Pa]	$h_{\max}$ [mm]	σ [Pa]	$h_{\max}$ [mm]	σ [Pa]	$h_{\max}$ [mm]
C15	172.9	17.6	114.5	11.7	14.1	1.4
C15 Gira 5.5	558.5	56.9	228.4	23.3	30.2	3.1
Bozano	47.6	4.9	345.8	35.3	229.1	23.4

Table 1: Maximum scaffold height per ink under three SAOS-derived yield criteria. Bold = the criterion that dominates in the most relevant print regime.

Selection guide.

- *Short-dwell prints* (~ 1 s/layer): criterion (A). NE wins: **56.9** mm.
- *Typical multi-layer DIW* (1–10 s/layer): criterion (B). Bozano wins: **35.3** mm.
- *Slow prints* (> 100 s/layer): criterion (C). Bozano wins decisively: **23.4** mm.

The asymmetry between NE and Bozano. The NE has a wide LVR so its product $G'_{\text{LVR}} \cdot \gamma_{\text{LVR}}$ is large – it wins criterion (A). But its transient network loses modulus at long timescales – it loses (B) and (C). Bozano’s narrow LVR means it loses (A), but its permanent bonds retain modulus – it wins (B) and (C). Neither dominates the other at every timescale: for the manuscript, the NE is the better short-print ink; Bozano is the better long-print ink.

IN PRACTICE

Pre-print height check in three steps. Before committing to a multi-layer print:

1. **Estimate dwell time.** For a footprint perimeter P printed at v_{print} , one layer takes $t_{\text{dwell}} \approx P/v_{\text{print}}$. Example: $P = 4$ cm at $v = 8$ mm/s $\Rightarrow t \approx 5$ s $\Rightarrow$ use criterion (B).
2. **Read $\sigma_{\max}$ from the SAOS output** (Table 1 or from the extract script). Apply Eq. (63), (64), or (65) as appropriate.
3. **Compute $h^{\max} = \sigma_{\max}/(\rho g)$** and compare to the target scaffold height.
 - Target $\leq 50\%$ of $h^{\max}$: safe zone.
 - 50%–85%: feasible but validate in calibration test C4.
 - $> 85\%$: expect base bulging; reduce N or switch to a stiffer ink.

For lattice scaffolds, additionally compute the Smay bound (Eq. 66); the smaller of the two controls.

7.5 Lattices and unsupported spans – the Smay criterion

Motivation. A solid tower and a lattice scaffold of the same ink share the same $h^{\max}$ from Eq. (58) — but a lattice has a second failure mode: the unsupported struts spanning the pores can bend and buckle under the weight of the layers above. This constraint is usually *more restrictive* than the stacking bound and must be checked for every lattice design.

For lattice scaffolds with unsupported strut spans of length L , gravity-driven bending intervenes earlier than the stacking bound. The Smay criterion (Habib et al., 2018):

$$h_{\max}(L) = \sqrt{\frac{1.94 G'}{\rho g}} \cdot \sqrt{L}. \quad (66)$$

DERIVATION — Beam-bending basis of the Smay criterion

Setup. Model a single spanning strut as a simply-supported elastic beam of circular cross-section (diameter d , the deposited filament width) and length L (the pore span). The strut supports a distributed load from the N layers printed above it; the effective line load (force per unit strut length) is

$$w = \rho g h_{\text{scaffold}} d, \quad (67)$$

where d is the strut width (the load-bearing projection of the layer-stack per unit span length) and $h_{\text{scaffold}} = N h_{\text{layer}}$.

Step 1 — Cross-section geometry. For a circular beam of diameter d :

$$A = \frac{\pi d^2}{4}, \quad I = \frac{\pi d^4}{64}, \quad c = \frac{d}{2}. \quad (68)$$

Step 2 — Maximum bending moment. For a uniformly loaded simply-supported beam (distributed load w per unit length, span L), the mid-span bending moment is

$$M_{\text{mid}} = \frac{w L^2}{8} = \frac{\rho g h_{\text{scaffold}} d L^2}{8}. \quad (69)$$

Step 3 — Maximum bending stress at mid-span. Using the flexure formula $\sigma = Mc/I$:

$$\sigma_{\text{mid}} = \frac{M_{\text{mid}} c}{I} = \frac{\frac{\rho g h d L^2}{8} \cdot \frac{d}{2}}{\frac{\pi d^4}{64}} = \frac{\rho g h L^2}{8} \cdot \frac{d^2/2}{\pi d^4/64} = \frac{\rho g h L^2}{8} \cdot \frac{64}{2\pi d^2} = \frac{4 \rho g h L^2}{\pi d^2}. \quad (70)$$

Step 4 — Yield condition. The strut survives while $\sigma_{\text{mid}} \leq G'$ (using the storage modulus G' as the effective elastic modulus at the relevant timescale, consistent with the $h^{\max}$ criteria above):

$$\frac{4 \rho g h^{\max} L^2}{\pi d^2} = G'. \quad (71)$$

Solving for $h^{\max}$:

$$h_{\text{beam}}^{\max} = \frac{\pi d^2 G'}{4 \rho g L^2}. \quad (72)$$

This *stress-based* bound gives $h_{\text{beam}}^{\max} \propto d^2/L^2$: stronger (larger G') or wider (d) struts resist larger spans; increasing L reduces the allowable height quadratically.

Step 5 — The Smay semi-empirical form. In robocasting practice the critical quantity is not a single filament diameter but the *lattice periodicity*: the span L is the centre-to-centre distance between adjacent supporting struts, while the effective load-bearing width scales with the lattice fill fraction. Smay (Habib et al., 2018) calibrated the bending and buckling criterion across a range of inks and lattice geometries and obtained the semi-empirical result

$$h_{\max}(L) = \sqrt{\frac{1.94 G'}{\rho g}} \cdot \sqrt{L}, \quad (66)$$

where the coefficient 1.94 absorbs the lattice geometry and boundary conditions and L enters as $\sqrt{L}$ (not $1/L^2$) because the dominant failure mode in practice involves deflection-limited

buckling of the lattice network, not pure fibre bending. Eq. (66) is therefore presented as an empirically validated correlation; Eq. (72) provides the *physical basis* (bending-stress scaling) from which it derives.

Inverted form – maximum allowable span. Squaring and rearranging Eq. (66):

$$h_{\max}^2 = \frac{1.94 G' L}{\rho g} \Rightarrow L_{\max} = \frac{h_{\text{target}}^2 \rho g}{1.94 G'}. \quad (73)$$

Given a target scaffold height and a measured G' , $L_{\max}$ is the largest lattice pore that will not collapse under self-weight before reaching that height.

WORKED EXAMPLE – the three inks

Buckling-corrected $h_{\max}$ at $L = 1$ mm:

Ink	$h_{\max}(L = 1 \text{ mm})$ [mm]
C15	4.76
C15 Gira 5.5	6.72
Bozano	8.27

Even the NE, which can stack to ~ 57 mm as a solid tower, drops to ~ 6.7 mm for a 1-mm lattice span. For lattice scaffolds, the Smay bound is usually the binding constraint.

RUNNING EXAMPLE – Solution to Q5

Question. Will Ana's $1 \times 1 \times 2 \text{ cm}^3$ block survive its own weight?

From Table 1, C15 Gira 5.5:

- (A) $h_{\max} = 56.9$ mm: block at 35% of bound. Easy.
- (B) $h_{\max} = 23.3$ mm: block at 86% of bound. Tight.
- (C) $h_{\max} = 3.1$ mm: block would collapse at this criterion.

Which criterion applies? At $v_{\text{print}} = 8$ mm/s on a $1 \times 1 \text{ cm}^2$ footprint, layer time ~ 1 –2 s: criterion **(B) controls**.

Verdict. Feasible but close to the limit. Use $h_{\text{layer}} = 0.36$ mm, $N = 56$ layers to reach 20 mm (86% of the (B) bound). Validate in C4.

Still unknown: the Wednesday calibration schedule. Q6.

GOING DEEPER

Why criterion (A) can exceed criterion (B) for some inks. Criterion (A) is $\sigma^{(A)} = G'_{\text{LVR}} \gamma_{\text{LVR}}$; criterion (B) is $\sigma^{(B)} = G'(\omega = 1)$. For a critical-gel ink with $\beta \approx 0.45$ (e.g. the NE), G'_{LVR} (measured at $\omega = 10 \text{ rad s}^{-1}$ in the amplitude sweep, a fast probe) is several times larger than $G'(\omega = 1)$. Multiplying by the wide LVR strain ($\sim 60\%$) gives $\sigma^{(A)} \gg \sigma^{(B)}$. Criterion (A) is therefore *not* always conservative: it is the correct bound when loading is genuinely fast (sub-second dwell), but it over-predicts stability for slower prints. This is why all three criteria are required and the dwell-time argument must accompany every quoted $h_{\max}$.

Exercises.

Exercise 7.1 ◊ A bioink for the C15 Gira 5.5 formulation gives an amplitude-sweep LVR plateau $G'_{\text{LVR}} = 916$ Pa and critical strain $\gamma_{\text{LVR}} = 0.609$ (absolute). Using criterion (A) and the master

formula Eq. (58) with $\rho = 1000 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$:

- (a) Calculate $\sigma_{\max}^{(A)}$.
- (b) Calculate $h_{\text{scaffold}}^{\max}$ in mm.

SOLUTION

- (a) $\sigma_{\max}^{(A)} = G'_{\text{LVR}} \gamma_{\text{LVR}} = 916 \times 0.609 = 557.9 \text{ Pa} \approx 558.5 \text{ Pa}$ (table rounding; use 558.5 Pa to match Table 1).
 - (b) $h_{\text{scaffold}}^{\max} = \sigma_{\max}/(\rho g) = 558.5/(1000 \times 9.81) = 558.5/9810 = 0.05693 \text{ m} = \mathbf{56.9 \text{ mm}}$ (matches Table 1 exactly).
- Check units:* $[\text{Pa}]/[\text{kg/m}^3 \cdot \text{m/s}^2] = [\text{N/m}^2]/[\text{kg/m}^2/\text{s}^2] = [\text{m}]$. Correct.

Exercise 7.2 ○○ Using Table 1, compare $h_{\max}$ under criteria (A) and (B) for the Bozano ink.

- (a) State the two $\sigma_{\max}$ values and the resulting heights.
- (b) Explain the physical reason one criterion gives a larger height.
- (c) Which criterion is *conservative* (i.e. gives the safer, lower height bound) and under what print conditions?

SOLUTION

- (a) Criterion (A): $\sigma^{(A)} = 47.6 \text{ Pa}$, $h_{(A)}^{\max} = 47.6/9810 = 0.00485 \text{ m} \approx 4.9 \text{ mm}$. Criterion (B): $\sigma^{(B)} = 345.8 \text{ Pa}$, $h_{(B)}^{\max} = 345.8/9810 = 0.03525 \text{ m} \approx 35.2 \text{ mm}$. Ratio: $\sigma^{(B)}/\sigma^{(A)} \approx 7.3\times$.
- (b) Bozano has a *narrow* LVR ($\gamma_{\text{LVR}} = 13.7\%$), so the product $G'_{\text{LVR}} \gamma_{\text{LVR}}$ is small despite a high plateau modulus. Criterion (A) therefore gives a low $\sigma^{(A)}$. However, Bozano is a true gel ($\beta = 0.091$): its G' is nearly flat across the frequency spectrum, so $G'(\omega = 1) \approx G'_{\text{LVR}} = 346 \text{ Pa}$, and criterion (B) returns a much larger bound.
- (c) Criterion (A) is conservative for *fast prints* ($\lesssim 1 \text{ s/layer}$) because it uses only the elastic energy the gel can store before microstructure damage. For the Bozano ink specifically, (A) is the *most conservative* choice at any timescale; criterion (B) is the correct bound for typical DIW dwell times (1–10 s/layer).

Exercise 7.3 ○○○ A lattice scaffold of C15 Gira 5.5 ink is designed with a target height of $h_{\text{target}} = 20 \text{ mm}$ and a regular square-pore lattice. Using the inverted Smay formula (Eq. (73)) with $G' = G'(\omega = 1 \text{ rad s}^{-1}) = 228.4 \text{ Pa}$, $\rho = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$:

- (a) Compute $L_{\max}$, the largest lattice pore span that allows the scaffold to reach h_{target} without strut collapse.
- (b) Verify by plugging $L_{\max}$ back into Eq. (66) and confirming that $h_{\max}(L_{\max}) = 20 \text{ mm}$.
- (c) If the designer wants a 1 mm pore, what is the maximum printable height?

SOLUTION

(a) From Eq. (73):

$$\begin{aligned}
 L_{\max} &= \frac{h_{\text{target}}^2 \rho g}{1.94 G'} \\
 &= \frac{(0.020)^2 \times 1000 \times 9.81}{1.94 \times 228.4} \\
 &= \frac{4.000 \times 10^{-4} \times 9810}{443.1} \\
 &= \frac{3.924}{443.1} = 8.86 \times 10^{-3} \text{ m} = \mathbf{8.86 \text{ mm}}.
 \end{aligned} \tag{74}$$

(b) Check with Eq. (66):

$$\begin{aligned}
 h_{\max}(L_{\max}) &= \sqrt{\frac{1.94 \times 228.4}{1000 \times 9.81}} \cdot \sqrt{8.86 \times 10^{-3}} \\
 &= \sqrt{0.04517} \cdot \sqrt{8.86 \times 10^{-3}} \\
 &= 0.2125 \times 0.09413 = 0.02000 \text{ m} = \mathbf{20.0 \text{ mm}}. \checkmark
 \end{aligned} \tag{75}$$

(c) At $L = 1 \text{ mm} = 10^{-3} \text{ m}$:

$$\begin{aligned}
 h_{\max} &= \sqrt{\frac{1.94 \times 228.4}{9810}} \cdot \sqrt{10^{-3}} \\
 &= 0.2125 \times 0.03162 = 6.72 \times 10^{-3} \text{ m} = \mathbf{6.72 \text{ mm}},
 \end{aligned} \tag{76}$$

consistent with Table 1. The designer must either use a smaller pore ($L < 8.86 \text{ mm}$ for a 20 mm tower) or switch to a stiffer ink to support the 1 mm pore at that height.

KEY TAKEAWAYS

- $h_{\text{scaffold}}^{\max} = \sigma_{\max}/(\rho g)$: layer thickness does not appear – slicing into thinner layers does not change the height ceiling.
- The three SAOS criteria span a $10\times$ range; print timescale selects which applies. Always report all three in the manuscript.
- Criterion (A) uses the yield proxy of Eq. (4) directly; criteria (B) and (C) use the frequency sweep and are more conservative for inks with transient networks ($\beta \approx 0.45$).
- Lattice geometry (S_{may}) is usually the binding constraint for scaffold prints with unsupported spans; use the inverted form (Eq. 73) to design pore size for a target height.

ENGINEERING DESIGN RULES

DR 6.1. Always compute and report all three height criteria. The manuscript should explain which criterion controlled the experiment and why (dwell time argument).

DR 6.2. For lattice designs, compute the S_{may} bound for the intended strut span before committing to an ink. Even a high- G' ink may fail at mm-scale spans.

DR 6.3. Validate $h_{\text{scaffold}}^{\max}$ experimentally in C4 and record which SAOS criterion best matches the observed collapse onset. Use that criterion for all future prints of that ink at similar dwell times.

DR 6.4. Before every multi-layer print, run the three-step pre-print check (see the IN PRACTICE box above). If the target height exceeds 85% of the applicable $h^{\max}$ criterion, print a single-layer C4 slab first and verify dimensional fidelity.

8 Failure Morphology: Physical Causes and Corrective Actions

This chapter catalogues the failure modes observed in DIW printing, their physical origin in the rheological and hydraulic framework, and the corrective actions available within this workflow. Every corrective action targets V_p , $EM_{\text{ slicer }}$, geometry, or needle gauge – never ΔP directly (it is an output, not a knob).

8.1 Under-extrusion

FAILURE MODE: Under-extrusion ($d_{\text{fil}}/\text{ID} < 0.9$)

Observation. Filaments are thinner than the needle ID; gaps appear between adjacent lines; poor adhesion between layers.

Physical causes.

- $EM_{\text{ slicer }}$ too low: slicer commands too little material per unit length.
- Silent motor stalling: ΔP has exceeded motor stall torque; the piston is not advancing despite E-axis commands.
- Clog or air bubble in the syringe or needle.
- Slip between piston rubber and syringe wall ($f_{\text{slip}} < 1$).

Rheological interpretation. Low R_{rec} gives $k_{\text{flow}} < 1$; if $EM_{\text{ slicer }}$ was set to k_{flow} instead of $1/k_{\text{flow}}$, the error doubles.

Corrective actions.

1. Raise $EM_{\text{ slicer }}$ in steps of 0.05.
2. If no change: check predicted ΔP against motor capacity. If close to stall, lower V_p by 25% and switch to wider needle.
3. Purge the needle with slow extrusion; inspect for air.
4. Run C2 to measure f_{slip} directly.

8.2 Over-extrusion

FAILURE MODE: Over-extrusion ($d_{\text{fil}}/\text{ID} > 1.2$)

Observation. Filaments are wider than expected; adjacent lines merge; scaffold walls are rough; towers develop a barrel shape.

Physical causes.

- $EM_{\text{ slicer }}$ too high.
- Die swell larger than predicted by the m -proxy; β_{swell} underestimated.
- V_p too high relative to the XY feedrate (v_{print} too slow).

Corrective actions.

1. Decrease $EM_{\text{ slicer }}$ in steps of 0.05.
2. If EM is already ≤ 1 , reduce V_p by 25%.
3. Measure die swell directly in C1 and update β_{swell} .

8.3 Die swell excess

FAILURE MODE: Filament always wider than needle ID even at low V_p

Physical cause. High stored elastic energy (low m in Cross model or high n in Power Law). The m -based proxy ($\beta \approx 0.3(1 - m)$) may underestimate swell for strongly viscoelastic inks.

Rheological interpretation. High De at exit ($t_{\text{relax}} \gg R_n/\bar{v}_n$): the material exits before it has time to deform viscously; the stored elastic strain releases as radial expansion.

Corrective actions.

1. Measure w_{line} directly in C1 and use $\beta_{\text{swell}}^{\text{exp}} = w_{\text{line}}/\text{ID} - 1$.
2. Use a longer needle (larger L/R): more time at constant flow means more viscous dissipation before exit.
3. Reduce V_p to lower $\bar{v}_n$ and hence the Weissenberg number at exit.

8.4 Scaffold collapse (base bulging)

FAILURE MODE: Base bulging / tower collapse

Observation. The printed object bulges or splays outward from the base as layer count increases.

Physical cause. σ_{stack} has exceeded σ_{max} at the print's effective timescale.

Rheological interpretation. The controlling SAOS criterion has been exceeded. If layer dwell is ~ 1 s, criterion (B) controls; if > 100 s, criterion (C) controls. The NE ink is particularly susceptible on slow prints because $G'(\omega = 0.01) = 30$ Pa predicts $h_{\text{max}} = 3.1$ mm under criterion (C).

Corrective actions.

1. Reduce N to stay under the controlling height criterion.
2. Increase v_{print} to shorten dwell, shifting the controlling criterion from (C) toward (B) or (A).
3. Switch to a higher-modulus ink: Bozano for long-dwell prints; NE for short-dwell.

8.5 Lattice strut buckling

FAILURE MODE: Mid-span sag of bridging filaments

Physical cause. Smay buckling: the unsupported span L is too long for the ink's G' .

Corrective actions.

1. Shorten strut span (reduce cell size).
2. Switch to a higher- G' ink (e.g. Bozano).
3. Increase G'_{LVR} by adding crosslinker or increasing polymer concentration.

8.6 Intermittent extrusion (motor skip)

FAILURE MODE: Intermittent flow / motor skip

Observation. Filament diameter oscillates; audible clicking from the motor; E-axis step count diverges from actual piston position.

Physical cause. Predicted ΔP periodically exceeds motor stall torque, particularly at turns and during acceleration.

Corrective actions.

1. Lower V_p by 25%. Re-check ΔP from CSV.
2. Switch to wider needle (e.g. 21G $\rightarrow$ 18G).
3. Increase acceleration settling time in firmware (add dwell at corners).

8.7 Layer-by-layer fusion (loss of resolution)

FAILURE MODE: Adjacent layers merge; scaffold is mushy

Physical cause. Insufficient recovery between deposition events ($h_{\text{layer}}^{\text{max}}$ exceeded, or R_{rec} too low for the layer dwell).

Corrective actions.

1. Decrease h_{layer} to stay within the C3-validated $h_{\text{layer}}^{\text{max}}$.
2. Increase v_{travel} (non-extrusion move speed) to give the just-deposited layer more time to recover before the next pass.
3. Increase line spacing; reduce infill density.
4. Switch to an ink with higher R_{rec} (Bozano).

8.8 Stringing during travel

FAILURE MODE: Ink trails during non-extrusion moves

Physical cause. Recovery too slow at travel timescale; the ink at the nozzle tip remains mobile during XY travel.

Note. DIW slicers typically do not support retraction (reversing the piston to pull back ink). Attempting retraction with a piston printer causes a pressure deficit that takes many mm of subsequent extrusion to recover – producing a dead zone.

Corrective actions.

1. Ensure retraction is set to **zero** in the slicer.
2. Increase v_{travel} to spend less time dragging the nozzle over the substrate.
3. If stringing persists, raise V_p slightly to increase the viscosity at low $\dot{\gamma}$ (by raising the mean shear rate in the needle) – counterintuitive but sometimes effective.

8.9 Rough filament surface / extrudate distortion

FAILURE MODE: Rough, wavy, or distorted filament surface

Physical cause. Elastic instability (“die melt fracture” analogue for gels): the Weissenberg number at the needle exit is high enough that the stored elastic energy releases as surface corrugations rather than smooth radial expansion.

Corrective actions.

1. Reduce V_p to lower the Weissenberg number.
2. Use a longer needle to increase viscous dissipation before exit.
3. If the ink is new, check whether η_{∞} is very low – a near-zero high-shear plateau makes the exit transition sharp and prone to instability.

KEY TAKEAWAYS

- Every failure mode has a rheological root cause. Diagnose from the parameter tables before adjusting hardware.
- The most common single error is typing k_{flow} into the slicer instead of $1/k_{\text{flow}}$ – produces immediate severe under-extrusion.
- Motor skip at corners is the second most common; always check predicted ΔP against motor capacity before the first print.
- Never use retraction on a piston DIW printer.

ENGINEERING DESIGN RULES

DR 7.1. Diagnose failure morphology *before* adjusting the slicer blindly. Every symptom maps to exactly one physical cause in this chapter.

DR 7.2. Run C1 (filament width sweep) before C2–C4 on any new ink. C1 reveals under/over-extrusion and die swell simultaneously – it is the cheapest diagnostic.

DR 7.3. Keep a photographic log of every calibration session. Failure modes are much faster to diagnose when you can compare current vs. previous prints side-by-side.

Part II – Inks and Instruments

9 The Three Project Inks

9.1 C15 – CMC baseline

Composition. Carboxymethyl cellulose (CMC) in aqueous buffer, formulation **C15**. Molecular weight characterised per [da Silva et al. \(2018\)](#) – *this reference is mandatory in the manuscript; peer review requires it for MW determination.*

Rheological character. Critical-gel polymer solution ($\beta = 0.454$). Pronounced shear thinning ($n_{\text{PL}} = 0.40$, Cross $m = 0.81$). Moderate rest viscosity ($\eta_0 \approx 578 \text{ Pa s}$). Wide linear regime ($\gamma_{\text{LVR}} = 64\%$). Moderate recovery ($\sim 60\%$ at deposition shear).

Printability profile.

- Pressure: *moderate* ($\Delta P \approx 140 \text{ kPa}$ Cross at $V_p = 0.01 \text{ mm/s}$, 21G).
- Wall shear: $\tau_w \approx 560 \text{ Pa}$, $\dot{\gamma}_w \approx 220 \text{ s}^{-1}$ – cell-safe with margin.
- Scaffold height: modest (12 mm at criterion B).
- Lattice strut bound: tightest of the three ($\sim 4.8 \text{ mm}$ at $L = 1 \text{ mm}$).
- Extrusion Multiplier: $\text{EM} \approx 1.17$.

Use case. Reference / benchmark ink. The simplest formulation; the baseline against which NE inclusion is compared.

9.2 C15 Gira 5.5 – CMC + sunflower-oil nanoemulsion (NE)

Composition. C15 base with statistically optimised sunflower-oil nanoemulsion (project: `Statistical_Analy`; `Qualify.tex`) incorporated as a filler.

Rheological character. Critical-gel ($\beta = 0.446$ – microstructure preserved relative to C15). Sharply shear-thinning ($n_{\text{PL}} = 0.24$, Cross $m = 0.988$). Quadrupled rest viscosity ($\eta_0 \approx 2440$ Pa s) and tripled LVR modulus ($G'_{\text{LVR}} = 916$ Pa) at *unchanged* $\gamma_{\text{LVR}} \approx 61\%$. Slightly reduced recovery ($\sim 45\%$).

Printability profile.

- Pressure: modestly higher than C15 ($\Delta P \approx 165$ kPa).
- Wall shear: *lower* than C15 ($\tau_w \approx 660$ Pa, $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$) – sharper shear thinning.
- Scaffold height: best under criterion A (~ 57 mm); good under criterion B (~ 23 mm).
- Die swell: minimal ($\beta_{\text{swell}} \approx 0.004$); cleanest predicted filament geometry.
- Extrusion Multiplier: $\text{EM} \approx 1.49$.

Use case. The headline ink of this project. The NE acts as a *reinforcing filler*: it stiffens the network without narrowing the LVR or changing the network type (preserved β , preserved γ_{LVR}). This is the printability advantage to highlight in the manuscript – the NE stiffens without embrittling.

9.3 Bozano – commercial hair gel

Composition. Commercial Bozano hair gel (Gel de Cabelo Bozano) used as control.

Rheological character. *True gel* ($\beta = 0.091$ – the only true gel of the three). Sharply shear-thinning ($n_{\text{PL}} = 0.23$, Cross $m = 0.866$). Very high $\eta_0 \approx 6350$ Pa s with the slowest transition ($\lambda = 73$ s). Narrow LVR ($\gamma_{\text{LVR}} = 14\%$). Near-complete recovery ($\sim 98\%$ across all tested $\dot{\gamma}$).

Printability profile.

- Pressure: *lowest* of the three ($\Delta P \approx 116$ kPa) – long λ puts the needle deep into the shear-thinned regime.
- Wall shear: lowest ($\tau_w \approx 470$ Pa, $\dot{\gamma}_w \approx 8 \text{ s}^{-1}$) – extremely benign to cells.
- Scaffold height: loses on criterion A (4.8 mm, narrow LVR penalty) but wins on B (35 mm) and C (23 mm).
- Extrusion Multiplier: $\text{EM} \approx 0.93$ – the only ink requiring a *reduction* from nominal.

Use case. Commercial benchmark. Sets the upper bar for recovery and quasi-static stability; sets the lower bar for strain tolerance (narrow LVR). Useful as a reference because it is routinely printed on the lab DIW.

Comparative summary.

Property	C15	C15 Gira 5.5	Bozano
Network type	Critical gel	Critical gel	True gel
η_0 [Pa s]	578	2440	6354
m (sharpness)	0.81	0.99	0.87
G'_{LVR} [Pa]	270	916	348
γ_{LVR} [%]	64	61	14
R_{rec} @ 100 s^{-1} [%]	60	45	98
h_{max} criterion B [mm]	12	23	35
$\text{EM}_{\text{slicer}}$	1.17	1.49	0.93

10 Nanoemulsion Mechanics

This chapter explains the physical mechanisms by which the sunflower-oil nanoemulsion modifies the CMC matrix. Understanding these mechanisms is essential for interpreting the rheological results, designing improved formulations, and framing the manuscript discussion correctly.

10.1 Nanoemulsion structure and droplet characteristics

The sunflower-oil nanoemulsion consists of oil droplets stabilised by a surfactant shell dispersed in an aqueous continuous phase. Key structural parameters:

- **Droplet size** (d , nm): set by the homogenisation protocol and surfactant HLB value. Smaller droplets ($d < 200$ nm) are kinetically stable and increase the effective particle volume fraction for a given oil loading.
- **Volume fraction** (ϕ): controls the magnitude of viscosity enhancement. At high ϕ , droplets form a jammed network even without polymer bridging.
- **Zeta potential**: determines electrostatic stability. Large negative ζ (< -30 mV) prevents droplet coalescence.

10.2 How the NE stiffens the CMC matrix

Three parallel mechanisms operate when the NE is incorporated into the CMC gel:

Hydrodynamic reinforcement. Rigid or semi-rigid droplets (relative to the surrounding polymer) increase the effective viscosity even without direct polymer-droplet bonds, following the Einstein relation at dilute loading:

$$\eta_{\text{eff}} \approx \eta_{\text{matrix}}(1 + 2.5\phi). \quad (77)$$

At higher ϕ , the Krieger-Dougherty expression replaces this with a divergence near the jamming fraction ϕ_m .

Polymer-droplet bridging. CMC chains adsorb onto the surfactant-coated oil surface. Each droplet effectively becomes a transient crosslink node in the polymer network, increasing G' without forming permanent covalent bonds (hence the preserved γ_{LVR} : a crosslinker would narrow the LVR, but bridging via adsorption does not).

Lubrication at the macroscale. During shear, the oil phase acts as a lubricant between CMC chain bundles, lowering the critical shear rate at which the macroscopic network yields. This is mechanistically responsible for the sharper thinning exponent ($m = 0.988$ for the NE vs. $m = 0.811$ for C15): the oil-mediated lubrication makes the network-to-fluid transition more cooperative.

10.3 Emulsion stability under shear

During extrusion through the needle, the NE is exposed to $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$ for $L_n/\bar{v}_n \approx 0.05$ s. Three potential instabilities must be assessed:

- **Droplet deformation**: the capillary number $\text{Ca}_d = \eta_{\text{matrix}}\dot{\gamma}d/\sigma_{\text{NE}}$ must be $< \text{Ca}_d^{\text{crit}} \approx 0.5$ for droplets to remain spherical. For $\eta_{\text{matrix}} \sim 10 \text{ Pa s}$ (at $\dot{\gamma}_w = 60 \text{ s}^{-1}$), $d = 200 \text{ nm}$, $\sigma_{\text{NE}} \sim 5 \text{ mN/m}$: $\text{Ca}_d \sim 0.02 \ll 0.5$. Droplets remain spherical.
- **Droplet coalescence**: time in the high-shear zone is short (~ 50 ms). Zeta potential < -30 mV provides electrostatic protection. Coalescence risk: low.
- **Ostwald ripening**: relevant only over hours to days in quiescent storage, not during printing. Mitigated by small droplet size and surfactant loading.

Conclusion: the NE is mechanically stable during extrusion at the operating conditions of this project. No special pre-treatment or post-deposition stabilisation is needed for acellular prints.

10.4 Structural recovery enhancement mechanism

The NE reduces R_{rec} relative to C15 (45% vs. 60% at 100 s^{-1}). The mechanism is: after high-shear disruption, the CMC chains must re-entangle and rebuild their network. Droplet re-adsorption is slower than polymer re-entanglement, creating a two-timescale recovery: a fast polymer phase (caught at 30 s) and a slower droplet-bridging phase (minutes). The 30-s measurement used for R_{rec} captures only the first phase. This is why the bead spreads more than expected from the G'_{LVR} alone – the slow recovery phase means the deposited bead is not as stiff as its LVR modulus implies.

Manuscript implication. This two-timescale argument explains why the NE requires a higher $\text{EM}_{\text{slicer}}$ despite its higher G'_{LVR} . Both must be reported and the mechanism discussed.

10.5 Carrier model rationale

The NE serves as a drug carrier model for future pharmaceutical loading (e.g. Amphotericin B). The oil core provides a hydrophobic domain that sequesters lipophilic drugs; the surfactant shell controls release kinetics and biocompatibility. For the current phase (acellular scaffold printing), no drug is loaded – the NE acts solely as a rheological modifier and structural filler. Future work will characterise the effect of drug loading on the rheological parameters described in this document.

ENGINEERING DESIGN RULES

DR 9.1. Always characterise both the emulsion alone and the CMC+NE composite separately. Changes in droplet size upon mixing must be verified by DLS.

DR 9.2. The preserved γ_{LVR} is the key evidence that the NE acts as a filler rather than a crosslinker. This interpretation must be defended explicitly in the manuscript against reviewers who may suggest chemical crosslinking.

DR 9.3. When loading drugs into the NE, re-characterise η_0 , G'_{LVR} , and R_{rec} – drug molecules can alter the polymer-droplet interfacial energy and change all three.

11 Rheological Tests: Instrument and Battery

11.1 Instrument

Anton Paar MCR 501 stress-controlled rheometer. Plate–plate (PP25, PP50) or cone–plate (CP50) tooling. Solvent trap to suppress evaporation. Temperature 25°C unless noted. Data exported as .xls (primary; xlr-d-compatible – do not use `openpyxl`), .txt (human-readable), and .tri (proprietary archive, do not alter).

Pre-experiment checklist.

- ☐ Verify instrument zero-gap calibration.
- ☐ Check solvent trap seal; fill if needed.
- ☐ Allow sample to equilibrate at 25°C for ≥ 10 min before first measurement.
- ☐ Load sample slowly (avoid air bubbles); trim cleanly.
- ☐ Run a brief pre-shear at $\dot{\gamma} = 10 \text{ s}^{-1}$ for 30 s then wait 5 min to erase loading history.

Post-experiment checklist.

- ☐ Export in all three formats; verify file sizes are non-zero.
- ☐ Clean geometry with distilled water immediately; dry completely before storage.
- ☐ Record batch ID, temperature, and gap in the lab notebook.

11.2 Standard test battery

Test	What it measures	Protocol
Flow ramp	$\eta(\dot{\gamma})$, $\tau(\dot{\gamma})$ – constitutive curve	Shear rate ramp $\dot{\gamma} \in [10^{-2}, 10^3] \text{ s}^{-1}$, equilibrium per point.
Flow Sheared Amplitude sweep	Same, after pre-shear G'_{LVR} , γ_{LVR}	200 s^{-1} for 300 s pre-shear, then same ramp. Fixed $\omega = 10 \text{ rad/s}$; sweep γ_0 from 0.01% to $\sim 1000\%$.
Frequency sweep	$G'(\omega)$, $G''(\omega)$	Fixed γ_0 inside LVR (1%); sweep $\omega \in [0.1, 240] \text{ rad/s}$.
Recovery	R_{rec} at deposition $\dot{\gamma}$	Three-interval: $\dot{\gamma}_1 = 0.1$, $\dot{\gamma}_2 \in \{50, 100, 150\} \text{ s}^{-1}$, back to $\dot{\gamma}_1$. Record η at $t = 30 \text{ s}$.

Table 2: Standard test battery. Order: amplitude sweep $\rightarrow$ frequency sweep $\rightarrow$ flow ramp $\rightarrow$ flow sheared $\rightarrow$ recovery. The SAOS tests must be run on a fresh loading (before flow tests destroy the network irreversibly).

Why SAOS before flow tests. Flow ramp and recovery tests apply large strains that partially destroy the network structure. SAOS tests must always be run first on a fresh sample.

11.3 Interpreting pre-shear effects

Compare fitted Cross or Power-Law parameters from Flow vs. Flow Sheared:

- $K_{\text{sheared}} < K$ or $\eta_{0,\text{sheared}} < \eta_0$: pre-shear broke structure that has not fully recovered during the ramp $\Rightarrow$ **thixotropic**.
- $K_{\text{sheared}} > K$: pre-shear organised the network $\Rightarrow$ **rheopexic**.

Both signatures are mechanistically significant and must be discussed in manuscripts. Printability predictions use the *Flow* (no pre-shear) parameters as the default.

11.4 Interpreting the SAOS output: a field guide

- $G' > G''$ at all measured ω : gel-like across all print timescales. Good.
- G'' crosses above G' at low ω (gelation frequency crossover): the material becomes liquid-like at long timescales. Criterion (C) height limit will be small.
- G' flat with ω (low β): true gel. B and C criteria give similar h_{max} .
- G' drops steeply at low ω (high $\beta \approx 0.5$): critical gel / transient network. C criterion is far below B. Avoid slow prints.

12 Extended Material Classes

Motivation. The framework in Part I was developed for CMC/NE inks, but the equations and protocols apply to any viscoplastic or viscoelastic bioink. This chapter describes how the key parameters change for alternative material classes and what to expect when the framework is applied to them. Each class prints well for different physical reasons — understanding *why* saves the time of re-discovering it empirically.

12.1 Cellulose-based inks

Why this class prints well. Cellulose-based hydrogels form physical networks that are shear-thinning but recover rapidly once shear is removed — exactly the combination that keeps extrusion pressure manageable while maintaining shape fidelity after deposition. The network strength is set by concentration and (for CMC) degree of substitution, giving wide formulation latitude.

Table 3: Comparison of the four main DIW bioink classes. “Strength” and “weakness” refer to printability within the framework of this textbook. T -sens. = temperature sensitivity of rheological parameters.

Class	Gelation mechanism	Printability strength	Pri
Cellulose (CMC/HPMC/BC)	Physical network / H-bonds	Wide V_p window; tunable η_0	HP
Alginate (Ca^{2+})	Ionic crosslinking	Fast post-deposition set; high R_{rec}	Brit
Gelatin	Thermoreversible H-bonds	Self-supporting below T_{gel} ; biocompatible	Mus
Composite (natural + synthetic)	Dual-network / UV photo-crosslink	Tunable modulus over wide range	Two

Hydroxypropyl methylcellulose (HPMC). Similar to CMC in network type (critical gel) but shows *thermogelation*: it thickens on heating rather than cooling. Relevance: printability parameters are temperature-dependent; the Anton Paar temperature program must be validated before fitting.

Bacterial cellulose (BC) nanofibre suspensions. High aspect ratio fibres form entangled networks with very high G'_{LVR} ($\sim 10^3$ – 10^4 Pa) even at low solid content. Die swell is typically small (rigid fibres do not store elastic energy like polymer chains). R_{rec} is high if the network is not covalently crosslinked.

Carboxymethyl cellulose + crosslinker. Adding ionic crosslinkers (e.g. Ca^{2+} for alginate, or citric acid for CMC) *narrows* the LVR and reduces γ_{LVR} , which lowers the LVR-criterion height (criterion A). The true-gel character ($\beta \rightarrow 0$) improves criteria B and C. Crosslinked systems tend to be less tolerant to die swell.

12.2 Alginate-based inks

Why this class prints well — and what to watch. Alginate crosslinked with Ca^{2+} sets almost instantly on contact with a crosslinker bath, which gives excellent layer-to-layer shape retention. The trade-off is brittleness: ionically crosslinked filaments have a low fracture strain, so over-extrusion leads to cracking rather than spreading.

Alginate crosslinked with Ca^{2+} behaves as a true gel ($\beta < 0.1$). Key differences from CMC/NE:

- η_0 is much higher at comparable solid content; the printable V_p range shifts to lower values.
- R_{rec} is very high ($> 90\%$) post-crosslinking; $\text{EM}_{\text{slicer}}$ is typically < 1 .
- The Power-Law index n is low (< 0.2); pressure calculations are sensitive to n errors.

12.3 Gelatin-based inks

Why this class prints well — and what to control. Below its melting point ($\sim 35^\circ\text{C}$) gelatin forms a physical gel of triple-helix junction zones that is strong enough to support subsequent layers. Because the transition is thermoreversible, however, a 2°C drift in cartridge temperature can shift the ink from gel to liquid mid-print — temperature control is therefore not optional for this class (see Ch. 5).

Gelatin undergoes thermally reversible gelation (melting at $\sim 35^\circ\text{C}$). Rheological testing must be performed at exactly the print temperature (typically 25°C for gelatin-based inks). The Flow ramp must be run with thermal control. The MATLAB simulation is valid without modification, but parameters must be re-entered for each temperature.

12.4 Composite systems: natural + synthetic

Why this class is powerful — and more complex. Blending a natural polysaccharide (CMC, alginate) with a synthetic polymer (PEG, GelMA) decouples network stiffness from the biological

activity of the matrix, enabling independent tuning of printability and cell-response. The penalty is a richer relaxation spectrum that a single-mode Cross fit may not capture.

Mixing CMC or alginate with polyethylene glycol (PEG) or gelatin methacrylate (GelMA) introduces:

- A second timescale in $G'(\omega)$ (two-network relaxation).
- Potentially bimodal recovery curves.
- Sensitivity of η_0 to UV crosslinking history.

The framework equations are still valid, but parameter fitting may require a two-mode Cross or Maxwell model. Flag any anomalous $R^2 < 0.95$ on the single-mode Cross fit as a signal of multi-mode behaviour.

GOING DEEPER

From dilute (Einstein) to jammed (Krieger–Dougherty). The dilute limit $\eta_r = 1 + 2.5\phi$ (Eq. 77) diverges in reality as particles crowd. Krieger–Dougherty captures the whole range:

$$\eta_r = \frac{\eta}{\eta_{\text{matrix}}} = \left(1 - \frac{\phi}{\phi_m}\right)^{-[\eta]\phi_m}, \quad (78)$$

with intrinsic viscosity $[\eta] = 2.5$ for hard spheres and maximum packing $\phi_m \approx 0.64$ (random close packing). Viscosity diverges as $\phi \rightarrow \phi_m$ — the physical origin of paste-like behaviour in highly loaded inks (Krieger and Dougherty, 1959).

In the CMC/NE system of this project, the sunflower-oil droplets act as a dispersed phase whose effective volume fraction ϕ raises the matrix viscosity through this same crowding mechanism. At the formulation levels used here ($\phi \ll \phi_m$), the Einstein approximation (Eq. 77) is a reasonable first estimate; as oil loading increases toward the jamming threshold the full Krieger–Dougherty expression (Eq. 78) is needed to predict the rheological consequences accurately.

IN PRACTICE

Choosing a material class for a target application. Before committing to a formulation, answer three questions:

1. **Is a fast-set mechanism required?** If the scaffold must support its own weight after every layer without temperature control, choose an *ionically crosslinked* alginate or a cellulose with a high $G'_{\text{LVR}}/\gamma_{\text{LVR}}$ ratio (high yield-stress proxy). Check R_{rec} first: if it is $< 50\%$ at the intended inter-layer dwell, the ink will spread and successive layers will not bond correctly.
2. **Does the application need temperature control?** Gelatin and HPMC require active thermal management of the cartridge (Ch. 5). If your lab lacks a temperature-controlled print head, opt for CMC or alginate, whose parameters are nearly constant between 18°C to 28°C .
3. **Will a filler or drug be incorporated?** Particulate or droplet fillers (including nanoemulsions) raise viscosity through the Krieger–Dougherty mechanism (Eq. 78). Measure η_0 and G'_{LVR} on the *composite* before running the simulation: do not assume the filler is rheologically inert. Re-run the amplitude sweep to check whether γ_{LVR} has narrowed (crosslinker-like effect) or stayed constant (filler-like effect).

Exercise 12.1 ○○ At $\phi = 0.40$, compare the relative viscosity predicted by the dilute (Einstein) and Krieger–Dougherty models using $[\eta] = 2.5$ and $\phi_m = 0.64$. By what factor does crowding

increase the estimate?

SOLUTION

Einstein (Eq. 77):

$$\eta_r = 1 + 2.5 \times 0.40 = 2.0.$$

Krieger–Dougherty (Eq. 78):

$$\eta_r = \left(1 - \frac{0.40}{0.64}\right)^{-2.5 \times 0.64} = (0.375)^{-1.6}.$$

Compute: $\ln(0.375) = -0.9808$, so $(0.375)^{-1.6} = \exp(1.6 \times 0.9808) = \exp(1.569) \approx \mathbf{4.80}$.

Crowding factor: $4.80/2.0 \approx \mathbf{2.4}$ — the KD prediction is roughly $2.4\times$ higher than the dilute limit at this loading.

Exercise 12.2 ○ ○ ○ A colloidal suspension shows a relative viscosity $\eta_r = 10$ at $\phi = 0.50$ with $[\eta] = 2.5$. Starting from Eq. 78, estimate the maximum packing fraction ϕ_m by numerical iteration (show at least three trial values and bracket to two significant figures).

SOLUTION

Taking the natural log of Eq. 78:

$$\ln \eta_r = -[\eta] \phi_m \ln \left(1 - \frac{\phi}{\phi_m}\right).$$

With $\eta_r = 10$ and $[\eta] = 2.5$, the target is $\ln 10 = 2.3026$. Define the residual $f(\phi_m) = 2.5 \phi_m [-\ln(1 - 0.5/\phi_m)] - 2.3026$.

Iteration (all to 4 d.p.):

ϕ_m	$1 - 0.5/\phi_m$	$-\ln(1 - 0.5/\phi_m)$	RHS
0.64	0.2188	1.5198	2.432
0.65	0.2308	1.4663	2.383
0.66	0.2424	1.4171	2.338
0.67	0.2537	1.3715	2.297

The RHS crosses the target 2.3026 between $\phi_m = 0.66$ (RHS= 2.338, too high) and $\phi_m = 0.67$ (RHS= 2.297, too low), so $\phi_m \approx \mathbf{0.67}$ to two significant figures.

Verification: $(1 - 0.5/0.67)^{-2.5 \times 0.67} = (0.2537)^{-1.675} = \exp(1.675 \times 1.371) \approx \exp(2.297) \approx 9.95 \approx 10$. ✓

KEY TAKEAWAYS

- Each material class prints well for a *different* physical reason: fast ionic set (alginate), high-modulus fibrous network (BC), thermoreversible gel (gelatin), or tunable physical network (CMC).
- The same framework parameters (η_0 , G'_{LVR} , γ_{LVR} , R_{rec}) govern printability in all classes; only the values and their sensitivities change.
- At $\phi \ll \phi_m$ the Einstein relation (Eq. 77) is adequate; as filler loading approaches the jamming threshold use Krieger–Dougherty (Eq. 78) to avoid under-estimating viscosity.
- *Re-measure rheology on the composite.* Never assume filler additives are inert.

ENGINEERING DESIGN RULES

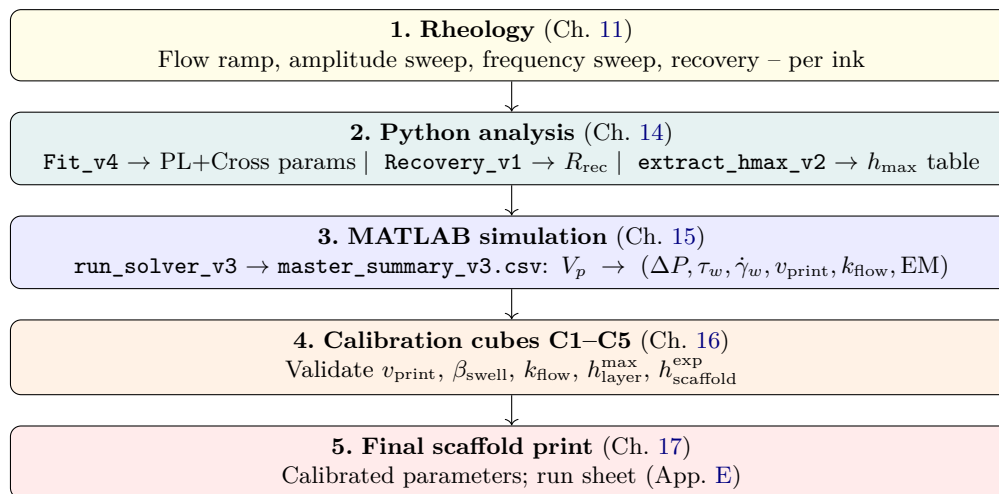
DR 11.1. Before printing a new material class for the first time, run the full SAOS battery (amplitude sweep + frequency sweep) and extract G'_{LVR} , γ_{LVR} , and $\tan \delta$ at $\omega = 1 \text{ rad s}^{-1}$. These three numbers set all downstream simulation inputs.

DR 11.2. For gelatin and HPMC, always quote the test temperature alongside every rheological parameter. A parameter reported without temperature is meaningless for simulation (see Ch. 5).

DR 11.3. If the single-mode Cross fit gives $R^2 < 0.95$, suspect multi-mode behaviour (composite system) and flag before using the parameters in the MATLAB simulation.

Part III – Protocols

13 Workflow Overview



The next sections describe steps 2–5 in operational detail. For step 1, see Ch. 11; for the theoretical rationale of each output, see Part I.

14 Step 2: Python Analysis

14.1 Pre-run checklist

- ☐ Activate the project Python environment.
- ☐ Verify `xlrd`, `scipy`, `numpy`, `pandas`, and `matplotlib` are installed.
- ☐ Check that the `.xls` files from the Anton Paar are in the expected folders (Reologia/Viscosity/, Reologia/Strain/, Reologia/Frequency/, Reologia/Recovery/).
- ☐ Confirm the active script is `Fit_Muitos_Modelos_v4.py` (see CLAUDE.md §6.9).

14.2 Flow-curve fitting – `Fit_Muitos_Modelos_v4.py`

```

cd Analises/Python
python3 Fit_Muitos_Modelos_v4.py
# Outputs: Results/FitAll-Ramp1-v4.txt
# Results/FitAll-Ramp2-v4.txt (if Flow Sheared run)
# Contains: PL (K, n) and Cross (eta_0, eta_inf, lambda, m)
# with R^2, AIC, BIC per model per ink.
  
```

What to check.

- $R^2 \geq 0.98$ for Cross across the full $\dot{\gamma}$ range.
- Cross η_∞ is finite and positive (not pinned to its lower bound – check that AIC/BIC improved over a 3-parameter fit).
- Cross AIC/BIC lower than PL on log-spaced data.
- PL parameters agree with the Cross slope in the printable $\dot{\gamma}_w$ range ($50\text{--}1000\text{ s}^{-1}$).
- If $R^2 < 0.95$ for Cross: suspect multi-mode behaviour or a data quality issue (bubble, edge effects); flag before proceeding.

14.3 Recovery – Recovery_v1.py

```
python3 Recovery_v1.py
# Output: Results/Relatorio_Recovery.txt
# Reports R_rec(%) at t=30s for each (ink, gamma_dot_2).
```

What to check. R_{rec} should decrease monotonically (or remain flat) with increasing $\dot{\gamma}_2$. A non-monotonic pattern (e.g. higher recovery at 100 than at 50 s^{-1}) typically means the recovery test was contaminated by the preceding amplitude sweep (network not fully rebuilt). Repeat with a fresh sample; wait ≥ 5 min between tests.

14.4 Scaffold-height extraction – extract_hmax_v2.py

```
python3 extract_hmax_v2.py
# Outputs:
# Results/SAOS_hmax_v2.txt
# Results/printing_parameters_per_ink.csv
# Results/Gprime_extrap_<ink>.png
# Reports h_scaffold_max under criteria A, B, C and N_max.
```

What to check. The $G'(\omega)$ log-log plot (Gprime_extrap_<ink>.png) should be visually linear at the low- ω end. If the lowest-frequency points scatter, trim the fit window upward by one point and re-run; the low- ω extrapolation is sensitive to noise.

15 Step 3: MATLAB Simulation – run_solver_v3.m

```
cd Analises/MatLab
matlab -batch "run_solver_v3"
# Loops: 3 inks x 2 needles (21G, 22G) x 8 piston velocities.
# Outputs:
# output_v3/slicer_lookup_<ink>_<needle>.csv
# output_v3/plots_<ink>_<needle>.png (4-panel summary)
# output_v3/master_summary_v3.csv (reference row Vp=0.01)
```

Reading master_summary_v3.csv. Columns (in order): sample, needle, V_p | v_{print} (PL and Cross) | ΔP | τ_w | $\dot{\gamma}_w$ | w_{line} | h_{layer} | k_{flow} | $\text{EM}_{\text{slicer}}$.

SAFETY / DATA INTEGRITY

The CSV column is k_{flow} , *not* $\text{EM}_{\text{slicer}}$. Read the $\text{EM}_{\text{slicer}}$ column (the last column in the lookup CSV, added in v3) for the number to type into the slicer. If you use an older CSV, compute $\text{EM} = 1/k_{\text{flow}}$ by hand. Never type k_{flow} directly into the slicer.

TIP

The default reference row is at $V_p = 0.01$ mm/s. For a different operating point, open the per-ink CSV `slicer_lookup_<ink>_<needle>.csv` which covers the full $V_p \in \{0.003 \dots 0.04\}$ mm/s sweep.

When to re-run the MATLAB simulation. Any time Python produces new Cross or Power-Law parameters, update the `samples` struct in `run_solver_v3.m` and re-run. Parameters must always be in sync between Python output and MATLAB input.

16 Step 4: Calibration-Cube Protocol (C1–C5)

Run all tests on the same day, with the same syringe load. Each test is a diagnostic for one physical question; if it fails, the failure mode points to one rheological input.

Pre-calibration checklist.

- ☐ Power up printer, home all axes, level bed.
- ☐ Pull ink from 4 °C storage; let it equilibrate at 25 °C for ≥ 30 min.
- ☐ Re-homogenise by slow inversion (30 times); inspect for phase separation or air bubbles.
- ☐ Load syringe; mount needle; prime until steady bead exits.
- ☐ Enter V_p , EM_{slicer} , and h_{layer} from the lookup CSV into the slicer.
- ☐ Verify predicted $\Delta P < 1500$ kPa from CSV.

16.1 C1 – print-speed sweep

C1: identify the optimum v_{print} at fixed V_p

Print five 30 mm straight filaments on a clean glass slide at five XY feedrates $v_{\text{print}} \in \{5, 10, 20, 30, 50\}$ mm/s, keeping V_p fixed at the lookup-CSV value. Wait 5 min. Under a microscope, measure filament width d_{fil} at three points per filament.

Optimum: the v_{print} at which $0.95 \leq d_{\text{fil}}/\text{ID} \leq 1.10$. Below 0.95: head outruns ink supply (under-extrusion or motor skip). Above 1.10: over-deposition or accumulated die swell.

Also read die swell: measure w_{line} at the optimum v_{print} ; compute $\beta_{\text{swell}}^{\text{exp}} = w_{\text{line}}/\text{ID} - 1$ and compare against the proxy from the lookup CSV. Update if $|\beta^{\text{exp}} - \beta^{\text{pred}}| > 0.05$.

16.2 C2 – mass-per-line (k_{flow}) calibration

C2: measure $k_{\text{flow}}^{\text{exp}}$ and update EM

At the C1-optimum v_{print} , print ten parallel 30 mm lines on a tared glass slide with 1 mm spacing. Time the print. Reweigh; convert to volume using $\rho_{\text{ink}} = 1.00$ g/cm³.

$$k_{\text{flow}}^{\text{exp}} = \frac{V_{\text{measured}}}{v_{\text{print}} \cdot w_{\text{line}} \cdot h_{\text{layer}} \cdot t_{\text{print}}}.$$

Invert and enter $EM^{\text{exp}} = 1/k_{\text{flow}}^{\text{exp}}$ as the slicer's Extrusion Multiplier for all subsequent prints with this ink and needle.

16.3 C3 – layer-height sweep

C3: find $h_{\text{layer}}^{\text{max}}$

Print a 3-layer 10×10 mm square at three layer heights $h_{\text{layer}} \in \{0.30, 0.40, 0.50\}$ mm. After 30 min of relaxation, measure the total stack height.

$h_{\text{layer}}^{\text{max}}$: the largest h_{layer} at which measured total height $\geq 90\%$ of commanded total $3h_{\text{layer}}$.

16.4 C4 – vertical-tower test

C4: validate $h_{\text{scaffold}}^{\text{max}}$ from SAOS

Print a 5×5 mm tower at $h_{\text{layer}}^{\text{max}}$ from C3, up to $N = 40$ layers (or the predicted N_{max} from `printing_parameters_per_ink.csv`, whichever is smaller). Photograph every 5 layers and at $t = 30$ min.

$h_{\text{scaffold}}^{\text{exp}}$: height at which base width grows by $> 10\%$. Record which SAOS criterion best matches the collapse onset. **Use that criterion for all future prints of this ink at similar dwell times.**

16.5 C5 (optional) – bridging test for lattice scaffolds

C5: unsupported-span sag test

Print three pairs of 20 mm pillars at gap lengths $L \in \{0.5, 1.0, 2.0\}$ mm. Print a single bridging filament over each gap; measure mid-span sag after 5 min. Compare against Smay prediction (Eq. 66). If sag exceeds the prediction, the strut span must be shortened.

RUNNING EXAMPLE – Solution to Q6

Question. What does Ana’s Wednesday morning look like?

Schedule ($\sim 4\text{--}5$ h):

08:30	Power up; home; level. Pull C15 Gira 5.5 from 4°C ; equilibrate 30 min.
09:00	Re-homogenise; load syringe; prime. Enter $V_p = 0.01$ mm/s, $\text{EM} = 1.49$, $h_{\text{layer}} = 0.36$ mm in slicer. Predicted $\Delta P = 164$ kPa: proceed.
09:30	C1. Five filaments at $v_{\text{print}} \in \{5, 10, 20, 30, 50\}$ mm/s. Microscope. Expected optimum near 10 mm/s.
10:30	C2. Ten lines on tared slide; weigh. Expected $k_{\text{flow}}^{\text{exp}} \in [0.6, 0.8]$; $\text{EM}^{\text{exp}} \in [1.25, 1.67]$.
11:30	C3. Three 3-layer squares at $h_{\text{layer}} \in \{0.30, 0.40, 0.50\}$ mm. Wait 30 min. Expected $h_{\text{layer}}^{\text{max}} \approx 0.40$ mm.
13:00	C4. Tower at $h_{\text{layer}}^{\text{max}}$; target $N = 56$ ($= 20$ mm). Stop early if base bulges.
15:00	Done. Log all calibrated parameters to <code>lab_calibration_log.csv</code> .

Go / No-go. If calibrated parameters match predictions within 30% and C4 tower clears ~ 50 layers cleanly: proceed to Thursday production print. Otherwise: escalate before burning more ink.

17 Step 5: Final Scaffold Print

After C1–C4, the slicer profile contains all five validated entries:

- V_p (E-axis feedrate) at the validated value.
- v_{print} (XY feedrate) at the C1 optimum.
- $\text{EM}^{\text{exp}} = 1/k_{\text{flow}}^{\text{exp}}$ from C2.

- $h_{\text{layer}} \leq h_{\text{layer}}^{\text{max}}$ from C3.
- Total scaffold height $\leq h_{\text{scaffold}}^{\text{exp}}$ from C4.

Save slicer profile as `<ink>_<needle>_<scaffold>_v<n>.ini`. Record the final print on the run sheet (App. E).

18 Troubleshooting: Failure-Mode Decision Tree

Observation	Physical root cause	Action
Motor hum / skip; no flow	ΔP exceeds motor stall, or clog	Lower V_p 25%. If still no flow: switch to wider needle; purge.
$d_{\text{fil}}/\text{ID} > 1.2$ consistently	Over-extrusion or β_{swell} underestimated	Lower EM in 0.05 steps; measure β^{exp} in C1.
$d_{\text{fil}}/\text{ID} < 0.9$ consistently	Under-extrusion or motor skip	Raise EM in 0.05 steps; verify ΔP vs. motor limit.
Tower base bulges at low N	σ_{stack} exceeded at effective timescale	Reduce target height; increase v_{print} ; switch ink.
Tower buckles sideways	Smay lattice buckling	Shorten span or use higher- G' ink.
Layer fusion / mushy scaffold	$h_{\text{layer}} > h_{\text{layer}}^{\text{max}}$ or low R_{rec}	Decrease h_{layer} ; increase v_{travel} .
Stringing on travel moves	Slow recovery at travel timescale	Set retraction to zero ; increase v_{travel} .
Rough filament surface	Elastic extrudate instability	Lower V_p ; use longer needle.
First-layer adhesion failure	Bed clean / Z-zero / temperature	Clean bed; raise first-layer h by 0.05 mm; re-home.

Table 4: Operator troubleshooting decision tree. All corrective actions target V_p , EM, geometry, or needle gauge. ΔP is an output and is never adjusted directly.

19 The Friday Print – Closing the Loop

By Wednesday 15:00, Ana has answered every question this book posed:

- **Q1.** The ink is a reinforced critical gel with wide LVR – excellent DIW candidate.
- **Q2.** $V_p = 0.01$ mm/s to reach $v_{\text{print}} \approx 8$ mm/s.
- **Q3.** $\Delta P \approx 164$ kPa – safe. $\tau_w = 663$ Pa – cell-safe.
- **Q4.** $k_{\text{flow}} \approx 0.67$; EM = 1.49 into the slicer.
- **Q5.** 20-mm block feasible under criterion B; validated by C4.
- **Q6.** Calibration completed 08:30–15:00; no surprises.

Thursday, 10:00. Ana exports G-code, mounts a fresh syringe of conditioned ink, and prints the $1 \times 1 \times 2$ cm³ block. Total print time ≈ 45 min. The block holds its shape. She photographs at $t = 0$, $t = 30$ min, and $t = 24$ h.

Friday, 15:30. Ana hands the supervisor the scaffold and the completed run sheet.

The pedagogical point. Every entry on Ana’s run sheet was computed from a rheological measurement and one equation chain. The slicer firmware was treated as a dumb FFF tool that needed translation. The rheometer told her what the ink wanted; the math told her what to type; the calibration cubes told her where the math was off and by how much. *This is the framework. The rest is execution.*

Part IV – Extended Topics

20 Biological and Regulatory Considerations

20.1 Cell viability and wall shear stress

Motivation. Cells fail at the wall, not at the average shear: the parabolic (or near-parabolic) velocity profile concentrates stress in the thin annular layer closest to the needle surface, while the central core experiences far lower shear. Moreover, damage is a cumulative dose, not just a peak: a modest τ_w sustained over a long needle transit can cause as much membrane fatigue as a brief, high-stress spike.

The primary mechanical threat to cells during DIW extrusion is the wall shear stress τ_w . Literature consensus (Blaeser et al., 2016) places the acute viability threshold at:

$$\tau_w^{\text{safe}} \lesssim 5 \text{ kPa}. \quad (79)$$

Above this threshold, cell membrane rupture and cytoskeletal damage increase sharply. For the inks in this project, τ_w at the nominal V_p is 460–660 Pa – one order of magnitude below the threshold. This provides substantial margin for cell-loaded future prints.

DERIVATION — Cumulative shear dose D_{shear}

Goal. Quantify the total mechanical dose experienced by a cell transiting the needle once.

Residence time. A cell entering the needle at the syringe–needle junction exits at the tip after travelling the full needle length L_n . Using the mean needle velocity $\bar{v}_n = (R_s/R_n)^2 V_p$ (Eq. 13), the transit time is:

$$t_{\text{res}} = \frac{L_n}{\bar{v}_n} = \frac{L_n}{(R_s/R_n)^2 V_p}. \quad (80)$$

For the 21G needle used in this project ($L_n = 31.75 \text{ mm}$, $(R_s/R_n)^2 = 771$) at $V_p = 0.01 \text{ mm s}^{-1}$:

$$t_{\text{res}} = \frac{31.75}{771 \times 0.01} = \frac{31.75}{7.71} \approx 4.12 \text{ s}.$$

Cumulative shear dose. The shear stress field $\tau(r)$ peaks at the wall ($r = R_n$) and decreases toward the centreline. As a conservative bound, every cell is exposed to τ_w throughout the transit (this overestimates damage for cells in the core, but is the safe design choice). The cumulative dose is then:

$$D_{\text{shear}} = \int_0^{t_{\text{res}}} \tau(t) dt \approx \tau_w t_{\text{res}} \quad [\text{Pa s}]. \quad (81)$$

Why both criteria matter. A wide-gauge, short needle may satisfy $\tau_w < 5 \text{ kPa}$ yet still deliver a damaging D_{shear} if the syringe-to-needle area ratio is small (low $\bar{v}_n$, long t_{res}). Conversely, a very fine needle at high V_p may have a large τ_w but negligible dose because t_{res} is extremely short. Both limits must be checked: peak $\tau_w < 5 \text{ kPa}$ and $D_{\text{shear}} = \tau_w t_{\text{res}}$ below the cell-type tolerance.

Cell type sensitivity. Different cell lines have different shear thresholds:

- Osteoblasts and fibroblasts: relatively resistant; tolerate $\tau_w \leq 10 \text{ kPa}$ for brief exposures.

- Stem cells (MSC, iPSC): more sensitive; threshold closer to 5 kPa.
- Chondrocytes: sensitive to both shear and compression history.

For any cell type, measure post-print viability (live/dead assay within 2 h of printing) and compare against unprinted controls. Do not assume compliance from literature thresholds alone.

20.2 Plug flow and cell protection

The radial stress profile $\tau(r) = \tau_w(r/R_n)$ vanishes at the centreline and rises to τ_w at the wall. For a yield-stress ink, the central region where $\tau < \tau_y$ is never sheared at all – it translates as a solid plug, and any cell inside it experiences *zero* shear stress during transit. Understanding the size of this plug is therefore the single most important geometric quantity for cell-viability engineering.

DERIVATION — Plug radius and shear-free cell fraction

Stress profile. For fully developed flow in a tube of radius R_n and length L_n , the axial pressure gradient $\Delta P/L_n$ drives flow and the local shear stress varies linearly with radius:

$$\tau(r) = \frac{\Delta P}{2L_n} r = \tau_w \frac{r}{R_n}. \quad (82)$$

At $r = R_n$ this gives $\tau_w = \Delta P R_n / (2L_n)$, confirming the wall stress formula used throughout Ch. 3.

Plug radius. The unyielded (plug) core exists wherever $\tau(r) \leq \tau_y$:

$$\tau_w \frac{r_0}{R_n} = \tau_y \implies \frac{r_0}{R_n} = \frac{\tau_y}{\tau_w}. \quad (83)$$

Substituting $\tau_w = \Delta P R_n / (2L_n)$:

$$r_0 = R_n \frac{\tau_y}{\tau_w} = R_n \cdot \frac{2L_n \tau_y}{\Delta P R_n} = \frac{2L_n \tau_y}{\Delta P},$$

which is the standard Bingham result: the plug radius grows with yield stress and needle length and shrinks with applied pressure.

Shear-free cell fraction. Cells are uniformly distributed across the cross-section (area πR_n^2). The protected (plug) area is πr_0^2 . Therefore the fraction of cells that experience *zero* shear during transit is:

$$f_{\text{core}} = \left(\frac{r_0}{R_n} \right)^2 = \left(\frac{\tau_y}{\tau_w} \right)^2. \quad (84)$$

Conversely, the fraction exposed to high shear is $f_{\text{wall}} = 1 - f_{\text{core}} = 1 - (\tau_y/\tau_w)^2$.

Consistency with the Bingham number. From the definition $\text{Bi} = \tau_y / (\eta_p \dot{\gamma}_w)$ and the Bingham constitutive law $\tau_w = \tau_y + \eta_p \dot{\gamma}_w$, one obtains:

$$\frac{\tau_y}{\tau_w} = \frac{\tau_y}{\tau_y + \eta_p \dot{\gamma}_w} = \frac{1}{1 + \eta_p \dot{\gamma}_w / \tau_y} = \frac{\text{Bi}}{\text{Bi} + 1}.$$

Substituting into Eq. 83:

$$\frac{r_0}{R_n} = \frac{\text{Bi}}{\text{Bi} + 1}. \quad (85)$$

This is exactly the expression derived in the Hydraulic-Loop plug-flow exercise (Ch. 3). As $\text{Bi} \rightarrow \infty$, $r_0/R_n \rightarrow 1$: the entire cross-section translates as a rigid plug and virtually all cells are shear-free. As $\text{Bi} \rightarrow 0$ (Newtonian limit), $r_0 \rightarrow 0$ and all cells experience the full parabolic shear field.

IN PRACTICE — Designing for cell viability in DIW

- **Choose the widest needle the resolution allows.** Switching from 22G to 21G at fixed V_p reduces τ_w by roughly half (recall $\tau_w \propto R_n^{-(3+n)}$ with $n < 1$) and increases t_{res} only modestly, giving a net reduction in both peak stress and dose.
- **Use the lowest V_p that clears the minimum dwell time.** Lowering V_p reduces $\bar{v}_n = (R_s/R_n)^2 V_p$, which lowers $\dot{\gamma}_w$ and τ_w . The penalty is a longer t_{res} , but for shear-thinning inks the τ_w reduction dominates, so the dose $D_{\text{shear}} = \tau_w t_{\text{res}}$ typically falls.
- **Maximise the Bingham number.** Formulate inks with as high a yield stress as possible while maintaining $\tau_w < 5 \text{ kPa}$: a large Bi means a large $r_0/R_n = \text{Bi}/(\text{Bi} + 1)$ and most cells ride the shear-free plug.
- **Compute D_{shear} before printing.** Use Eq. 81: $D_{\text{shear}} = \tau_w L_n / \bar{v}_n$. For multi-layer prints each cell passes the needle once, so cumulative dose equals D_{shear} per layer multiplied by the number of times that cell is re-extruded (usually once for scaffold prints).

KEY TAKEAWAYS

- Peak wall stress $\tau_w < 5 \text{ kPa}$ is necessary but not sufficient for cell safety; the cumulative dose $D_{\text{shear}} = \tau_w t_{\text{res}}$ must also be checked.
- A high Bingham number ($\text{Bi} \gg 1$) creates a large shear-free plug: $f_{\text{core}} = (\text{Bi}/(\text{Bi} + 1))^2 \rightarrow 1$. This is the principal rheological lever for protecting cells in DIW.
- For the project inks at $V_p = 0.01 \text{ mm s}^{-1}$ on a 21G needle, $\tau_w \approx 663 \text{ Pa}$ and $D_{\text{shear}} \approx 2.7 \text{ kPa s}$ – well inside acute safety margins, providing a comfortable baseline for future cell-loading studies.

Exercise 20.1 ○○ **Cumulative shear dose.** A 21G needle ($R_n = 0.2575 \text{ mm}$, $L_n = 31.75 \text{ mm}$) is used with $V_p = 0.01 \text{ mm s}^{-1}$. The syringe-to-needle area ratio is $(R_s/R_n)^2 = 771$ (standard 10 mL BD syringe).

- Compute the mean needle velocity $\bar{v}_n$.
- Compute the needle residence time t_{res} .
- The NE ink has wall shear stress $\tau_w = 663 \text{ Pa}$ at this operating point. Compute the cumulative shear dose $D_{\text{shear}} = \tau_w t_{\text{res}}$ in Pa s.
- Is τ_w within the acute viability threshold (Eq. 79)? Comment on whether D_{shear} raises additional concerns.

SOLUTION

- $\bar{v}_n = 771 \times 0.01 = 7.71 \text{ mm s}^{-1}$.
- $t_{\text{res}} = L_n / \bar{v}_n = 31.75 / 7.71 \approx 4.12 \text{ s}$.
- $D_{\text{shear}} = 663 \times 4.12 \approx 2732 \text{ Pa s} \approx 2.7 \text{ kPa s}$.
- Yes: $\tau_w = 663 \text{ Pa} \ll 5 \text{ kPa}$, so the acute peak criterion is comfortably satisfied (factor ~ 7.5 of safety margin). The dose $D_{\text{shear}} \approx 2.7 \text{ kPa s}$ is moderate; for most adherent cell types this single-transit dose is acceptable, but it should be recorded and compared against cell-type-specific tolerance values. Multi-pass prints (if any cell traverses the needle more than once) would multiply this figure and require re-evaluation.

Exercise 20.2 ○○○ **Plug-core fraction and the Bingham number.** A cell-loaded Bingham ink with $\text{Bi} = 12$ is extruded through a 21G needle ($R_n = 0.2575 \text{ mm}$) at the nominal print speed.

- Using $r_0/R_n = \text{Bi}/(\text{Bi} + 1)$ (Eq. 85), compute the plug radius as a fraction of R_n .

- (b) What fraction of the needle cross-sectional area is occupied by the shear-free plug?
- (c) If the same formulation is extruded through a 22G needle at the same V_p , qualitatively does Bi increase or decrease? Is the change beneficial for cell viability?

SOLUTION

- (a) $r_0/R_n = 12/(12 + 1) = 12/13 \approx 0.923$.
- (b) $f_{\text{core}} = (r_0/R_n)^2 = (12/13)^2 = 144/169 \approx 0.852$, i.e. approximately **85%** of the cross-sectional area is shear-free. Only the outermost 15% of the area (a thin annulus of thickness $\approx 0.077 R_n \approx 20 \mu\text{m}$) experiences wall-level shear stress.
- (c) Switching to 22G at fixed V_p : R_n decreases ($0.2575 \rightarrow 0.2065 \text{ mm}$), so $\dot{\gamma}_w \propto R_n^{-3}$ rises and $\tau_w \propto R_n^{-(3+n)}$ rises faster; τ_y is a material property and does not change. Therefore $\text{Bi} = \tau_y/(\eta_p \dot{\gamma}_w)$ *decreases*: the plug shrinks. The change is **detrimental** – a smaller f_{core} means more cells are exposed to the higher (and now larger) τ_w . For cell-loaded prints, the widest needle meeting the resolution requirement minimises damage.

20.3 Long-duration print considerations

For prints lasting $> 30 \text{ min}$ with cell-loaded inks:

- Keep the unused ink at 4°C or on ice; avoid temperature fluctuations that change rheology mid-print.
- Monitor filament width at intervals (every 10 layers); drift indicates ink dewatering or temperature change.
- Consider enclosing the printer in a humidified chamber to prevent evaporation from the deposited scaffold.

20.4 MAPA-oriented regulatory considerations

For healthcare applications in Brazil (pharmaceutical or dermocosmetic products), the National Health Surveillance Agency (ANVISA) and the Ministry of Agriculture, Livestock, and Food Supply (MAPA) impose requirements on the production process:

- **Raw material traceability.** The CMC grade, molecular weight determination method (cite da Silva et al. 2018), and NE formulation parameters (oil type, surfactant, batch number) must all be documented.
- **Process reproducibility.** Calibration data (C1–C4 outputs) must be recorded and archived per batch.
- **Biocompatibility testing.** For dermocosmetic applications (e.g. Amphotericin B carrier), ISO 10993-5 cytotoxicity testing is required. The current project phase is pre-regulatory (acellular scaffold characterisation only).
- **Scale-up validation.** The DIW parameters characterised on the lab-scale piston printer must be re-validated at pilot scale; scale-up introduces pressure gradients and syringe-wall effects not present at 10 mL scale.

TIP

When writing the manuscript, frame the DIW as a “proof-of-concept fabrication platform” rather than a production process. This framing keeps the paper in the research domain and defers the regulatory validation discussion to future work.

21 Statistical Optimisation: DOE and Response Surface Methodology

21.1 Overview

Design of Experiments (DOE) and Response Surface Methodology (RSM) are the standard tools for optimising the NE formulation (oil fraction, surfactant type, surfactant concentration, homogenisation protocol) with respect to measured responses (droplet size, η_0 , G'_{LVR} , printability). This section documents the critical decisions required before running any RSM, with emphasis on categorical variables – the most common source of invalid models in this type of study.

21.2 Continuous vs. categorical variables

RSM models assume all factors are *continuous*: they can take any value in a range, and the response varies smoothly with them. A factor is *categorical* if it takes discrete, unordered levels with no implied metric distance between them.

Continuous factors (safe for RSM quadratic terms):

- Oil mass fraction [%]
- Surfactant concentration [mg/mL]
- Homogenisation speed [rpm]
- Homogenisation time [min]

Categorical factors (cannot receive quadratic terms in RSM):

- Surfactant type (Tween 80 / Span 80 / Polysorbate 20 / ...)
- Emulsification method (probe sonication / high-pressure HPH / rotor-stator)
- Needle type (blunt / tapered)
- Pulse mode (on/off), if coded as 1/0

SAFETY / DATA INTEGRITY

Categorical variables must not appear in quadratic RSM terms. A quadratic term for “surfactant type” (e.g. $x_{\text{surfactant}}^2$ where surfactant type is coded as 1, 2, 3) imposes an implied quadratic distance between surfactant types that has no physical meaning. The resulting “optimal” surfactant is often a non-existent interpolation between two real options.

Correct handling: Run separate RSM surfaces for each level of the categorical factor, or use mixture designs, or test only one categorical level at a time with a fixed level for all others.

21.3 Model selection recommendations

- For 2–4 continuous factors: full Central Composite Design (CCD) or Box-Behnken Design. Minimum 5 centre-point replicates for pure error estimation.
- For > 4 continuous factors: use a D-optimal or I-optimal custom design to keep run count manageable.
- For one categorical factor: use a split-plot design or run separate CCD blocks per categorical level.
- Always report the lack-of-fit test. A significant lack of fit ($p < 0.05$) means the quadratic surface does not capture the response – add cross terms or increase the polynomial degree.

21.4 Responses and their transformations

- **Droplet size** (d , nm): typically log-normally distributed. Fit after $\log_{10}$ transformation; check residuals.
 - **Viscosity** (η_0 , Pa s): strongly right-skewed. Use $\ln(\eta_0)$ as the response.
 - **Modulus** (G'_{LVR} , Pa): similarly skewed. Use $\ln(G'_{\text{LVR}})$.
 - **Recovery** (R_{rec} , %): bounded $[0, 100]$. If values cluster near the bounds, use logit transformation: $\text{logit}(p) = \ln(p/(1 - p))$.
-

A Glossary

V_p – piston velocity

Linear advance speed of the syringe piston in mm/s. The single flow input on this printer.

Q – volumetric flow rate

$$Q = \pi R_s^2 V_p \text{ (mm}^3/\text{s)}.$$

$\bar{v}_n$ – mean nozzle velocity

$$Q/(\pi R_n^2): \text{ average velocity inside the needle (mm/s)}.$$

v_{print} – print speed

$$Q/(w_{\text{line}} h_{\text{layer}}): \text{ head XY translation speed during deposition}.$$

ΔP – extrusion pressure

Pressure drop across the needle. **Output** on this printer; diagnostic against the ~ 1500 kPa syringe ceiling.

τ_w – wall shear stress

Force per unit area at the needle wall. Cell-safe below ~ 5 kPa.

$\dot{\gamma}_w$ – wall shear rate

Strain rate at the needle wall; selects the recovery test shear rate.

K, n – Power-Law parameters

$$\tau = K \dot{\gamma}^n.$$

$\eta_0, \eta_\infty, \lambda, m$ – Cross-model parameters

Eq. 7.

G' – storage modulus

Elastic part of SAOS response (Pa).

G'' – loss modulus

Viscous part of SAOS response (Pa).

$\tan \delta$

G''/G' : loss tangent. < 1 = gel-like.

γ_{LVR} – LVR endpoint strain

Strain at which G' has dropped 10% from its plateau in an amplitude sweep [%].

β – low- ω slope

$d \ln G' / d \ln \omega$ at low frequency. Gel-character diagnostic: $\beta \rightarrow 0$ = true gel, $\beta \approx 0.5$ = critical gel.

R_{rec} – structural recovery

% return of η at 30 s after high-shear disruption.

 h_{layer}

Layer thickness chosen by the operator.

 $h_{\text{layer}}^{\text{max}}$

Largest h_{layer} validated in C3 (per-layer sag bound).

 $h_{\text{scaffold}}^{\text{max}}$

Total height ceiling from $\sigma_{\text{max}}/(\rho g)$.

 N_{max}

$h_{\text{scaffold}}^{\text{max}}/h_{\text{layer}}$.

 k_{flow} – deposition efficiency

Fraction of the slicer-commanded volume that is locked in the bead (Eq. 52). **Not** the number to type into the slicer.

 $\text{EM}_{\text{slicer}}$ – Extrusion Multiplier

$1/k_{\text{flow}}$ (Eq. 53). This is the number entered into the slicer (labelled “Flow”, “Flow Rate”, or “Volumetric Flow Compensation” depending on the slicer package).

 β_{swell} – die-swell ratio

$w_{\text{line}}/\text{ID} - 1$. Estimated from Cross m or measured in C1.

LVR

Linear viscoelastic regime.

SAOS

Small-amplitude oscillatory shear.

Critical gel

Transient network on the edge of percolation ($\beta \approx 0.5$).

True gel

Permanent network ($\beta \rightarrow 0$).

Rabinowitsch–Weissenberg integral

Generalises Hagen–Poiseuille to any $\eta(\dot{\gamma})$. Eq. 37.

Smay criterion

Buckling bound for unsupported lattice struts. Eq. 66.

Die swell

Radial expansion of the extrudate on exit from the needle, driven by elastic stress release.

Deborah number De

$t_{\text{relax}}/t_{\text{process}}$; governs whether elastic or viscous behaviour dominates.

Bingham number Bi

Ratio of yield stress to viscous stress; governs whether flow is plug-like or Newtonian-like.

Capillary number Ca

Ratio of viscous to surface tension forces.

DOE

Design of Experiments.

RSM

Response Surface Methodology.

B Notation and Unit Reference

Symbol	Name	SI unit	Practical unit
V_p	Piston velocity	m/s	mm/s
Q	Volumetric flow rate	m ³ /s	mm ³ /s
$\bar{v}_n$	Mean nozzle velocity	m/s	mm/s
v_{print}	Print speed	m/s	mm/s
R_s	Syringe inner radius	m	mm
R_n	Needle inner radius	m	mm
L_n	Needle length	m	mm
τ	Shear stress	Pa	Pa
τ_w	Wall shear stress	Pa	Pa
$\dot{\gamma}$	Shear rate	s ⁻¹	s ⁻¹
η	Viscosity	Pa s	Pa s
η_0	Zero-shear viscosity	Pa s	Pa s
ΔP	Pressure drop	Pa	kPa
G'	Storage modulus	Pa	Pa
G''	Loss modulus	Pa	Pa
K	Power-Law consistency	Pas ⁿ	Pas ⁿ
n	Power-Law index	–	–
λ	Cross relaxation time	s	s
m	Cross exponent	–	–
ρ	Ink density	kg/m ³	g/cm ³
h_{layer}	Layer thickness	m	mm
$h_{\text{scaffold}}^{\text{max}}$	Max scaffold height	m	mm
k_{flow}	Deposition efficiency	–	–
EM	Extrusion Multiplier	–	–
β_{swell}	Die-swell ratio	–	–
R_{rec}	Structural recovery	–	%

Hardware defaults (override in `CLAUDE.md` §6.5 if changed):

Component	ID (mm)	Length (mm)
Syringe (BD 10 mL)	14.30 ($R_s = 7.15$)	90
Needle 21G blunt	0.515 ($R_n = 0.258$)	31.75
Needle 22G blunt	0.413 ($R_n = 0.207$)	25.40

C Worked End-to-End Example: C15 Gira 5.5 on 21G

Step 0 – starting from rheology.

- Cross: $\eta_0 = 2440$, $\eta_\infty = 0.001$, $\lambda = 4.10$ s, $m = 0.988$.
- PL: $K = 267$, $n = 0.24$.
- SAOS: $G'_{\text{LVR}} = 916$ Pa, $\gamma_{\text{LVR}} = 60.9\%$, $G'(\omega = 1) = 228$ Pa, $G'(\omega = 0.01) = 30$ Pa.
- Recovery at 100 s^{-1} : $R_{\text{rec}} = 45\%$.

Step 1 – $V_p \rightarrow Q$. $V_p = 0.01$ mm/s; $Q = \pi(7.15)^2 \cdot 0.01 = 1.6$ mm³/s.

Step 2 – Cross-model ΔP . Rabinowitsch–Weissenberg with 21G ($R_n = 0.258$ mm, $L_n = 31.75$ mm): $\tau_w \approx 660$ Pa, $\dot{\gamma}_w \approx 60 \text{ s}^{-1}$, $\Delta P \approx 164$ kPa.

Step 3 – safety check. $\Delta P = 164$ kPa $\ll 1500$ kPa. $\tau_w = 660$ Pa $\ll 5$ kPa. Print is mechanically and biologically feasible.

Step 4 – slicer parameters. $\beta_{\text{swell}} = 0.3(1 - 0.988) = 0.004$; $w_{\text{line}} = 0.52 \text{ mm}$; $h_{\text{layer}} = 0.7 \cdot 0.515 = 0.36 \text{ mm}$; $k_{\text{flow}} = (1.004)^2 \cdot 1 \cdot \sqrt{0.45} = 0.67$; $\text{EM}_{\text{slicer}} = 1/0.67 \approx 1.49$.

Step 5 – $h_{\text{scaffold}}^{\text{max}}$. Criterion (B) at typical 1-s/layer dwell: $h_{\text{max}} = 228/(1000 \cdot 9.81) \cdot 10^3 = 23.3 \text{ mm}$ (65 layers at $h_{\text{layer}} = 0.36 \text{ mm}$).

Step 6 – expected calibration results. C1: $v_{\text{print}} \approx 8\text{--}10 \text{ mm/s}$. C2: $k_{\text{flow}}^{\text{exp}} \approx 0.60\text{--}0.80$; $\text{EM}^{\text{exp}} \approx 1.25\text{--}1.67$. C3: $h_{\text{layer}}^{\text{max}} \approx 0.40 \text{ mm}$. C4: tower to $N \approx 60$ layers; best-matching criterion: (B).

Step 7 – final scaffold. $5 \times 5 \times 20 \text{ mm}$ solid block at 21G is comfortably within all bounds and is the recommended first production print for this ink.

D Engineering Design Rules – Consolidated Reference

- DR 1.** Before reading any DIW paper, identify the control mode (pneumatic, screw, or piston). Parameters are not transferable without going through the constitutive model.
- DR 2.** Never enter ΔP into the slicer – it is an output on this printer.
- DR 3.** Always report both G'_{LVR} and γ_{LVR} ; their product controls scaffold height.
- DR 4.** Prefer the Cross model for printability prediction. Use the Power Law only for single-decade fitting.
- DR 5.** Structural recovery is more important than static viscosity for scaffold quality.
- DR 6.** Measure β_{swell} directly for any ink with $m < 0.7$.
- DR 7.** $\text{EM}_{\text{slicer}} = 1/k_{\text{flow}}$. Never type k_{flow} into the slicer.
- DR 8.** Verify $\text{Re}_{\text{gen}} \ll 2100$ before accepting any ΔP calculation.
- DR 9.** Do not change needle gauge without recomputing all downstream quantities; $\Delta P \propto R_n^{-4}$.
- DR 10.** For cell-loaded inks, prefer high-Bi inks (approaching plug flow).
- DR 11.** Always report ΔP , τ_w , and $\dot{\gamma}_w$ as a trio.
- DR 12.** Always compute and report all three h_{max} criteria.
- DR 13.** For lattices, always compute the Smay bound for the intended strut span.
- DR 14.** Validate $h_{\text{scaffold}}^{\text{max}}$ in C4 and record which criterion best matches.
- DR 15.** Diagnose failure morphology from first principles before adjusting the slicer blindly.
- DR 16.** Run C1 before C2–C4 on any new ink.
- DR 17.** Set retraction to zero on any piston DIW printer.
- DR 18.** Update MATLAB parameters immediately when Python produces new Cross/PL parameters.
- DR 19.** Categorical variables must not appear in quadratic RSM terms.
- DR 20.** Preserve the daSilva2018 citation for CMC molecular weight in every manuscript.

E Single-Page Run Sheet

DIW Printing Run Sheet – Textbook v2.0					
Operator:_____		Date:_____		Ink batch:_____	
Ink: ☐ C15 ☐ C15 Gira 5.5 ☐ Bozano		Needle: ☐ 21G ☐ 22G		Room T (°C):_____	
Predicted (from CSV):					
V_p =_____ mm/s		v_{print} =_____ mm/s		ΔP =_____ kPa	
τ_w =_____ Pa		k_{flow} =_____		EM = $1/k$ =_____	
h_{layer} =_____ mm		$N_{\text{max}}(\text{B})$ =_____		$N_{\text{max}}(\text{C})$ =_____	
$\Delta P < 1500$ kPa?		☐ Yes ☐ No		(stop)	
C1 line widths (d_{fil}/ID) at 5:_____ 10:_____ 20:_____ 30:_____ 50:_____					
Optimum v_{print} :_____ mm/s		$\beta_{\text{swell}}^{\text{exp}}$:_____			
C2: m_{dep} =_____ mg		t_{print} =_____ s		$k_{\text{flow}}^{\text{exp}}$ =_____ EM ^{exp} = $1/k$ =_____	
C3: stack @ 0.30:_____		@ 0.40:_____		@ 0.50:_____ $h_{\text{layer}}^{\text{max}}$ =_____ mm	
C4: N^{exp} =_____		$h_{\text{scaffold}}^{\text{exp}}$ =_____ mm		Best criterion: ☐ A ☐ B ☐ C	
Motor stall events: ☐ None ☐ _____					
Final slicer profile:		V_p =_____ v_p =_____ EM=_____ h =_____		Profile file:_____	
Notes:_____					

F Software Setup Guide

F.1 Python environment

```
# Create environment
python3 -m venv .venv
source .venv/bin/activate # macOS/Linux
# .venv\Scripts\activate # Windows

# Install dependencies
pip install numpy scipy pandas xlrd matplotlib

# Verify xlrd version (must be 1.x for .xls; 2.x removed .xls support)
pip show xlrd # should show Version: 1.x.x
# If 2.x: pip install xlrd==1.2.0
```

Key dependency constraint. The Anton Paar **.xls** files use the legacy binary format. The xlrd library dropped support for this format in version 2.0. Always use **xlrd==1.2.0**. Never substitute **openpyxl** – it reads **.xlsx** only.

F.2 MATLAB

MATLAB R2021a or later is required. The following toolboxes are used:

- Optimization Toolbox (**fzero**)
- Statistics and Machine Learning Toolbox (optional, for AIC/BIC)

No additional file-path configuration is needed; the driver scripts use relative paths from **Analises/MatLab/**.

F.3 LaTeX

This document was compiled with pdfLaTeX (TeX Live 2023+). Required packages: **amsmath**, **amssymb**, **booktabs**, **siunitx**, **tcolorbox**, **tikz**, **natbib**, **lastpage**, **titlesec**, **fancyhdr**,

`enumitem`. Install via `tlmgr` or use an Overleaf project with TeX Live 2023 compiler.

References

- H. A. Barnes. Thixotropy—a review. *Journal of Non-Newtonian Fluid Mechanics*, 70(1–2):1–33, 1997. doi: 10.1016/S0377-0257(97)00004-9.
- A. Blaeser, D. F. Duarte Campos, U. Puster, W. Richtering, M. M. Stevens, and H. Fischer. Controlling shear stress in 3D bioprinting is a key factor to balance printing resolution and stem cell integrity. *Advanced Healthcare Materials*, 5(3):326–333, 2016. doi: 10.1002/adhm.201500677.
- W. P. Cox and E. H. Merz. Correlation of dynamic and steady flow viscosities. *Journal of Polymer Science*, 28(118):619–622, 1958. doi: 10.1002/pol.1958.1202811812.
- G. M. da Silva, R. F. P. da Rocha, M. P. M. da Costa, I. L. d. M. Ferreira, and M. C. Delpech. Evaluation of viscometric properties of carboxymethylcellulose and gellan. *Journal of Molecular Liquids*, 268:201–205, 2018. doi: 10.1016/j.molliq.2018.07.062.
- A. Habib, V. Sathish, S. Mallik, and B. Khoda. 3D printability of alginate-carboxymethyl cellulose hydrogel. *Materials*, 11(3):454, 2018. doi: 10.3390/ma11030454.
- I. M. Krieger and T. J. Dougherty. A mechanism for Non-Newtonian flow in suspensions of rigid spheres. *Transactions of the Society of Rheology*, 3(1):137–152, 1959. doi: 10.1122/1.548848.
- A. B. Metzner and J. C. Reed. Flow of non-newtonian fluids—correlation of the laminar, transition, and turbulent-flow regions. *AIChE Journal*, 1(4):434–440, 1955. doi: 10.1002/aic.690010409.
- J. Mewis and N. J. Wagner. Thixotropy. *Advances in Colloid and Interface Science*, 147–148: 214–227, 2009. doi: 10.1016/j.cis.2008.09.005.
- M. Mooney. Explicit formulas for slip and fluidity. *Journal of Rheology*, 2(2):210–222, 1931. doi: 10.1122/1.2116364.
- N. Paxton, W. Smolan, T. Bock, F. Melchels, J. Groll, and T. Jungst. Proposal to assess printability of bioinks for extrusion-based bioprinting and evaluation of rheological properties governing bioprintability. *Biofabrication*, 9(4):044107, 2017. doi: 10.1088/1758-5090/aa8dd8.
- A. Schwab, R. Levato, M. D’Este, S. Piluso, D. Eglin, and J. Malda. Printability and shape fidelity of bioinks in 3D bioprinting. *Chemical Reviews*, 120(19):11028–11055, 2020. doi: 10.1021/acs.chemrev.0c00084.
- R. I. Tanner. A theory of die-swell. *Journal of Polymer Science Part A-2: Polymer Physics*, 8(12):2067–2078, 1970. doi: 10.1002/pol.1970.160081203.
- M. L. Williams, R. F. Landel, and J. D. Ferry. The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids. *Journal of the American Chemical Society*, 77(14):3701–3707, 1955. doi: 10.1021/ja01619a008.
- H. H. Winter and F. Chambon. Analysis of linear viscoelasticity of a crosslinking polymer at the gel point. *Journal of Rheology*, 31(8):683–697, 1987. doi: 10.1122/1.549955.